

Modelos Fundacionales y Multimodales

Modelos Multimodales

Pablo Hernández Cámara
pablo.hernandez-camara@uv.es

2/3 Septiembre 2026

Índice

1. Introducción
2. Texto-imagen:
 - 2.1 Tareas
 - 2.2 Datasets
 - 2.3 Modelos
 - 2.3.1 Contrastivos
 - 2.3.2 MLLM
3. Multimodal image generation
4. Otras modalidades

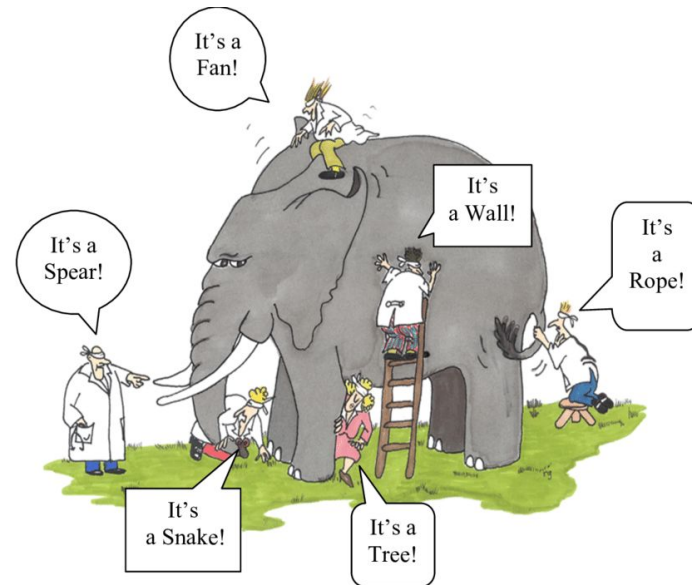
Introducción

Modelos multimodales: Introducción

Vivimos en un mundo multimodal

5 sentidos, 5 modalidades de percepción

¿Por qué utilizar sólo datos de una modalidad?



Regla de comunicación 7-38-55 ([Silent messages, Albert Mehrabian, 1972](#))

“In communication, 7% of the overall meaning is conveyed through verbal mode (spoken words), 38% through voice and tone and 55% through body language and facial expressions.”

- Modalidad textual 7%
- Modalidad audio 38%
- Modalidad vision 55%



Una conversación también es multimodal

Modelos multimodales: Introducción

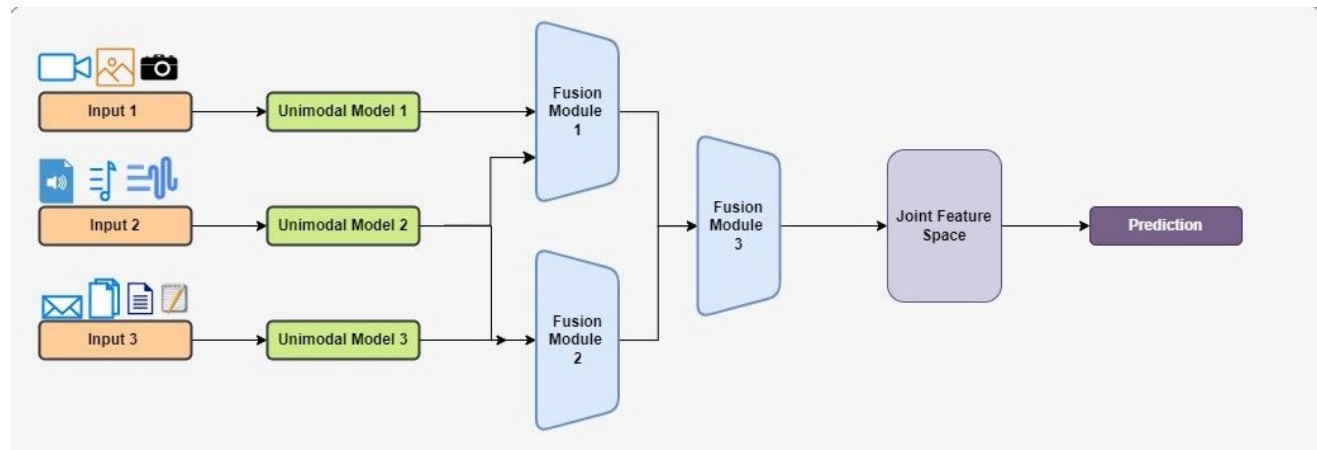
Ejemplos:

- **Vision + Texto:** Memes, Articulos con imagenes...
- **Vision + Audio:** Videollamada...
- **Audio + Texto:** Notas de voz, Canciones con la letra...
- **Vision + Audio + Texto:** Video subtitulado, tiktoks (y otras redes sociales)...

Multimodal deep learning

¿Qué significa realmente que un modelo sea multimodal?

Aquel capaz de procesar, comprender y combinar simultáneamente diferentes tipos de datos dentro de una misma arquitectura.



Multimodal deep learning

¿Por qué es importante?

- **Uni-modal:** Limitado por el uso de datos de un solo tipo (image classification, sentiment analysis...).
- **Multimodal:**
 - a) mejor representación del mundo
 - b) aumenta la información disponible
 - c) modelos más completos y con mejores capacidades

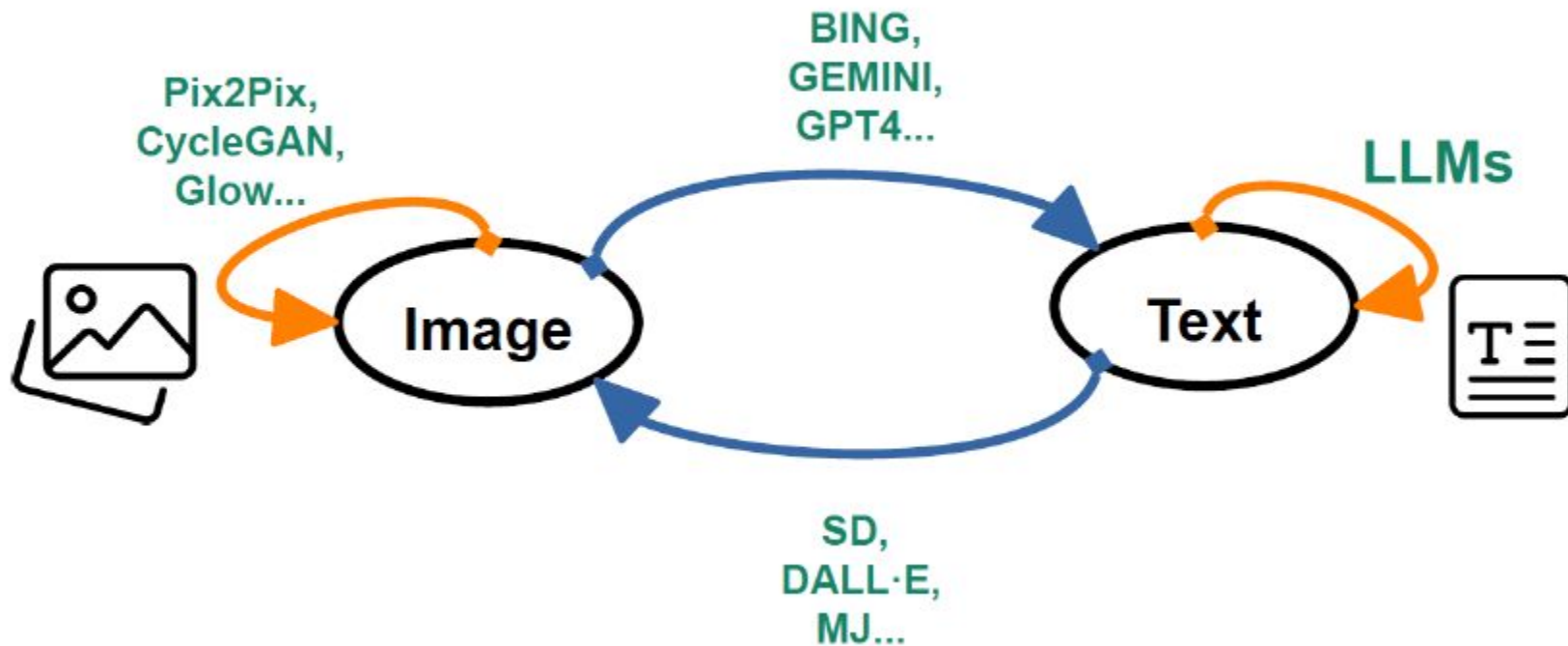
Modelos multimodales: Introducción

Multimodal Deep Learning

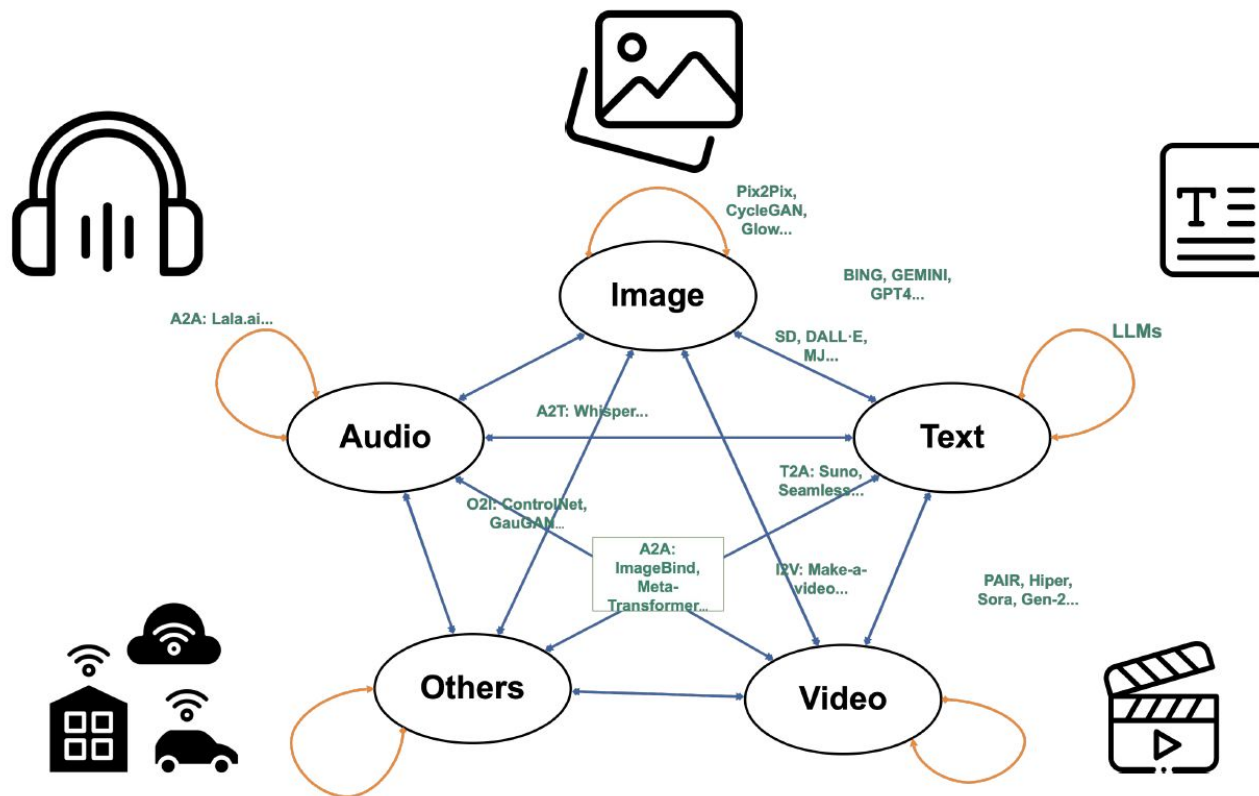
- Modalidades básicas: texto, imagen, audio
- Otras modalidades: LIDAR, profundidad, EEG...



Modelos multimodales: Introducción



Modelos multimodales: Introducción



Dificultades de la multimodalidad:

- Una modalidad puede dominar al resto (número de datos, importancia...)
- Modalidades adicionales pueden introducir ruido
- Relaciones complicadas entre distintas modalidades

Texto e imagen

- Tareas
- Datasets
- Tipos de modelos
 - Contrastive models
 - Vision Language Models (VLM)
 - Image Generation

Texto e imagen

Tareas

4 familias de tareas:

- a) Imagen $\leftrightarrow$ texto: Ejemplo, matching/retrieval/alignment.
 - i)  $\leftrightarrow$ "Un perro corriendo"
- b) Imagen $\rightarrow$ texto: Ejemplo, image captioning.
 - i)  $\rightarrow$ "Un perro corriendo por un parque"
- c) Imagen + texto $\rightarrow$ respuesta: Ejemplo, VQA/Multimodal reasoning
 - i)  + "¿De qué color es el coche?" $\rightarrow$ "Rojo"
- d) Texto $\rightarrow$ imagen: Ejemplo, Text-to-image generation.
 - i) "Un perro corriendo por un parque" $\rightarrow$ 

Image-Text Retrieval: Hacer “match” entre textos e imágenes.

Image-text Retrieval (Text-to-Image Retrieval)

Text Query: A dog lying on the grass next to a frisbee

Match



Not Match



Visual grounding: Localizar los objetos descritos en la imagen.



A dog is lying on the grass next to a frisbee.

(a) Phrase grounding.



The red frisbee next to the dog.

(b) Referring expression comprehension.

Image Captioning: Describir imágenes.



Image Captioning (Paragraph)

Caption: There is a white dog lying on a grass field. There are a lot of leaves on the grass field. There is a chain-link fence next to the dog. There is a red frisbee under the dog's left-front paw.



Image Captioning (Single Sentence)

Caption: A dog tries to catch a yellow, flying frisbee.

Visual Question Answering (VQA) and Visual Reasoning: Responder preguntas y razonar sobre una imagen.



Visual Question Answering

Q: What is the dog holding with its paws?

A: Frisbee.



Visual Reasoning

Q: Is the dog in the air **AND** is the frisbee in the air?

A: Yes

Texto e imagen: Tareas

Document Visual Question Answering (DocVQA): Responder preguntas sobre documentos (parecido a OCR).

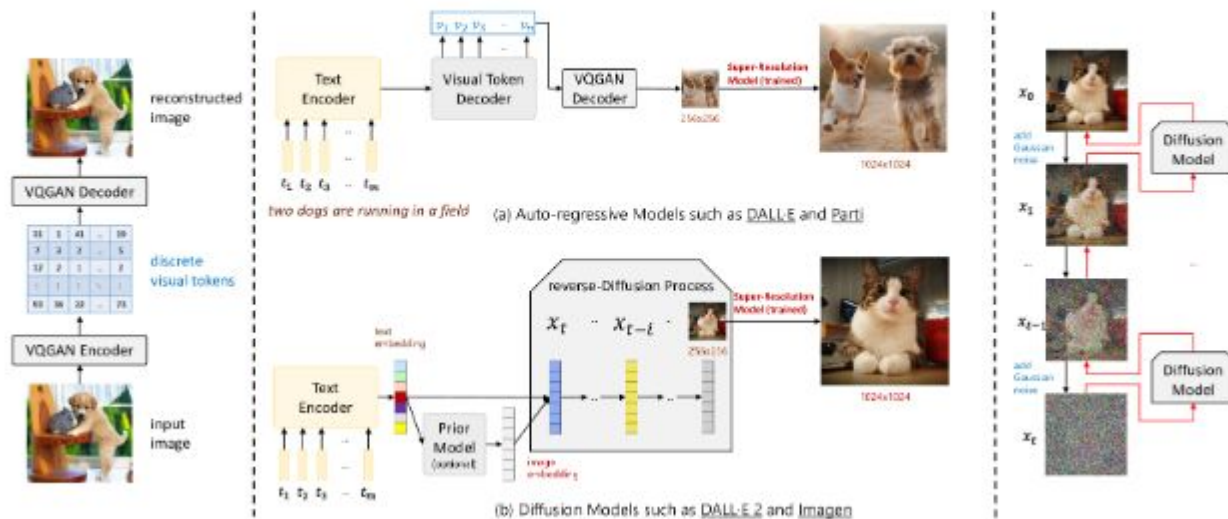
PUBLIC DISCLOSURE COMMISSION		Candidate Registration	C1
 775 CAPITAL WAY STE 100 PO BOX 9049 OAKLAND CA 94616-0049 (510) 764-8100 FAX (510) 764-8101			
Candidate's Name (Use candidate's full name.) ANNA M. RIVERS			
Candidate's Committee Name (Do not abbreviate.) FRIENDS OF ANNA RIVERS			
Mailing Address PO BOX 9047			
City LA CENTER County CLATSOP Zip + 4 97123			
1. What office are you running for? STATE SENATOR Legislative District, County or City LEG DISTRICT 17 - SENATE Precinct No. N/A			
2. Political party (if partisan office) REP/INDEP 3. Date of general or special 11-02-2016			
4. How much do you plan to spend during your entire election campaign, including the primary and general elections? See the reporting options below. If no box is checked you are obligated to use Option B, Full Reporting. See instruction manuals for and changing reporting options.			
☐ Option I: MINI REPORTING : In addition to my filing fee of \$_____, I will raise and spend no more than \$5,000, including any loan interest payments. I will not accept more than \$500 in the aggregate from any contributor except "in-kind." ☒ Option II: FULL REPORTING : I will use the Full Reporting system. I will file the frequent, detailed campaign reports required by law.			
5. Treasurer's Name and Address. Does treasurer perform only ministerial functions? Yes ___ No ___ See WAC 206-05-045 and note page for details. List deputy treasurer on attached sheet. ☐ Treasurer is officer or agent			
ANNA M. RIVERS PO BOX 9047, LA CENTER WA 99129			

PUBLIC DISCLOSURE COMMISSION		Candidate Registration	C1
 775 CAPITAL WAY STE 100 PO BOX 9049 OAKLAND CA 94616-0049 (510) 764-8100 FAX (510) 764-8101			
Candidate's Name (Use candidate's full name.) ROBERT REEDY			
Candidate's Committee Name (Do not abbreviate.) CLATSOP COUNTY CHARTER			
Mailing Address PO Box 61			
City Mountlake Terrace County Sno Zip + 4 98043			
1. What office are you running for? CLATSOP COUNTY CHARTER Legislative District, County or City SNO Precinct No. 4			
2. Political party (if partisan office) NIP 3. Date of general or special 11-02-2016			
4. How much do you plan to spend during your entire election campaign, including the primary and general elections? See the reporting options below. If no box is checked you are obligated to use Option B, Full Reporting. See instruction manuals for and changing reporting options.			
☒ Option I: MINI REPORTING : In addition to my filing fee of \$_____, I will raise and spend no more than \$5,000, including any loan interest payments. I will not accept more than \$500 in the aggregate from any contributor except "in-kind." ☐ Option II: FULL REPORTING : I will use the Full Reporting system. I will file the frequent, detailed campaign reports required by law.			
5. Treasurer's Name and Address. Does treasurer perform only ministerial functions? Yes ___ No ___ See WAC 206-05-045 and note page for details. List deputy treasurer on attached sheet. ☐ Treasurer is officer or agent			
ROBERT REEDY			

Q: In which years did Anna M. Rivers run for the State senator office?

A: [2016, 2020]

Text-to-Image Generation: Creación de imágenes descritas.



Texto e imagen

Datasets

Texto e imagen: Datasets

Los datasets han cambiado de escala:

Datasets “curados”: datos seleccionados y anotados manualmente por humanos, de miles o pocos de millones de datos



Datasets de web: datos extraídos de la web mediante filtrado automatizado y heurísticas, de millones de datos.



Datasets de web de gran escala: extracción masiva sin filtrado humano profundo, confiando en descripciones alt e índices web públicos: miles de millones de datos.

Para entrenar modelos multimodales hoy en día se usan los datasets más grandes.

Ejemplos de datasets típicos (curated):

- [MSCOCO](#): 328k imagenes + 5 captions/imagen + otras tareas.
- **Conceptual Captions**: 3M imagenes + captions.

Para entrenar modelos multimodales hoy en día se usan los datasets más grandes.

Ejemplos de datasets típicos (curated):

- [MSCOCO](#): 328k imágenes + 5 captions/imagen + otras tareas.
- **Conceptual Captions**: 3M imágenes + captions.

Ejemplos de datasets típicos (web):

- **ALIGN**: 1.8B imágenes + captions.

Para entrenar modelos multimodales hoy en día se usan los datasets más grandes.

Ejemplos de datasets típicos (curated):

- [MSCOCO](#): 328k imágenes + 5 captions/imagen + otras tareas.
- **Conceptual Captions**: 3M imágenes + captions.

Ejemplos de datasets típicos (web):

- **ALIGN**: 1.8B imágenes + captions.

Ejemplos de datasets típicos (web gran escala):

- [CommonPool](#): desde 12.8M a 12.8B pares texto-imagen.
- [LAION](#) **datasets**: principal referencia de datasets web-scale de imagen-texto de acceso abierto. Hasta 5B.

Texto e imagen: Datasets

Pero...Y como se descargan? Hay que ir imagen por imagen dándole a “descargar”?

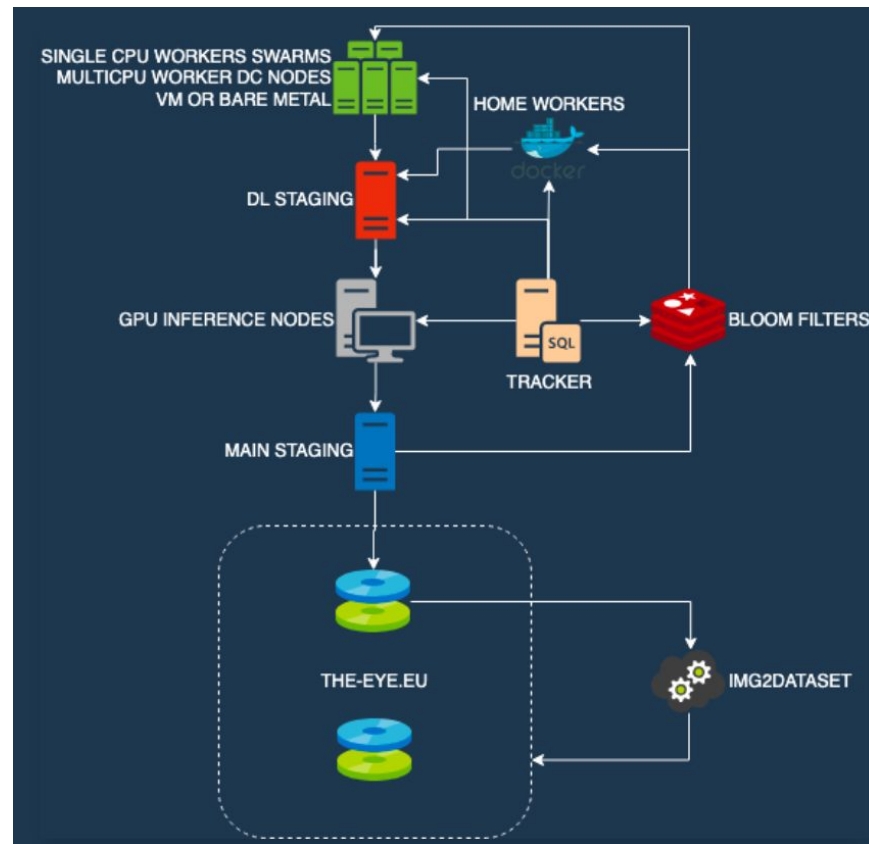
Por suerte existe [img2dataset](#):

Librería que permite descargar fácil (ojo, que no rápido) muchos de estos datasets. También permite filtrar al descargar los datos.

Texto e imagen: Datasets

Normalmente los pasos (para crear el dataset y para descargarlo) son:

- 1) URL + caption
- 2) Descarga
- 3) Obtención imagen
- 4) Validación
- 5) Filtrado
- 6) Dataset final



Si tengo varios miles de millones de URLs...qué puede salir mal?

Problemas usuales: Ruido, duplicados, copyright/licencias, contenido inapropiado, imágenes que desaparecen, calidad desigual, coste computacional, deduplicación, filtrado...

El dataset que se publica no es necesariamente el dataset que usamos:

Dataset publicado

- URLs muertas
- contenido eliminado
- errores descarga
- filtros
- deduplicación = Dataset final

Ejemplo, hasta el 30% son dead links y URLs que ya no funcionan!

Training dataset & objective. We pre-trained CLIP and MaMMUT models on DataComp-1.4B [19], Re-LAION-1.4B [21] and DFN-1.4B [20] datasets. Re-LAION-1.4B was obtained by downloading the Re-LAION-2B-en-research subset of Re-LAION [21] which contained $\approx 30\%$ dead links, resulting in a total of 1.4B image-caption pairs. DFN-1.4B was obtained by downloading DFN-2B dataset [20], which also resulted in discovering $\approx 30\%$ link rot, again providing 1.4B image-text

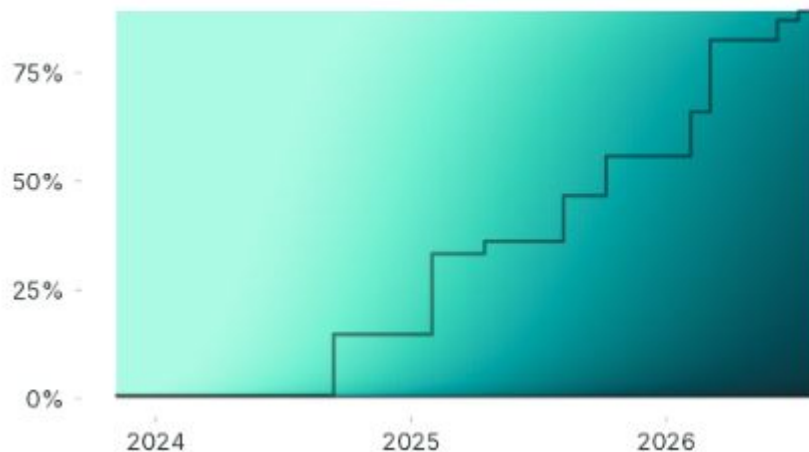
Datasets de evaluación: suelen ser mucho más pequeños y se usan para medir las capacidades de los modelos.

Un benchmark no es una propiedad intrínseca del modelo; es una medición bajo un protocolo concreto.

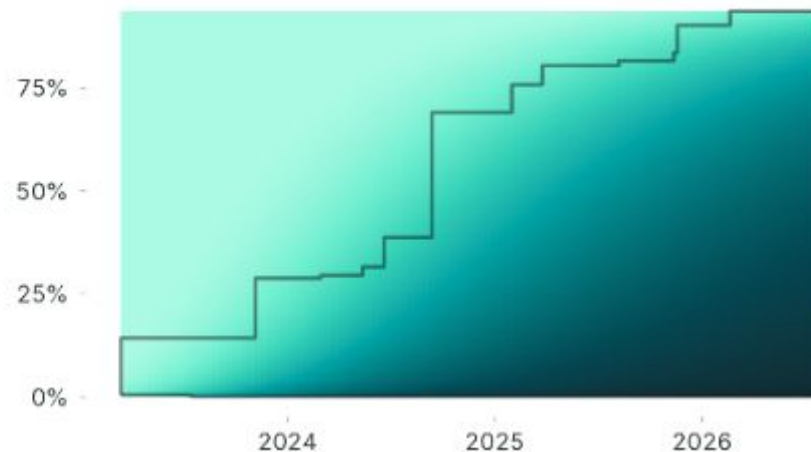
También puede haber problemas: contaminación, saturación de benchmarks, evaluación automática vs humana...

Datasets de evaluación: saturación de benchmarks.

FrontierMath, tiers 1-3



GPQA Diamond



Hay datasets de muchos tipos (conocimiento general, razonamiento matemático, razonamiento científico, generación de código...).

Algunos de los más famosos:

- [MMMU](#): 11.5K preguntas multimodales nivel universidad.
- [VQA: Visual Question Answering](#): preguntas y respuestas de sentido común.
- [Humanity Last Exam](#): actual, “frontier of human knowledge”.
- [Furniture Assembly Benchmark](#): Errores montando muebles de Ikea.
- [Arena.ai](#): Para evaluar nosotros mismos o ver el ranking.
- [ArtificialAnalysis.Ai](#): Evalúan modelos de forma independiente.

¿Cómo se evalúa un modelo multimodal moderno? Normalmente en múltiples áreas (conocimiento, razonamiento, visión, coding, agentic use...). En los model card suele haber decenas de benchmarks.

Ejemplos:

- <https://huggingface.co/Qwen/Qwen3.5-397B-A17B>
- <https://blog.google/innovation-and-ai/models-and-research/gemini-models/gemini-3-5/>
- <https://www.anthropic.com/claude-fable-and-mythos-5-1>

Texto e imagen: Datasets

Y los videos también ayudan: Son múltiples imágenes.
Como? Extraer N frames y hacer la media

Image QA		Q: Which Place is shown in the image? A: A serene lake scene with a monument featuring a man on horseback.	Image OCR		Q: What location does this map represent? A: The map is of Kansai International Airport (KIX), which is located in Japan.	Multi Image		Q: Spot the difference between these two images. A: The color of the pole on the dog's right changes from black to light red.
Video QA		Q: What is the person doing in the video? A: Person is wearing ring in left hand and cutting the onions with a knife.	Video Cap.		Q: Briefly caption the video. A: The video shows a person making avocado sandwich by mashing the avocado and spreading it on bread.	PLM-FGQA		Q: How does a person pour the tea into the mug? A: The person moves the teapot up and down three times while pouring the tea into the mug.
PLM-RTLLoc		Q: Localize the event. "Person facing towards the camera is dancing." for the shaded region? A: [0, 13]	PLM-RCap		Q: Describe the highlighted subject in the video between frames [0, 13]. A: A person facing towards the camera is dancing.	PLM-RDcap		Q: Provide a dense caption for the shaded region. A: [0, 12] Kid is playing hide and seek and hides behind a tree. [13, 16] The kid reappears.

Texto e imagen: Datasets

El más reciente y grande: [LAION Big Video Dataset](#)



Texto e imagen: Datasets

Un modelo puede entrenarse en una combinación de muchos datasets. Ejemplo:

[Meta PerceptionLM](#)

Image

Video

Text

Image QA				Grounding		Video Temporal Loc.	
Dataset	Size	Dataset	Size	Dataset	Size	Dataset	Size
DVQA [22]	222222	STAR [496]	3032	VisualGenome [62]	154792	HiREST [214]	7919
PlotQA [122]	157070	NEaT-QA [200]	3870	Fllicker Entities [147]	296332	Charades [162]	7566
MapQA [21]	42761	VISION [10]	9900	DCI (Region Caption) [177]	304912	DeDeMo [2]	435
OCRQA [134]	167646	FlintstonesSV [63]	22341	ReCOCO [8/1-22]	212923	Total	15920
Localized Narratives [148]	199998	ImageCoDe [84]	16594	VCR [210]	855577		
FigureQA [23]	119999	ViaWiz [13]	4900	Total	1398690		
Hateful Memes [80]	9713	MIT-States (State Coherence) [46]	1900			Video Region Captioning	
CLEVR [21]	73181	MIT-States (Prop. Coherence) [46]	1900	Image Synth.		Dataset	Size
CLEVR v1.0 [71]	70000	WebQA [22]	9338	Dataset	Size	HC-STVG [172]	10131
IconQA [113]	116514	Birds to Words [40]	14281	DocVQA [124]	50170	VidILN (UVO subset) [173]	5296
TextVQA [105]	21953	AESOP [151]	6915	InfographicVQA [125]	21660	Total	15427
GeoVerse [74]	11162	RecipeQA (Img. Coherence) [201]	8699	PDFAcc [Cap.][141]	12024670		
RobuT (wikispl) [228]	80757	CLEVR-Change [142]	3885	PDFAcc (QA) [141]	12024670	Video Dense Cap.	
WebSight [225]	10000	IEdu [114]	3456	UCSF [140]	5953490	Dataset	Size
VisualFW [226]	15961	ChartQA [123]	73888	ArxivCap [25]	1859680	ActivityNet [48]	8859
TallyQA [3]	100050	DocVQA [124]	109023	SA1B [34]	9834573	YouCook [233]	1039
RobuT (WTQ) [228]	42495	InfographicVQA [125]	56599	Object365 [169]	3484584	Total	9898
DaTikz [12]	47974	TextVQA [105]	69170	OpenImages [83]	1740864		
CocoQA [154]	46287	TextCaps [62]	21324	PixmoCap [11]	584650		
ChartQA [123]	27395	VisualMRC [160]	45526	Total	47579011	Video Synth.	
VQA v2 [60]	82772	WTQ [143]	228994			Dataset	Size
Chart2Text [136]	35946			Video QA		Spoken Moments [132]	449044
VisText [170]	35995	HME100k [211]	74492	Dataset	Size	Charades [162]	7919
FinQA [41]	5276	chrome_writing [184]	8825	EgoQA [134]	7813	Kinetics710 [22]	39949
DocVQA [124]	12089	OK-VQA [121]	27536	NExT-QA (Instruct) [200]	34114	DDMe [8]	7566
STVQA [14]	18684	Geometry3k [117]	4802	NExT-QA (MCQ) [200]	34114	Ego4D (Cap.) [69]	183029
TAT-QA [224]	2199	VQA-RAD [86]	1793	PerceptionTest [145]	2403	Ego4D (QA) [69]	703935
RenderedText [194]	10435	Total	3141377	ActivityNetQA [211]	23530	YT-1B (Cap.) [212]	14792983
RAVEN [215]	31418			VideoInstruct (human) [118]	25803	YT-1B (QA) [212]	3383670
IAM [122]	7549	Image Cap.		CLEVRER (MC) [208]	42620	Total	19568095
A-OKVQA [156]	17720	Dataset	Size	CLEVRER (QA) [208]	40006		
TabMWP [116]	45439	DOCCI [137]	13362	Kinetics710 [22]	39949	Text-QA	
CocoQA [154]	9009	DCI [177]	7599	SSv2 (classification) [58]	40000	Dataset	Size
TextCaps [62]	21953	Altogether [201]	15166	VidILN [179]	43126	no robots [159]	9485
Screen2Words [180]	16713	Total	36127	VidILN (QA) [179]	45731	MathQA [2]	29837
VSR [102]	2157			How2QA [26]	35297	LIMA [230]	1030
TQA [23]	9742	Image Misc.		STAR [106]	35297	GSM8k (socratic) [26]	7473
RobuT (SQ) [228]	12769	Dataset	Size	Memento [189]	40060	GSM8k [26]	7473
VisualMRC [160]	3027	AI2d [22]	12413	Memento-MultiImage [189]	40060	FLAN [104]	1366050
ScienceQA [114]	9947	COCO cap. [101]	414113	Total	569710	Dolly15k [167]	15011
VQA-RAD [86]	313	GQA-Balanced [85]	943000			Maggie Pro (MT) [203]	300000
InfographicVQA [125]	2118	Total	36127	Video Cap.		Maggie Pro [203]	300000
Hitab [37]	4995			Dataset	Size	Total	2056359
AI2D [78]	4863			VATEX (en caption) [188]	259910		
Inter-GPS [112]	2555			Charades (caption) [162]	11593		
diagram image to text [4]	595			ActivityNet (captions) [48]	33375		
MIMIC-IT (CGD) [80]	70939			YouCook2 [232]	10337		
MultiHertL [222]	15233						
NLVR2 [165]	136799						
RAVEN (Multi-image) [219]	56081						
SpotTheDiff [42]	19340						

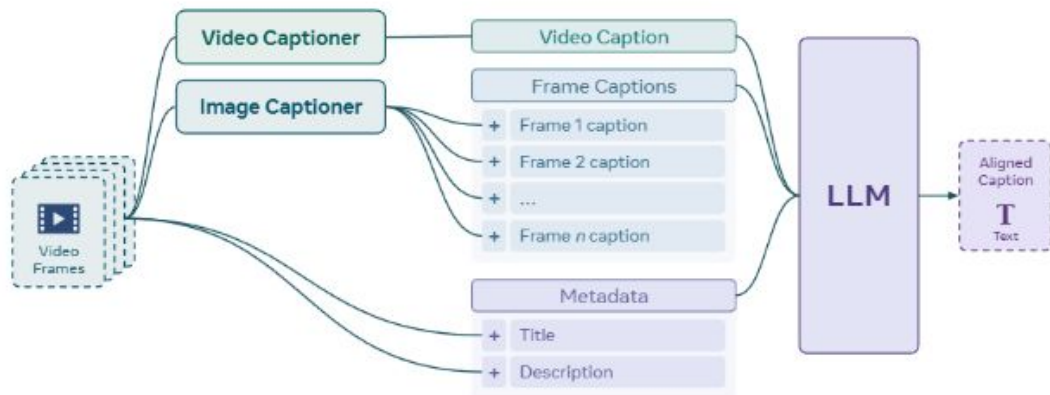
Texto e imagen: Datasets

Hacen falta muchísimos datos, pero... las anotaciones humanas son costosas (caras)

Synthetic Data

“The synthetic data can be noisy, but is available at large scale; on the other hand, human-annotated data provides rich, high-quality supervision for image and video tasks.”

Meta PerceptionLM

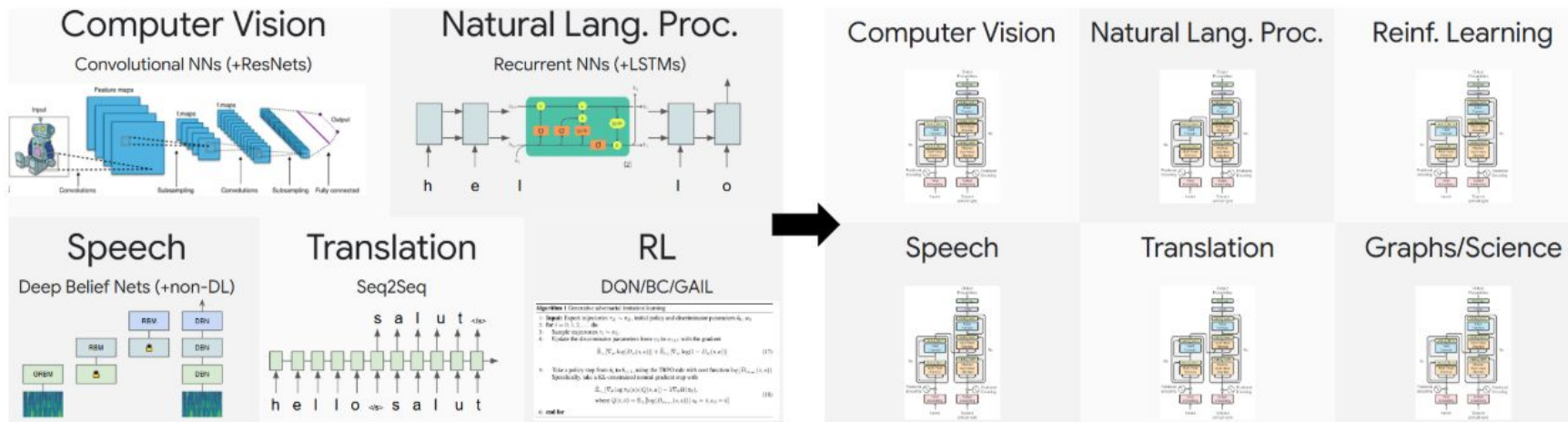


Texto e imagen

Modelos

Texto e imagen: Modelos

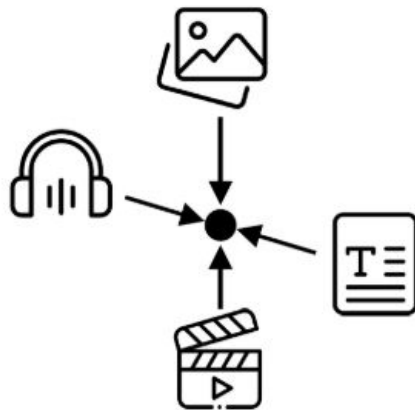
Todos basados en Transformers:



Texto e imagen: Modelos

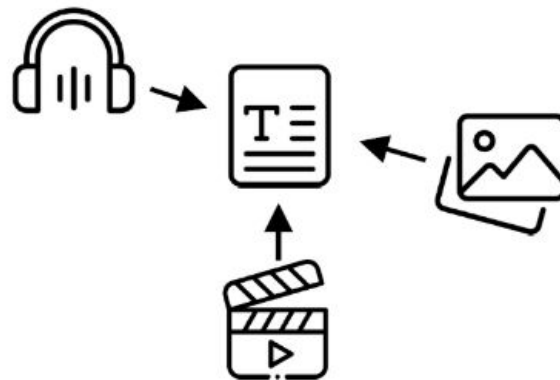
Principalmente 2 tipos de modelos

Common inner domain



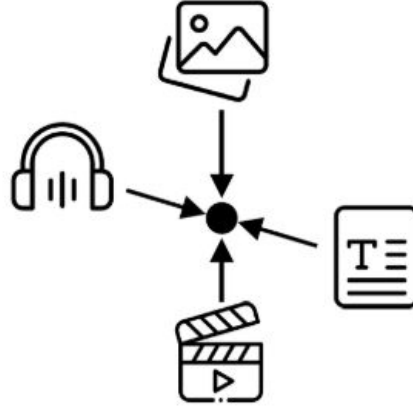
Contrastive models (CLIP)

An anchoring modality



MLLM (VLM)

Common inner domain



Contrastive models (CLIP)

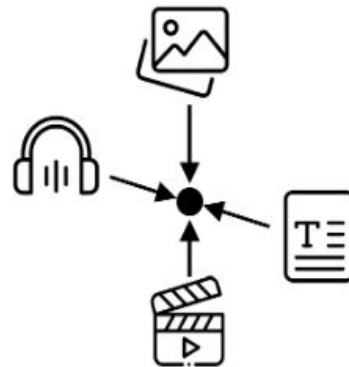
Texto e imagen: Modelos contrastivos

Hay un momento que cambia el desarrollo de este tipo de modelos:
5 de enero de 2021, OpenAI presenta [CLIP](#)

Trabajos AC (Antes CLIP):

- [Multimodal Deep Learning by Ngiam et al. \(2011\)](#)
- [Deep Visual-Semantic Alignments for Generating Image Descriptions by Karpathy and Fei-Fei \(2015\)](#)
- [Show and Tell: A Neural Image Caption Generator by Vinyals et al. \(2015\)](#)

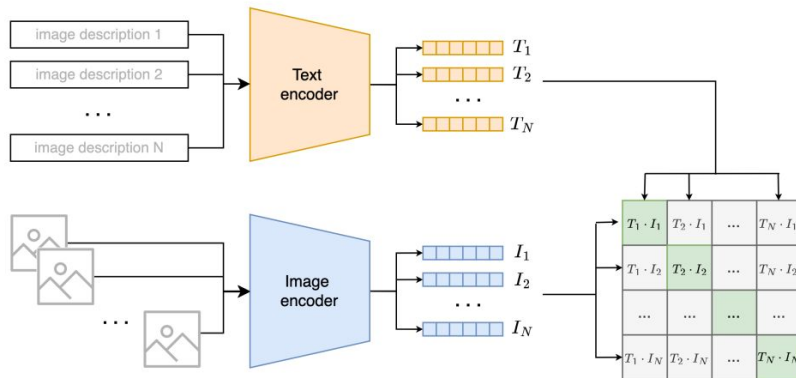
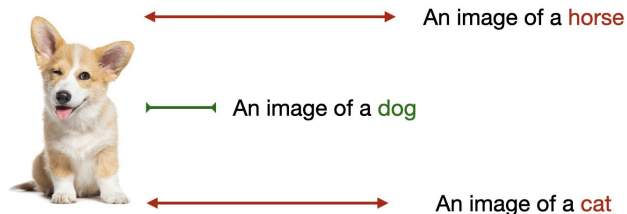
Common inner domain



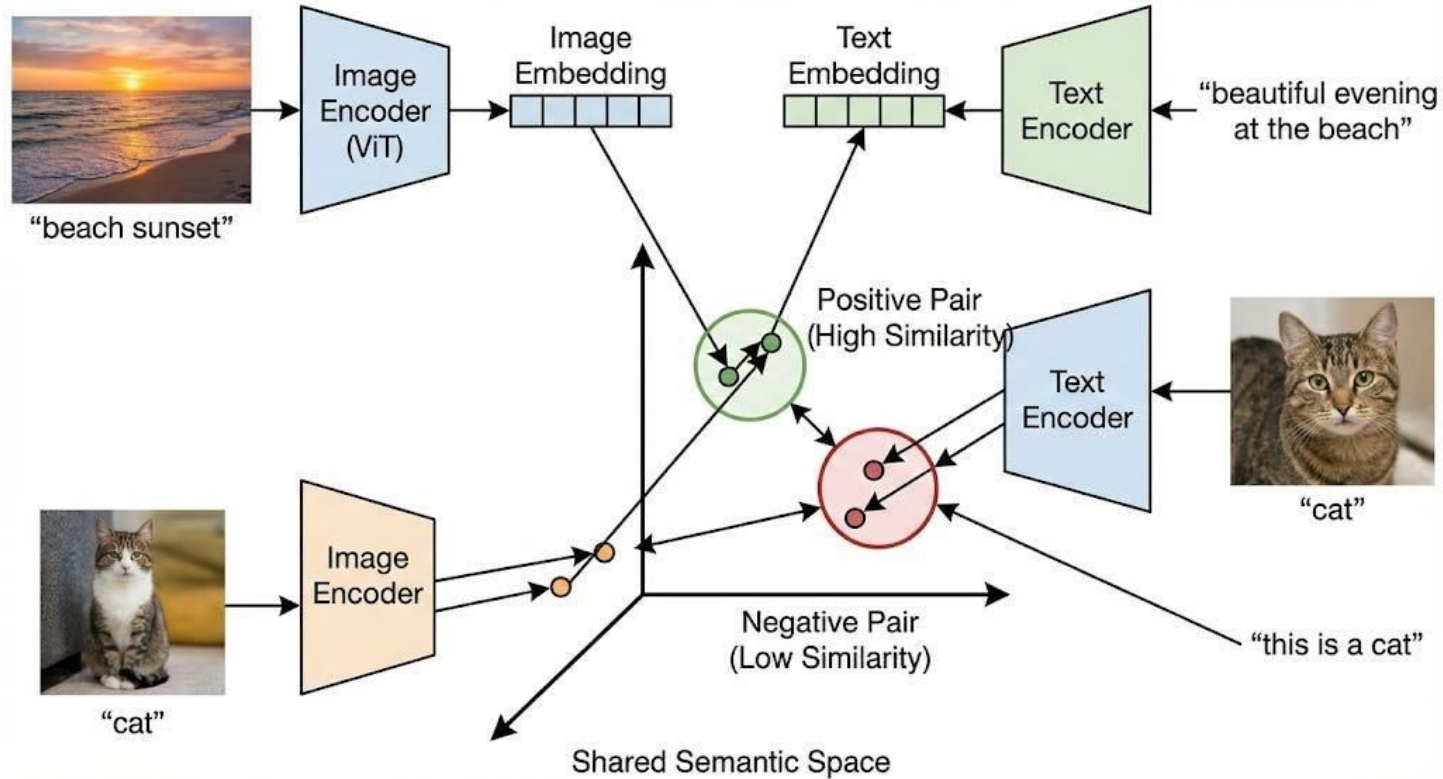
Texto e imagen: Modelos contrastivos

CLIP:

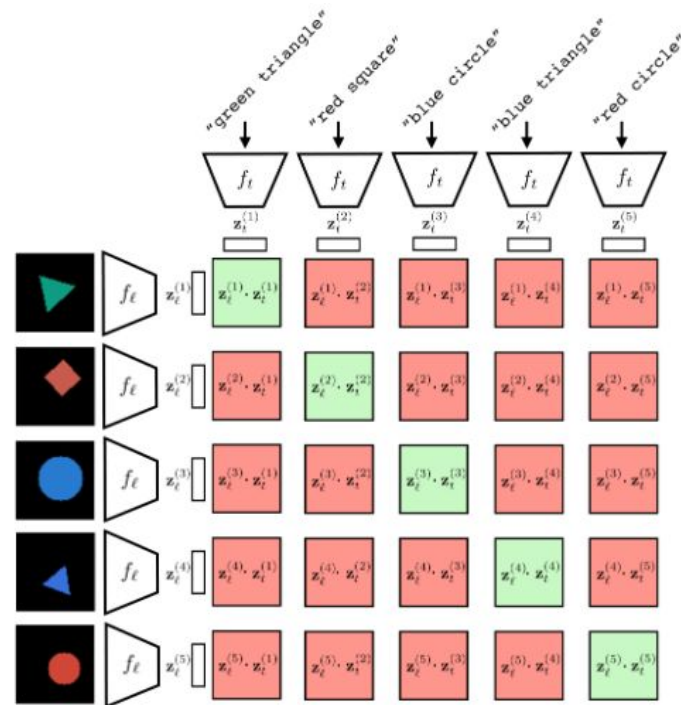
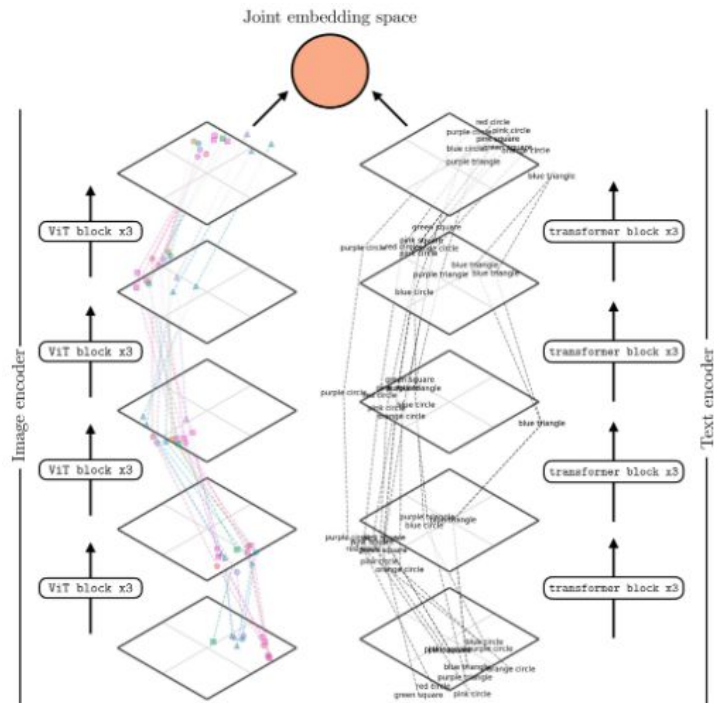
Son las siglas de: **C**ontrastive **L**anguage - **I**mage **P**retraining.
El objetivo era entrenar un modelo de imágenes usando texto.
Se hicieron su propio dataset: 400M, que no es publico.



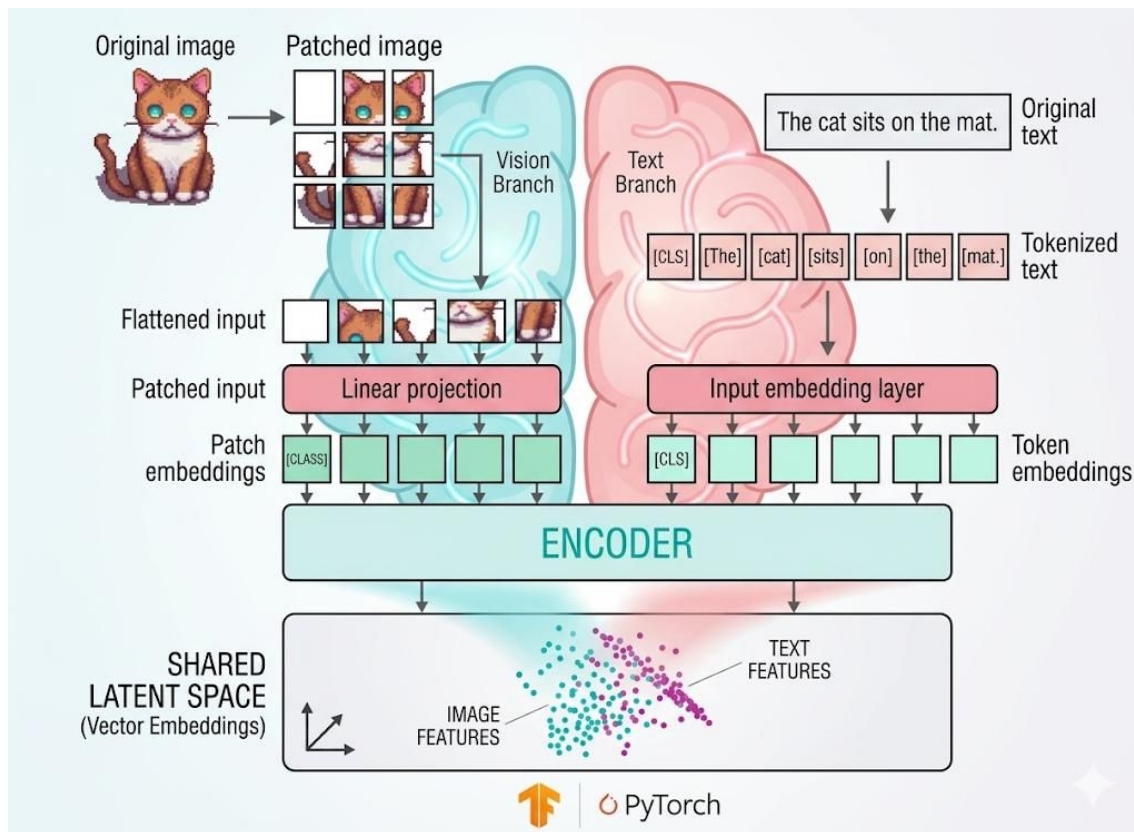
Texto e imagen: Modelos contrastivos



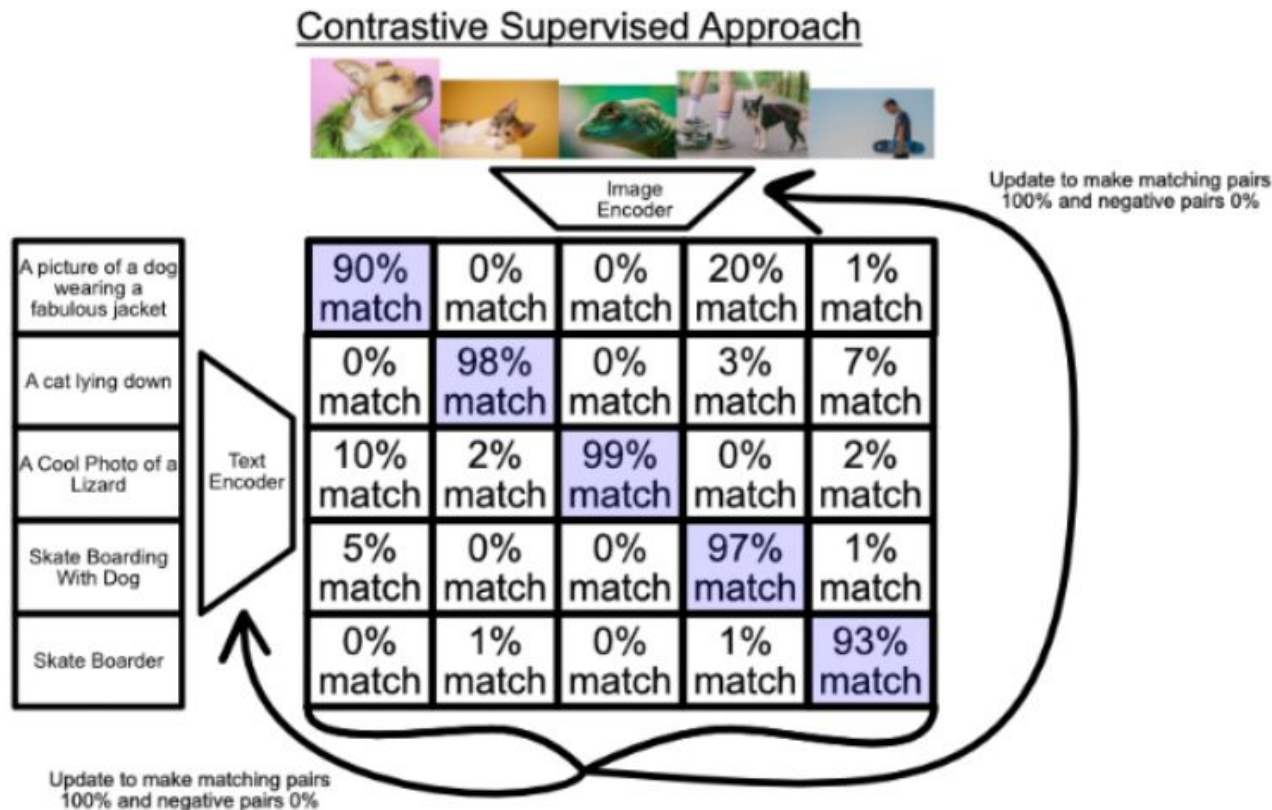
Texto e imagen: Modelos contrastivos



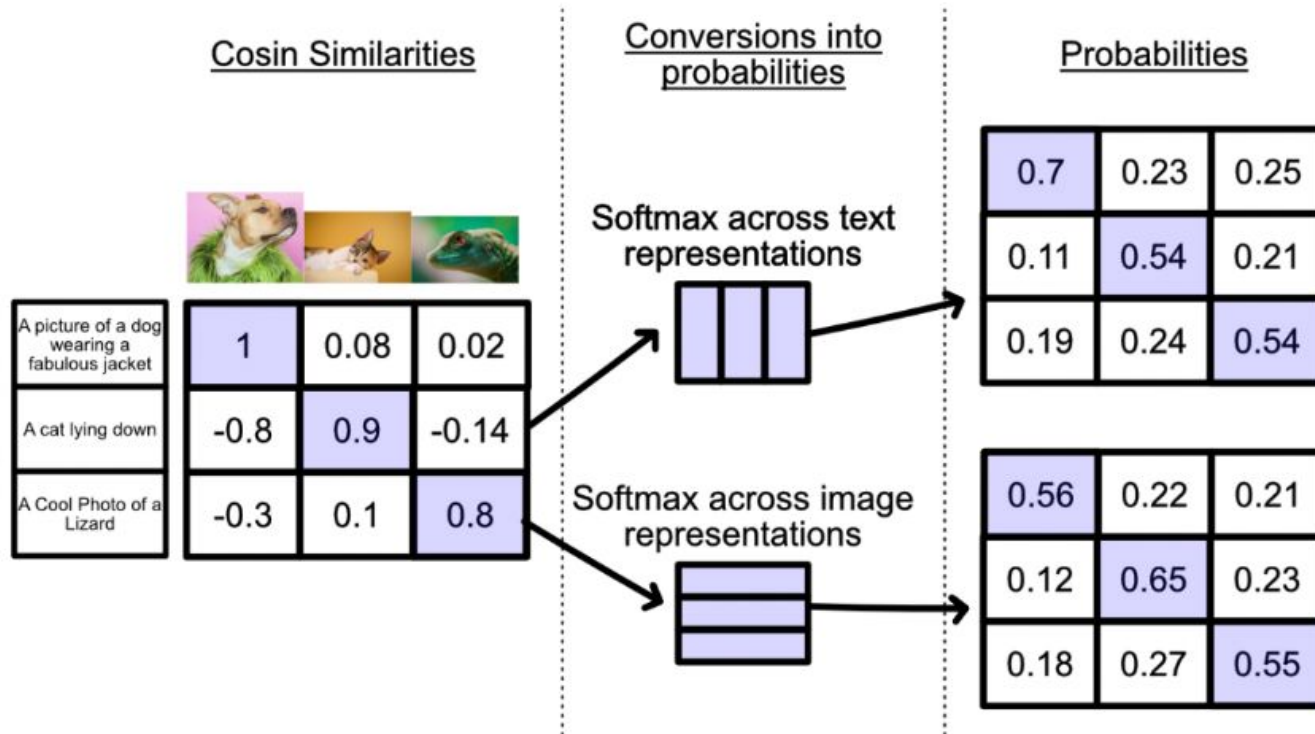
Texto e imagen: Modelos contrastivos



Texto e imagen: Modelos contrastivos



Texto e imagen: Modelos contrastivos



InfoNCE Loss

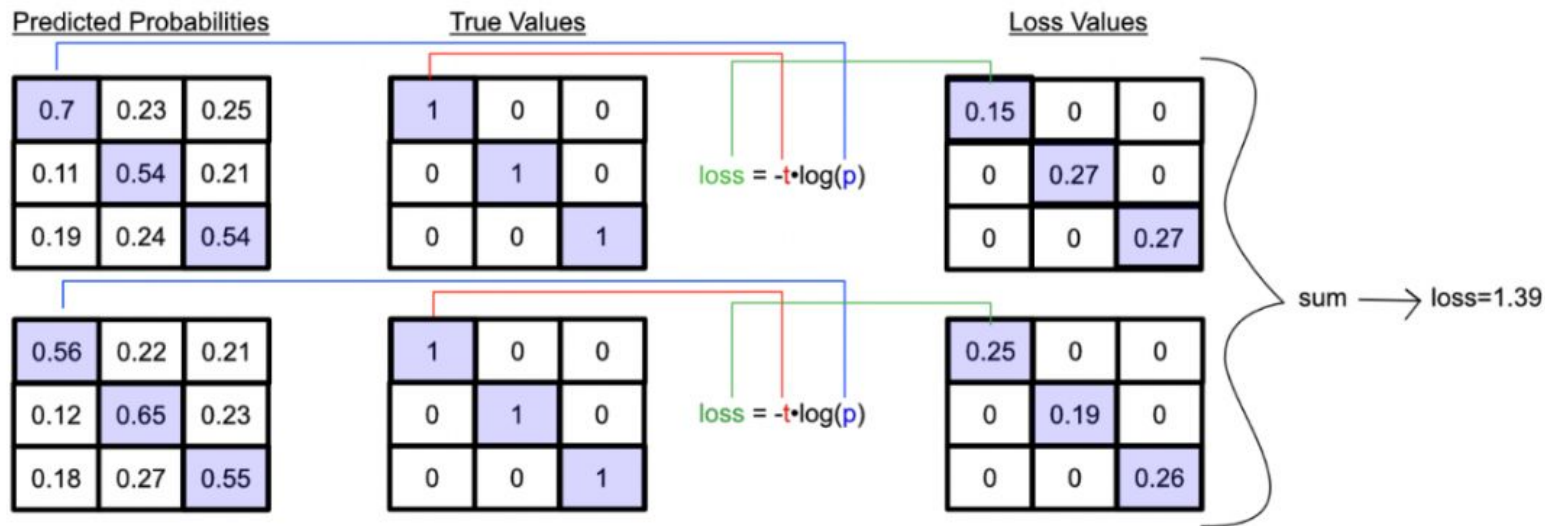
For a batch of N pairs, the diagonal $S_{i,i}$ represents positive pairs, and off-diagonals $S_{i,j}$ ($i \neq j$) represent negatives.

$$L_{I \rightarrow T} = -\frac{1}{N} \sum_{i=1}^N \log \frac{\exp(S_{i,i})}{\sum_{j=1}^N \exp(S_{i,j})}$$

$$L_{T \rightarrow I} = -\frac{1}{N} \sum_{j=1}^N \log \frac{\exp(S_{j,j})}{\sum_{i=1}^N \exp(S_{i,j})}$$

$$L_{\text{CLIP}} = \frac{1}{2} (L_{I \rightarrow T} + L_{T \rightarrow I})$$

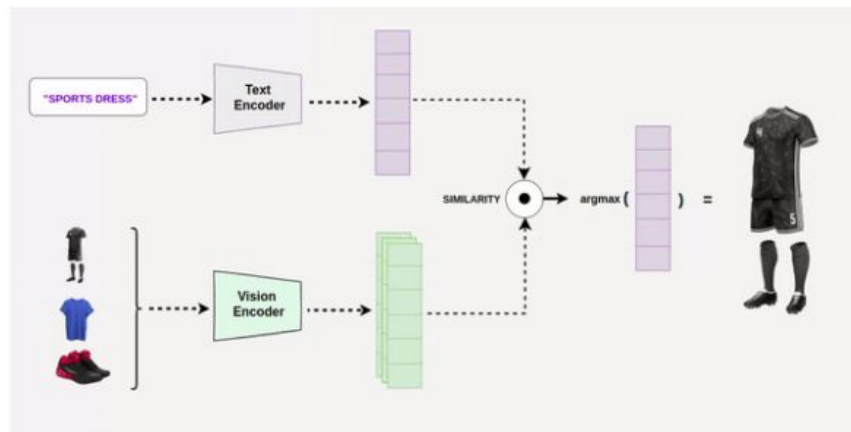
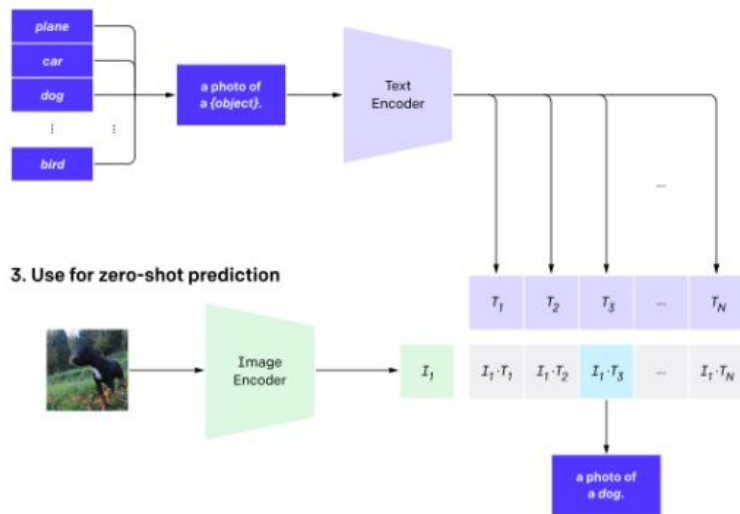
Texto e imagen: Modelos contrastivos



Y el modelo se entrena para minimizar el loss, como cualquier modelo de ML.

Texto e imagen: Modelos contrastivos

Una vez el modelo está entrenado podemos usarlo para clasificación de imágenes, para image retrieval o simplemente como extractor de features.



Texto e imagen: Modelos contrastivos

Para clasificación:

Zero-Shot Image Classification Prompts - ImageNet

a bad photo of a {c}. a photo of many {c}. a sculpture of a {c}. a photo of the hard to see {c}. a low resolution photo of the {c}. a rendering of a {c}. graffiti of a {c}. a bad photo of the {c}. a cropped photo of the {c}. a tattoo of a {c}. the embroidered {c}. a photo of a hard to see {c}. a bright photo of a {c}. a photo of a clean {c}. a photo of a dirty {c}. a dark photo of the {c}. a drawing of a {c}. a photo of my {c}. the plastic {c}. a photo of the cool {c}. a close-up photo of a {c}. a black and white photo of the {c}. a painting of the {c}. a painting of a {c}. a pixelated photo of the {c}. a sculpture of the {c}. a bright photo of the {c}. a cropped photo of a {c}. a plastic {c}. a photo of the dirty {c}. a jpeg corrupted photo of a {c}. a blurry photo of the {c}. a photo of the {c}. a good photo of the {c}. a rendering of the {c}. a {c} in a video game. a photo of one {c}. a doodle of a {c}. a close-up photo of the {c}. a photo of a {c}. the origami {c}. the {c} in a video game. a sketch of a {c}. a doodle of the {c}. a origami {c}. a low resolution photo of a {c}. the toy {c}. a rendition of the {c}. a photo of the clean {c}. a photo of a large {c}. a rendition of a {c}. a photo of a nice {c}. a photo of a weird {c}. a blurry photo of a {c}. a cartoon {c}. art of a {c}. a sketch of the {c}. a embroidered {c}. a pixelated photo of a {c}. itap of the {c}. a jpeg corrupted photo of the {c}. a good photo of a {c}. a plushie {c}. a photo of the nice {c}. a photo of the small {c}. a photo of the weird {c}. the cartoon {c}. art of the {c}. a drawing of the {c}. a photo of the large {c}. a black and white photo of a {c}. the plushie {c}. a dark photo of a {c}. itap of a {c}. graffiti of the {c}. a toy {c}. itap of my {c}. a photo of a cool {c}. a photo of a small {c}. a tattoo of the {c}.

Texto e imagen: Modelos contrastivos

Al entrenarlo con unos 400M de pares texto-imagen OpenAI obtuvo un muy buen modelo de clasificación.

	Dataset Examples	ImageNet	Zero-Shot	Δ Score
		ResNet101	CLIP	
ImageNet		76.2	76.2	0%

Texto e imagen: Modelos contrastivos

Al entrenarlo con unos 400M de pares texto-imagen OpenAI obtuvo un muy buen modelo de clasificación. La diferencia no está en lo bien que clasifica imágenes “normales”, si no en lo robusto/general que es.

	Dataset Examples						ImageNet ResNet101	Zero-Shot CLIP	Δ Score
ImageNet							76.2	76.2	0%
ImageNetV2							64.3	70.1	+5.8%
ImageNet-R							37.7	88.9	+51.2%
ObjectNet							32.6	72.3	+39.7%
ImageNet Sketch							25.2	60.2	+35.0%

Texto e imagen: Modelos contrastivos

OpenAI no compartió el dataset ni el código. La gente de LAION sí!

Welcome to an open source implementation of OpenAI's [CLIP](#) (Contrastive Language-Image Pre-training).

OpenCLIP

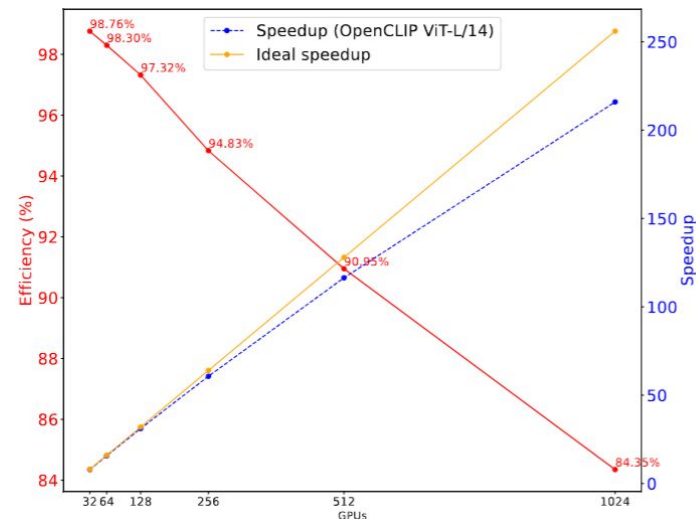
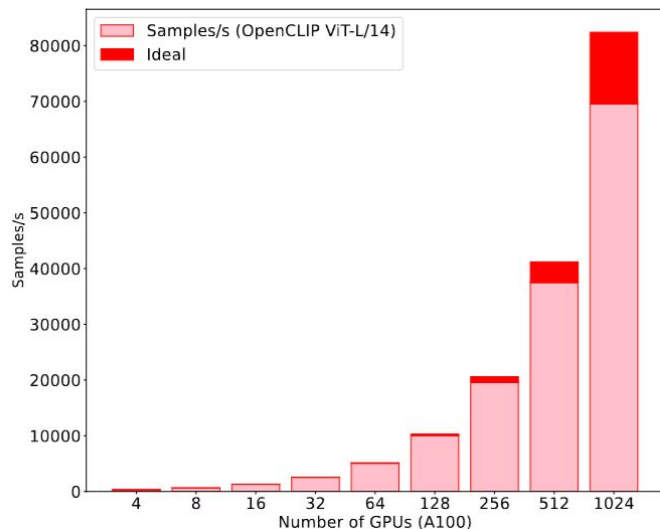
Model	Training data	Resolution	# of samples seen	ImageNet zero-shot acc.
ConvNext-Base	LAION-2B	256px	13B	71.5%
ConvNext-Large	LAION-2B	320px	29B	76.9%
ConvNext-XXLarge	LAION-2B	256px	34B	79.5%
ViT-B-32-256	DataComp-1B	256px	34B	72.8%
ViT-B-16	DataComp-1B	224px	13B	73.5%
ViT-L-14	LAION-2B	224px	32B	75.3%
ViT-H-14	LAION-2B	224px	32B	78.0%
ViT-L-14	DataComp-1B	224px	13B	79.2%
ViT-bigG-14	LAION-2B	224px	34B	80.1%

Texto e imagen: Modelos contrastivos

Si quereis probar a entrenarlo:

https://github.com/mlfoundations/open_clip#training-clip

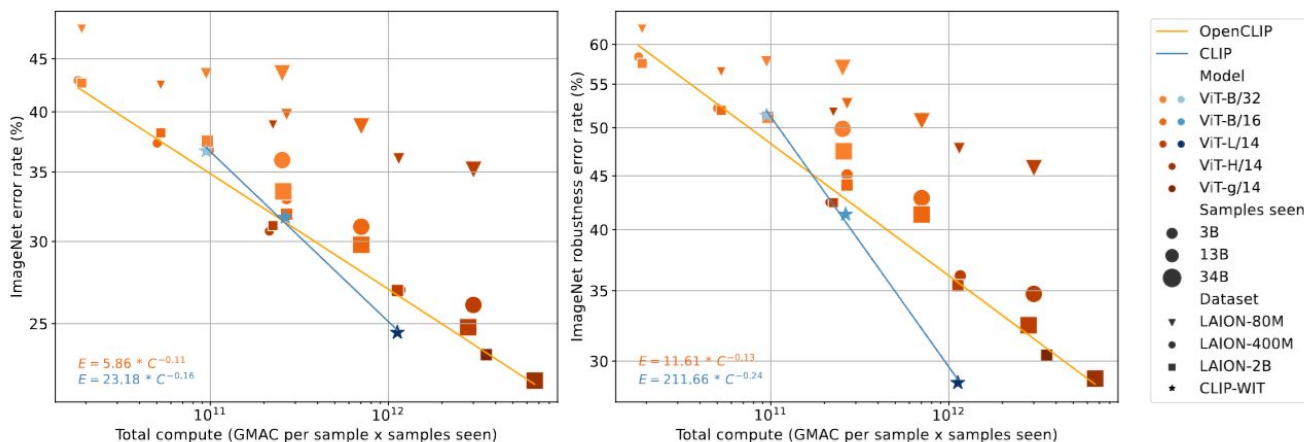
Eso si, casi seguro que ellos tienen más GPUs.



Texto e imagen: Modelos contrastivos

La gente de OpenCLIP tiene 2 trabajos impresionantes:

[Reproducible scaling laws for contrastive language-image learning](#), CVPR 2023.



(a) Relationship between total training compute and zero-shot classification performance on downstream tasks. **Left:** ImageNet performance. **Right:** average performance on five ImageNet robustness datasets (ImageNet-V2 [60], ImageNet-R [22], ImageNet-Sketch [75], ObjectNet [5], and ImageNet-A [24]). Scaling model size, data size, and samples seen leads to better performance on zero-shot classification. Models trained on OpenAI's WebImageText (WIT) show a stronger scaling than models trained on LAION.

Texto e imagen: Modelos contrastivos

La gente de OpenCLIP tiene 2 trabajos impresionantes:

[Reproducible scaling laws for contrastive language-image learning](#), CVPR 2023.

A Further details on distributed training

A.1 Supercomputer specifications

The JUWELS Booster [34] supercomputer used for training consists of 936 compute nodes that host four NVIDIA A100 GPUs each, providing 3744 GPUs in total. The installed A100 Tensor Core GPUs (40 GB) provide 19.5 TFLOP/s of FP64_{TC} computing performance each. The GPUs are hosted by AMD EPYC 7402 CPUs with 2×24 cores (SMT-2) per node, clocked with 2.8 GHz. Each node is diskless and is equipped with 512 GB of RAM. The network is based on Mellanox HDR200 InfiniBand, with four Mellanox ConnectX 6 devices per node, each providing 200 Gbit/s bandwidth per direction.

The NVIDIA A100 GPUs reach peak efficiency of 48.75 GFLOP/(s W) when utilizing the FP64 Tensor Cores. This made the employed machine rank highest in the Green500 list as of November 2020 as the most energy efficient supercomputer among the first 100 machines of the Top500 list with 25 GFLOP/(s W).

Model	Dataset	BS. (global)	LR.	Warm.	#samples.	#GPUs	Time (hrs.)	GPU-h/MW
B/32	LAION-80M	256(32768)	5e-4	2K	3B	128	7	836/0.29
B/32	LAION-80M	256(32768)	5e-4	2K	13B	128	33	4181/1.46
B/32	LAION-80M	256(88064)	1e-3	10K	34B	344	96	32953/11.53
B/32	LAION-400M	256(88064)	1e-3	10K	3B	344	3	1063/0.37
B/32	LAION-400M	672(86016)	1e-3	2K	13B	128	70	8912/3.12
B/32	LAION-400M	256(32768)	5e-4	2K	34B	128	87	11177/3.91
B/32	LAION-2B	256(88064)	1e-3	10K	3B	344	3	1121/0.39
B/32	LAION-2B	256(32768)	5e-4	2K	13B	128	39	4954/1.73
B/32	LAION-2B	96(79104)	1e-3	2K	34B	824	51	42307/14.81
B/16	LAION-80M	256(88064)	1e-3	10K	3B	344	6	1900/0.66
B/16	LAION-80M	512(90112)	1e-3	10K	13B	176	71	12518/4.38
B/16	LAION-80M	256(88064)	1e-3	10K	34B	344	70	24032/8.41
B/16	LAION-400M	256(88064)	1e-3	10K	3B	344	5	1713/0.60
B/16	LAION-400M	512(90112)	1e-3	10K	13B	176	61	10736/3.76
B/16	LAION-2B	256(88064)	1e-3	10K	3B	344	5	1822/0.64
B/16	LAION-2B	512(90112)	1e-3	10K	13B	176	66	11675/4.09
B/16	LAION-2B	256(88064)	1e-3	10K	34B	344	121	41726/14.60
L/14	LAION-80M	224(89600)	1e-3	10K	3B	396	18	7243/2.54
L/14	LAION-80M	448(89600)	1e-3	10K	13B	200	102	20393/7.14
L/14	LAION-80M	224(89600)	1e-3	10K	34B	400	227	90647/31.73
L/14	LAION-400M	224(88704)	1e-3	10K	3B	396	17	6717/2.35
L/14	LAION-400M	112(86016)	1e-3	2K	13B	768	61	46735/16.36
L/14	LAION-400M	84(86016)	1e-3	10K	34B	1024	122	124727/43.65
L/14	LAION-2B	224(88704)	1e-3	10K	3B	396	18	7055/2.47
L/14	LAION-2B	84(86016)	1e-3	10K	13B	1024	52	53599/18.76
L/14	LAION-2B	224(86016)	1e-3	10K	34B	384	319	122509/42.88
H/14	LAION-2B	96(79104)	5e-4	2K	34B	824	279	229665/80.38
g/14	LAION-2B	80(64000)	5e-4	2K	13B	800	137	109392/38.29
Total								1058318/370.41

Texto e imagen: Modelos contrastivos

La gente de OpenCLIP tiene 2 trabajos impresionantes:

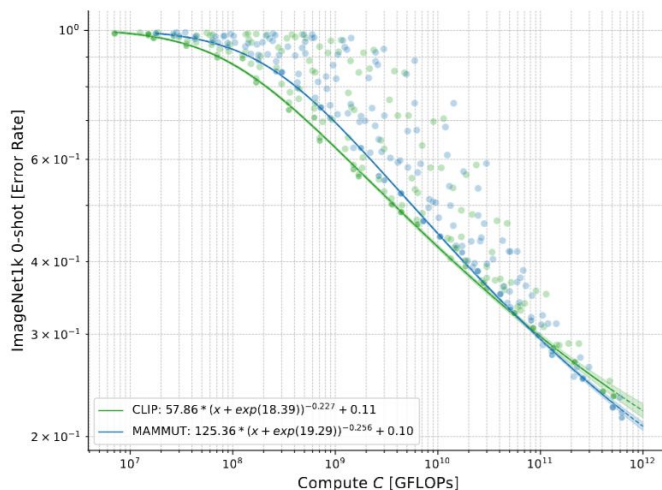
[Reproducible scaling laws for contrastive language-image learning](#), CVPR 2023.

Model	Samples seen	LAION-80M	LAION-400M	LAION-2B
ViT-B/32	3B	51.94	57.12	57.36
	13B	56.46	63.23	62.53
	34B	56.43	64.06	66.47
ViT-B/16	3B	57.55	62.68	61.82
	13B	60.24	67.00	68.13
	34B	61.28	69.00	70.22
ViT-L/14	3B	61.14	69.31	68.93
	13B	63.96	73.06	73.10
	34B	64.83	73.94	75.20
ViT-H/14	34B	-	-	77.97
ViT-g/14	13B	-	-	76.66

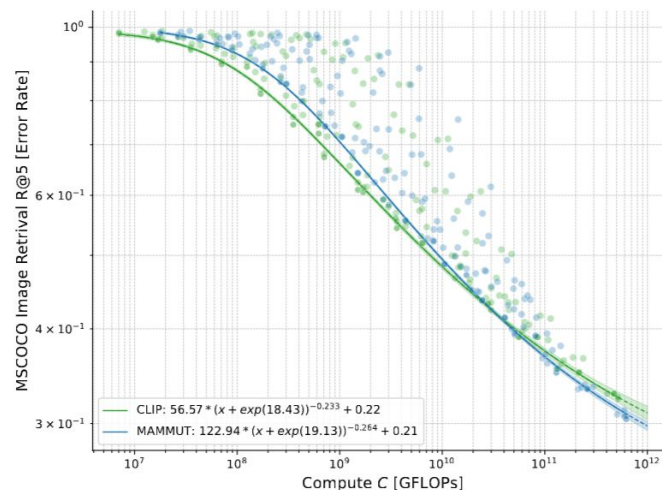
Texto e imagen: Modelos contrastivos

La gente de OpenCLIP tiene 2 trabajos impresionantes:

[Scaling Laws for Robust Comparison of Open Foundation Language-Vision Models and Datasets](#), Neurips 2025.



(a) ImageNet-1k 0-shot classification



(b) MS-COCO image R@5

Texto e imagen: Modelos contrastivos

La gente de OpenCLIP tiene 2 trabajos impresionantes:

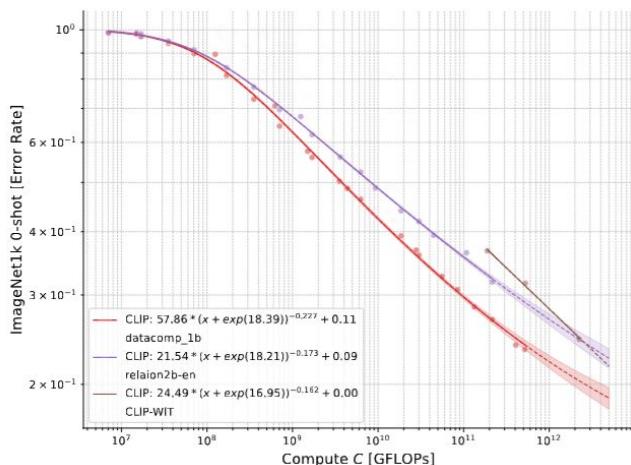
[Scaling Laws for Robust Comparison of Open Foundation Language-Vision Models and Datasets](#), Neurips 2025.

LR Scheduler	GPU	Dataset	MWh	GPU Hours
NVIDIA A100				
cosine	NVIDIA-A100	DataComp-1.4B	2.59e+05	1.03e+06
const-cooldown	NVIDIA-A100	DataComp-1.4B	1.43e+05	5.72e+05
const	NVIDIA-A100	DataComp-1.4B	9.30e+04	3.72e+05
cosine	NVIDIA-A100	Re-LAION-1.4B	3.91e+04	1.56e+05
const-cooldown	NVIDIA-A100	Re-LAION-1.4B	1.70e+04	6.79e+04
const	NVIDIA-A100	Re-LAION-1.4B	4.61e+03	1.84e+04
A100 subtotal:			5.56e+05	2.22e+06
NVIDIA H100				
cosine	NVIDIA-H100	DataComp-1.4B	2.09e+04	2.98e+04
cosine	NVIDIA-H100	Re-LAION-1.4B	1.06e+04	1.52e+04
H100 subtotal:			3.15e+04	4.50e+04
Total:			5.87e+05	2.27e+06

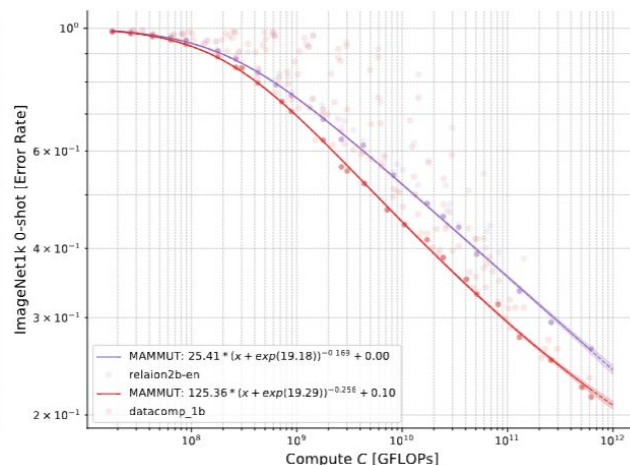
Texto e imagen: Modelos contrastivos

La gente de OpenCLIP tiene 2 trabajos impresionantes:

[Scaling Laws for Robust Comparison of Open Foundation Language-Vision Models and Datasets](#), Neurips 2025.



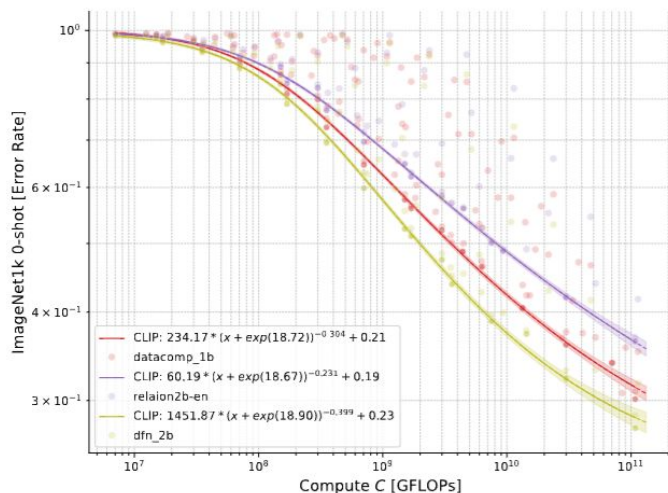
(a) CLIP on WIT, Re-LAION, DataComp



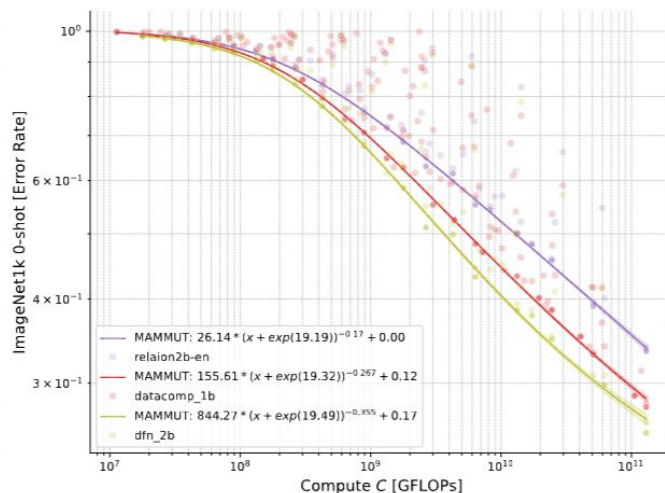
(b) MaMMUT on Re-LAION, DataComp

La gente de OpenCLIP tiene 2 trabajos impresionantes:

[Scaling Laws for Robust Comparison of Open Foundation Language-Vision Models and Datasets](#), Neurips 2025.



(a) CLIP on Re-LAION, DataComp, DFN



(b) MaMMUT on Re-LAION, DataComp, DFN

CLIP vs SIGLIP

Cambia la softmax por una sigmoide.

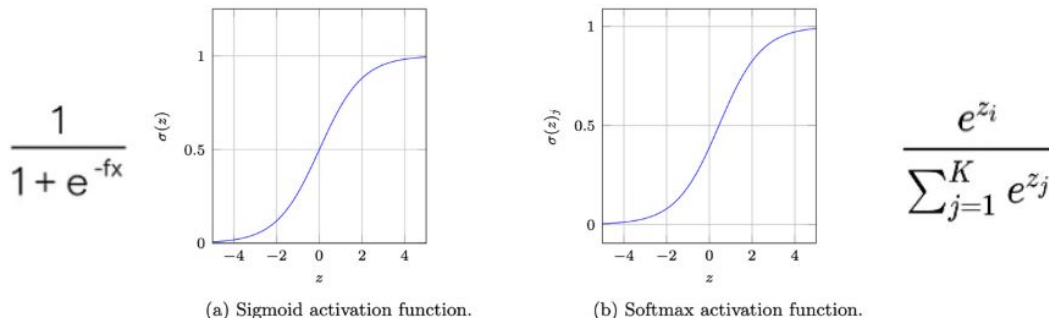


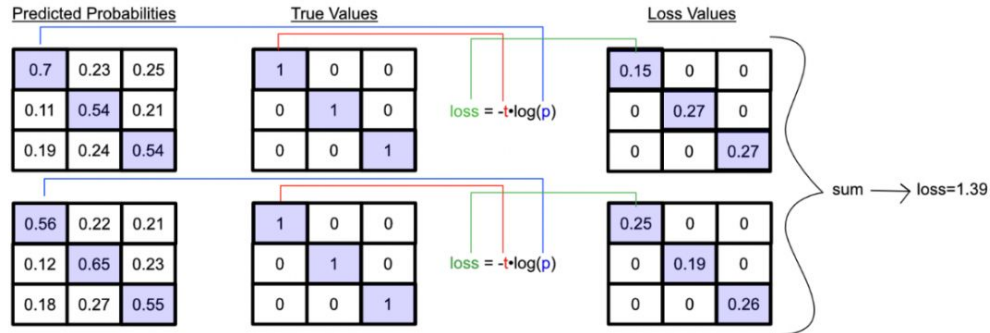
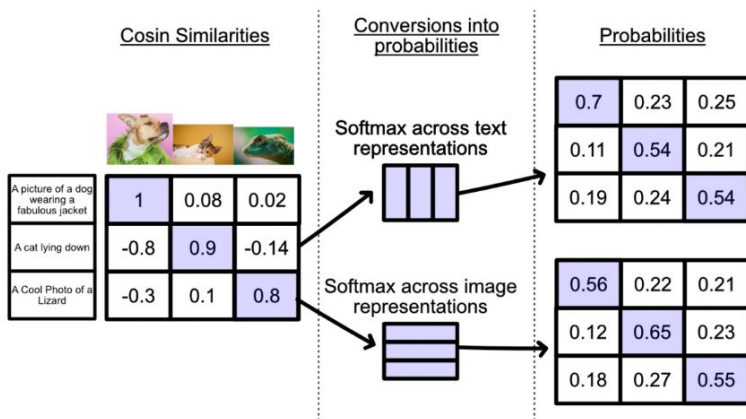
Figure 1: Sigmoid and Softmax activation functions

Vale pero... y que? Qué ventajas tiene?

Texto e imagen: Modelos contrastivos

CLIP vs SIGLIP

Qué ventajas tiene? Recordamos CLIP



CLIP vs SIGLIP

Qué ventajas tiene? Recordamos CLIP.

CLIP construye matriz de $N \times N$ similitudes. Al aplicar la softmax por filas y columnas hace que cada imagen compite contra los textos del batch (y viceversa).

Esto hace que CLIP necesite batches grandes (valor estandar ~ 32K to 86K).

CLIP vs SIGLIP

Qué ventajas tiene?

SigLIP aplica sigmoide a cada similitud de la matriz de forma independiente, de forma que cada par es una probabilidad. Es decir, muchas decisiones binarias imagen-texto en lugar de distribuciones softmax por filas/columnas.

En CLIP enseñas al modelo: "Aquí tienes una foto. ¿Cuál de estas 100 frases describe la imagen?" y tiene que elegir una entre todas. SigLIP convierte el problema en muchas decisiones independientes de "match/no match".

Texto e imagen: Modelos contrastivos

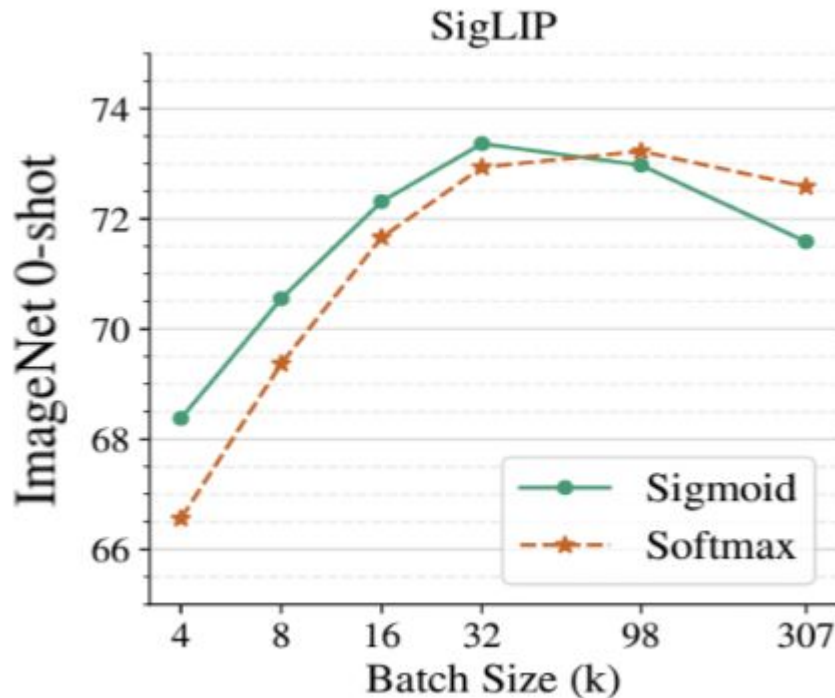
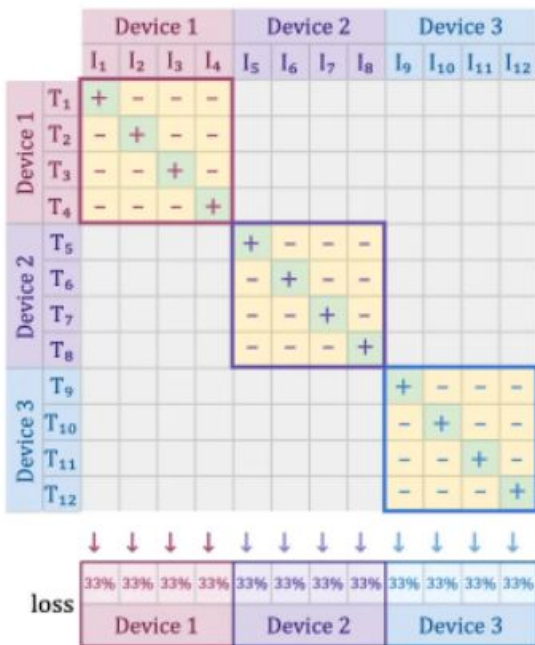
CLIP vs SIGLIP

Qué ventajas tiene?

	CLIP	SigLIP
Idea	Alinear imagen y texto	Alinear imagen y texto
Objetivo	Contrastivo	Contrastivo/binario
Loss	Softmax	Sigmoid
Pregunta	"¿Cuál es el texto correcto?"	"¿Este par hace match?"
Competición	Entre candidatos	Cada par independientemente
Dependencia del batch	Alta	Menor
Escalabilidad	Buena	Muy buena

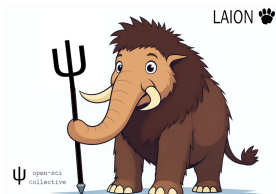
Texto e imagen: Modelos contrastivos

CLIP vs SIGLIP



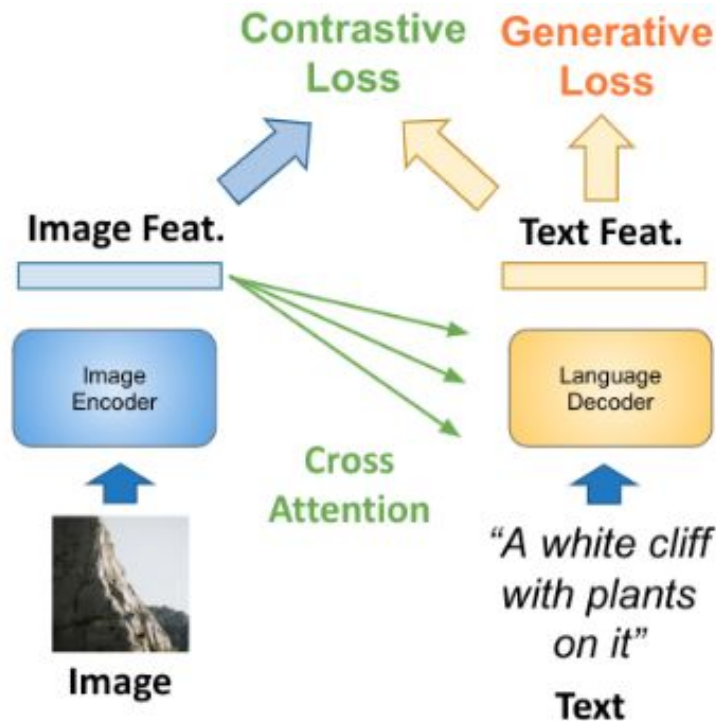
Texto e imagen: Modelos contrastivos

Mammut



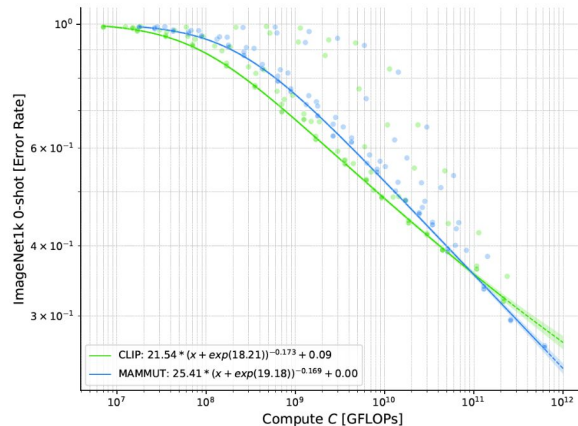
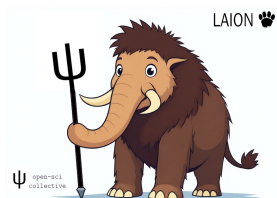
[MaMMUT: A Simple Architecture for Joint Learning for MultiModal Tasks](#)

Basicamente: CLIP + captioning

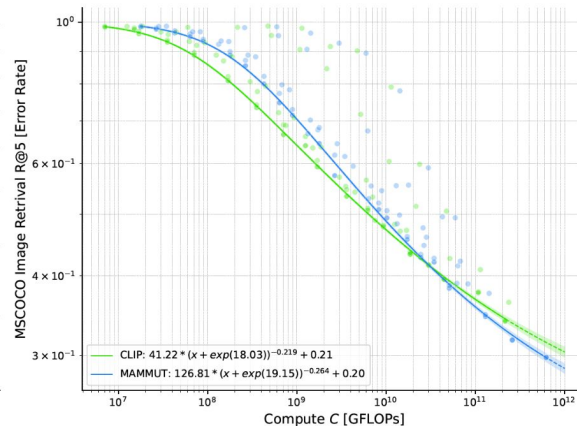


Texto e imagen: Modelos contrastivos

Mammut



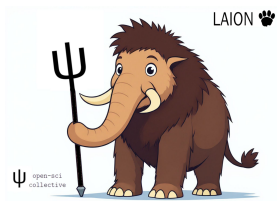
(a) ImageNet-1k 0-shot classification



(b) MS-COCO image R@5

Figure 2: **Scaling on Re-LAION-1.4B.** Comparison of CLIP and MaMMUT via scaling laws on Re-LAION-1.4B. Error rate on downstream tasks is plotted against compute. MaMMUT outperforms CLIP in terms of scalability, indicated by crossing scaling law fit lines, where MaMMUT takes over CLIP in performance from larger compute scale $> 10^{11}$ GFLOPS on, showing similar trends as on DataComp-1.4B.

Mammut



Model Candidate	Samples Seen Candidate	GFLOPs	IN1k acc1 Predicted (95% CI)
ViT-L-14	12.8B	2.14e+12	0.796 (0.788, 0.804)
ViT-L-14	15.5B	2.59e+12	0.800 (0.791, 0.808)
mammut-ViT-L-14	10.6B	2.14e+12	0.816 (0.811, 0.821)
mammut-ViT-L-14	12.8B	2.59e+12	0.820 (0.815, 0.826)

Table 1: Estimation of IN1k performance for CLIP and MaMMUT on unseen compute scales using our scaling laws fits on DataComp-1.4B. Additionally, for each compute scale, we provide possible models and samples seen (assuming unique samples) sizes.

Más versiones:

[AltCLIP: Itering the Language Encoder in CLIP for Extended Language Capabilities.](#)

Cambian el encoder de texto por uno multilingüe y ya tienen un CLIP que entiende distintos idiomas.

[Scaling Up Visual and Vision-Language Representation Learning With Noisy Text](#)

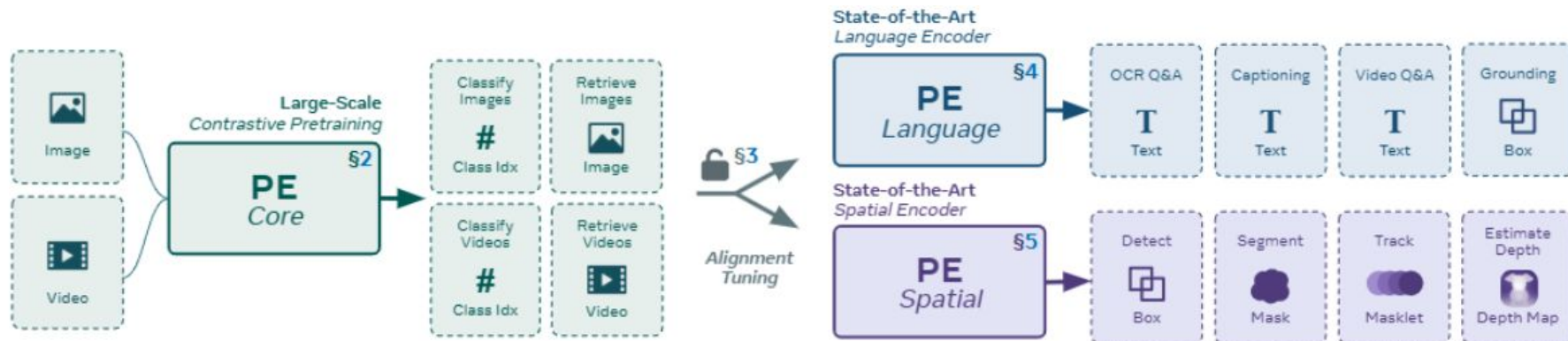
[Supervision.](#) Meter un poquito de ruido en el entrenamiento viene bien para evitar sobreajuste, solo hay que cambiar algún texto.

[Chinese CLIP: Contrastive Vision-Language Pretraining in Chinese.](#) Si esta esta en ingles, ¿por qué no en chino? Tambien esta en italiano ([clip-italian](#)), etc, etc.

[BioCLIP.](#) Y para cualquier dominio que se nos ocurra se puede entrenar uno siempre que tengamos bastantes datos!

Texto e imagen: Modelos contrastivos

Y en la práctica como se entrena un modelo de estos?
Vamos a fijarnos en el [Meta Perception Encoder](#)
Meta da muchísimos detalles de cómo es el entrenamiento.



Texto e imagen: Modelos contrastivos

Meta Perception Encoder training recipe:

1. Baseline:

Images at 224 pixels, batch size of 32k,
AdamW optimizer, trained until 12B
samples seen on the MetaCLIP-2.3B
dataset.



Texto e imagen: Modelos contrastivos

[Meta Perception Encoder](#) training
recipe:

2. Increase resolution with progress
resizing:

98 → 154 → 224 pixels

Con 4B samples en cada uno, mismo
batch size (32k) y optimizer (Adam).

	Robustness <i>avg of 6</i> / ImageNet <i>val</i>		Training <i>ZFLOPs</i>
1. Baseline	75.3	78.9	1.0
2. Prog. Res	75.1	78.9	0.5

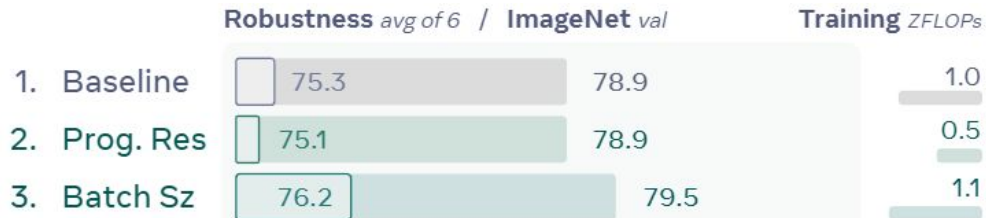
Que se gana al aumentar la resolución?

Texto e imagen: Modelos contrastivos

[Meta Perception Encoder](#) training
recipe:

3. Increase batch size:

Increase the batch size from 32k to 64k.



Ojo, también se aumenta el número de imágenes que ve el modelo!

Recordad que CLIP iba mejor con batches grandes.

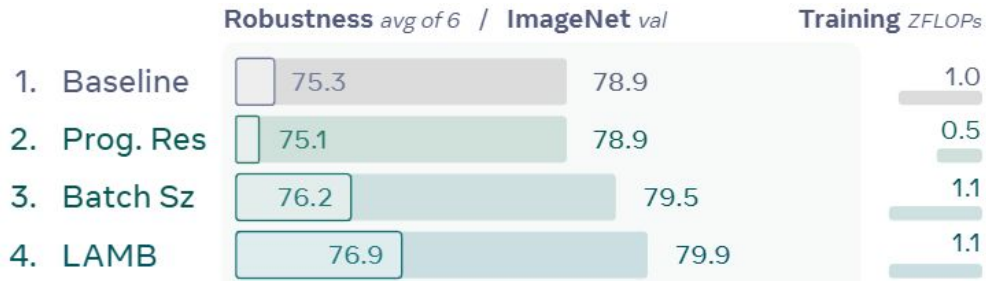
Texto e imagen: Modelos contrastivos

[Meta Perception Encoder](#) training
recipe:

4. Optimizer change:

Cambio de optimizador (AdamW por LAMB). LAMB es mejor para redes enormes y batch sizes grandes.

El optimizador “optimo” no tiene por que ser siempre el mismo, si cambiamos algo, puede que cambie.

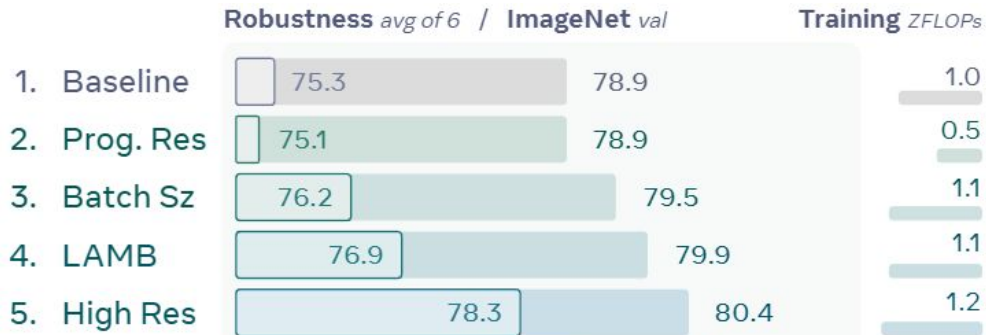


Texto e imagen: Modelos contrastivos

[Meta Perception Encoder](#) training
recipe:

5. Resolution + compute allocation:
Entrenar a 98 pixels (10B imágenes
vistas) → 154 pixels (8B) → 224 pixels
(4B) → 336 pixels (2B)

Por que reducimos samples seen?

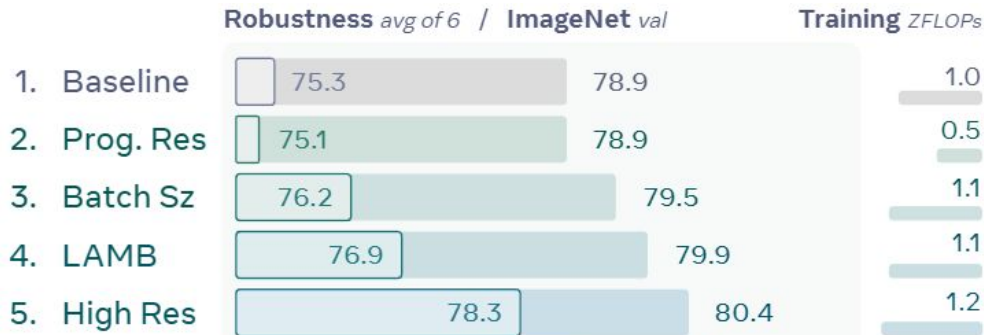


Texto e imagen: Modelos contrastivos

Meta Perception Encoder training
recipe:

5. Resolution + compute allocation:
Entrenar a 98 pixels (10B imágenes
vistas) → 154 pixels (8B) → 224 pixels
(4B) → 336 pixels (2B)

Por que reducimos samples seen?
Una imagen a 336px tiene un coste
computacional superior a una a 98px.

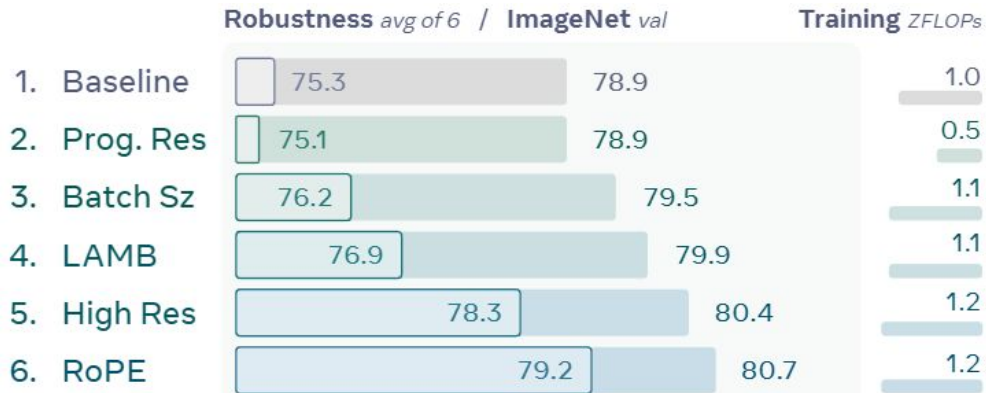


**La pregunta del millón en “scaling” es:
¿Cómo distribuyo mi presupuesto
computacional entre resolución y
número de imágenes?**

Texto e imagen: Modelos contrastivos

[Meta Perception Encoder](#) training
recipe:

6. 2D Rotary Position Embedding:
Introduce información espacial relativa
en la atención y mejorar la extrapolación
a distintas resoluciones .

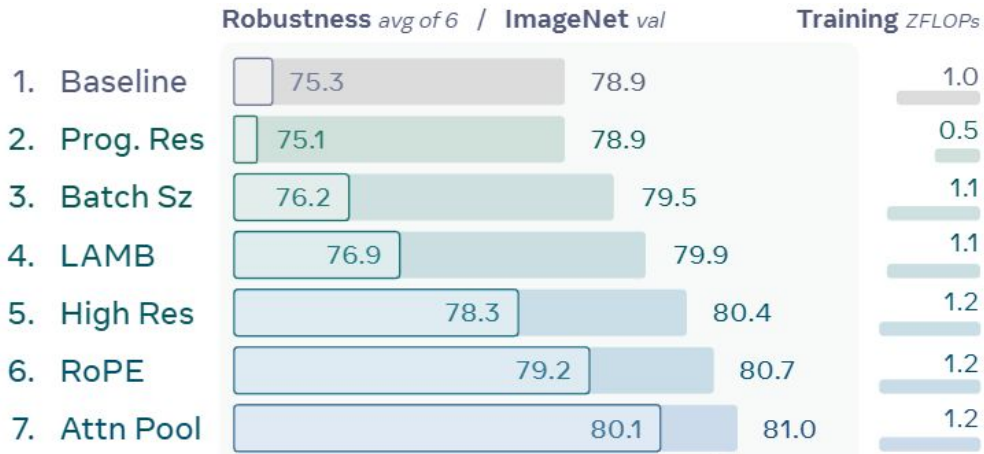


Texto e imagen: Modelos contrastivos

[Meta Perception Encoder](#) training
recipe:

7. Attention Pooling:

En lugar de quedarnos solo con el token de clasificación, añadimos un bloque de atención que puede consultar todas las features visuales para construir el embedding final.

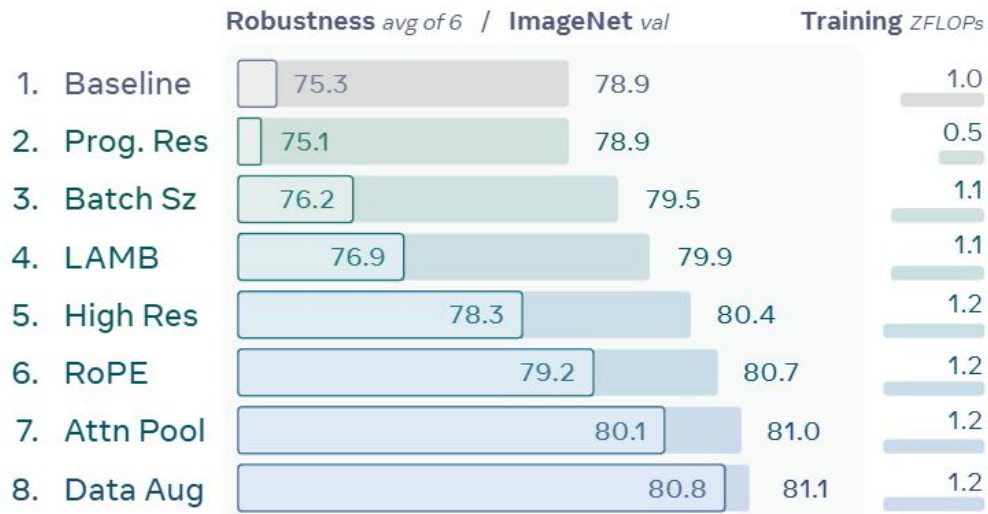


Texto e imagen: Modelos contrastivos

[Meta Perception Encoder](#) training
recipe:

8. Data augmentation:

Para aumentar la diversidad de las imágenes y obtener situaciones más diversas.



Texto e imagen: Modelos contrastivos

[Meta Perception Encoder](#) training
recipe:

9. Mask regularization:

Tapar de forma aleatoria unos pocos patches actúa como regularizador, aumentando la robustez del modelo y disminuyendo su dependencia espacial de los patches centrales.



Texto e imagen: Modelos contrastivos

Scaling behaviours!



Figure 3 Scaling Behavior (Model Size). Results before and after our recipe changes (Fig. 2) for S/14, B/14, and L/14 models. Our recipe improves scaling for difficult metrics like ObjectNet [4] and ImageNet Adeversarial [45].



Figure 4 Scaling Behavior (Training Steps). Results before and after our recipe changes for an L/14 model trained with 120K, 240K, and 360K steps, adjusting the learning rate and progressive resolution schedules accordingly. Despite our recipe being much stronger than the original, there is still room for further improvement by training longer.

Texto e imagen: Modelos contrastivos

Y una vez tienen un modelo bueno de imágenes, también usan video!

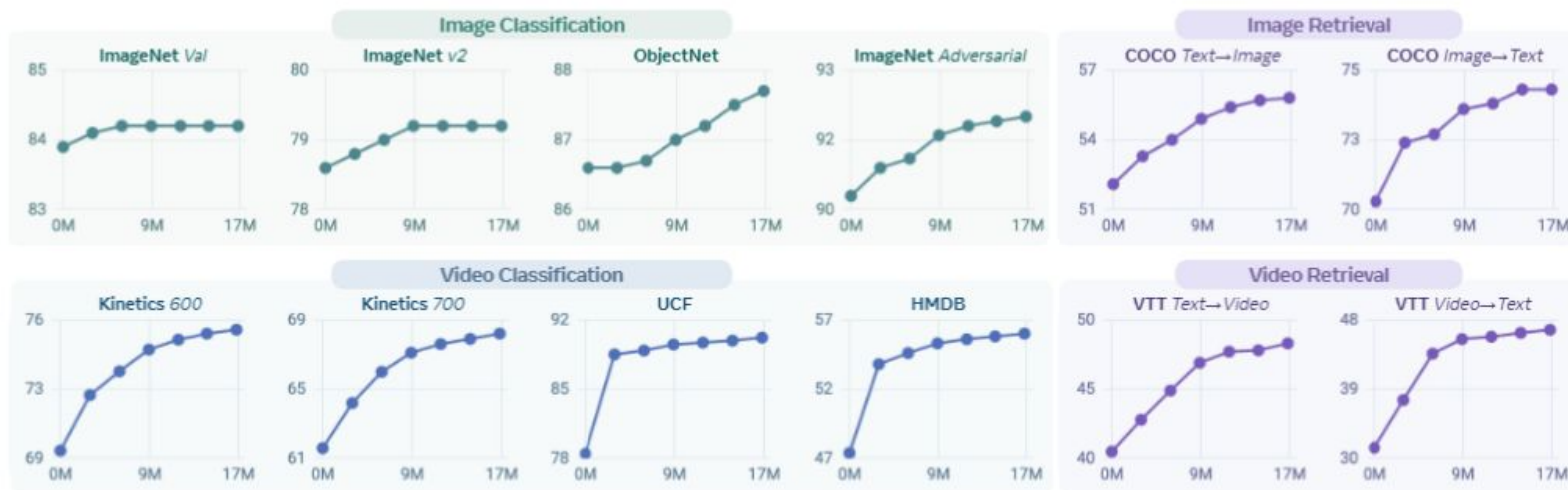
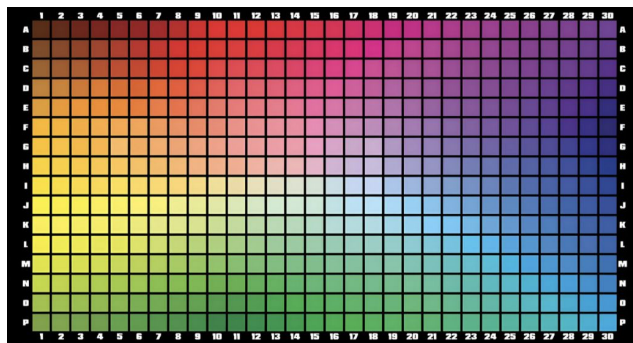


Figure 6 Video Data Scaling. Finetuning on videos recaptured by the PE video data engine from 0M (baseline image-only model) to 17M samples consistently improves both image and video performance, both classification and retrieval.

Texto e imagen: Modelos contrastivos

Pero... de verdad son tan buenos CLIP y el resto de modelos contrastivos?

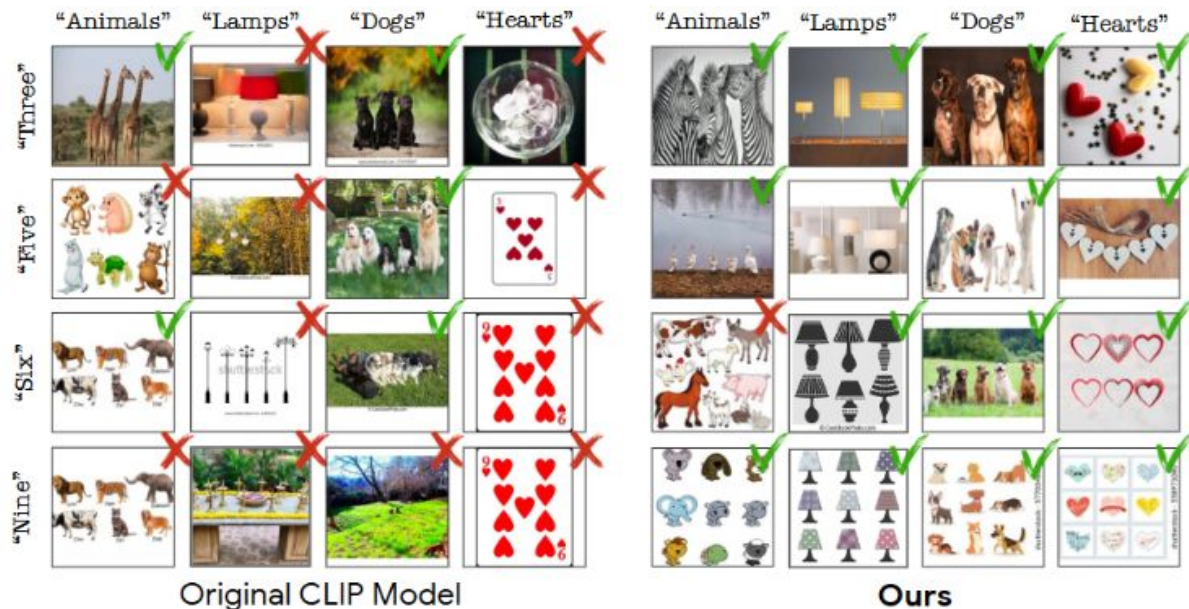
Aunque son muy usados, también tienen ventajas y problemas!



Texto e imagen: Modelos contrastivos

Pero... de verdad son tan buenos CLIP y el resto de modelos contrastivos?

Teaching CLIP to Count to Ten



Texto e imagen: Modelos contrastivos

Pero... de verdad son tan buenos CLIP y el resto de modelos contrastivos?



Human:
Elephant

Model:
Elephant



Human:
Cat

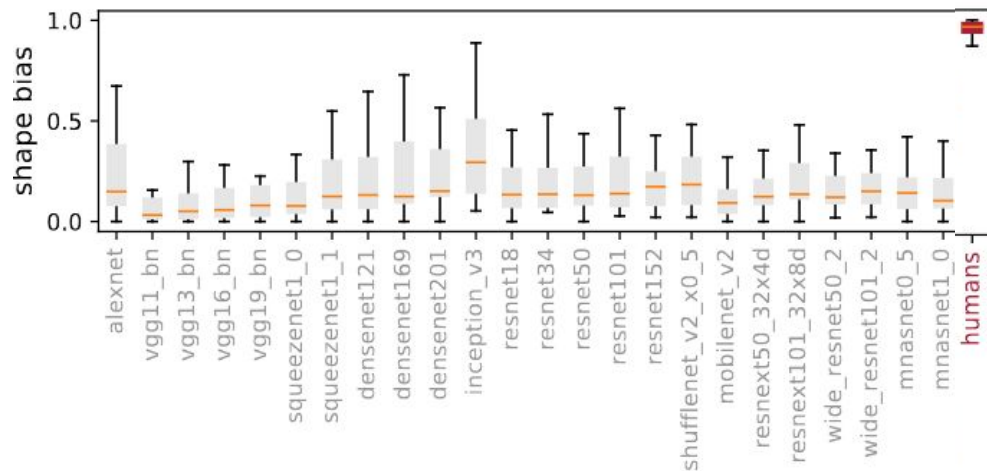
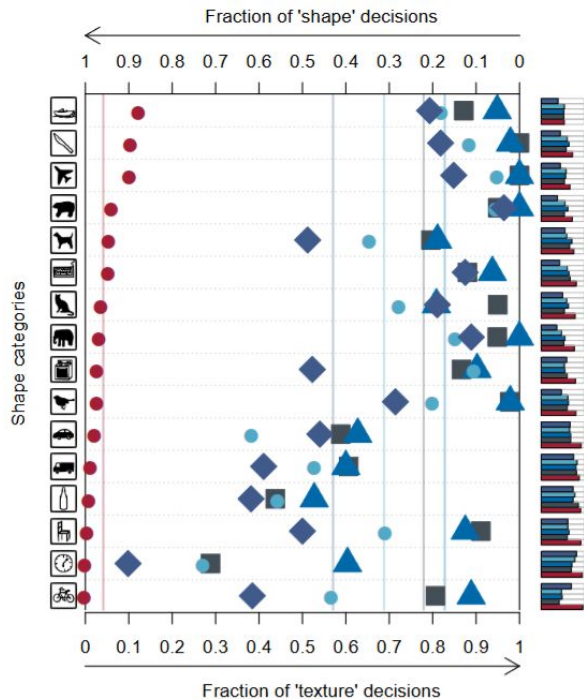
Model:
Cat



Human:
?

Model:
?

Pero... de verdad son tan buenos CLIP y el resto de modelos contrastivos?



Texto e imagen: Modelos contrastivos

Pero... de verdad son tan buenos CLIP y el resto de modelos contrastivos?

ImageNet-trained CNNs are biased towards texture; increasing shape bias improves accuracy and robustness, ICLR 2019.

Partial success in closing the gap between human and machine vision, NeurIPS 2021.

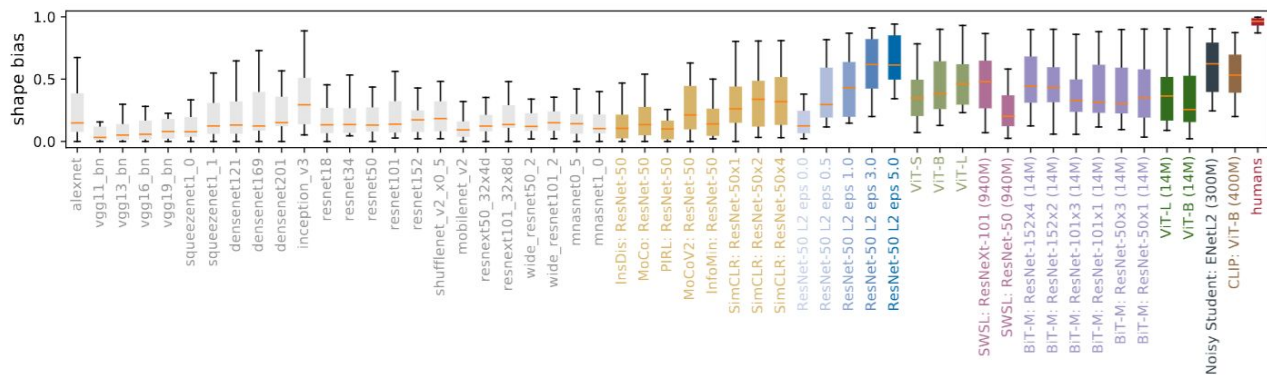
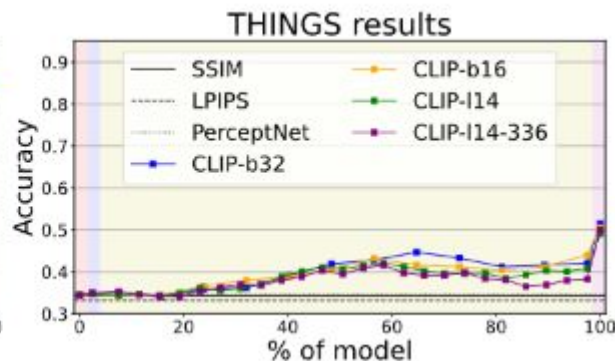
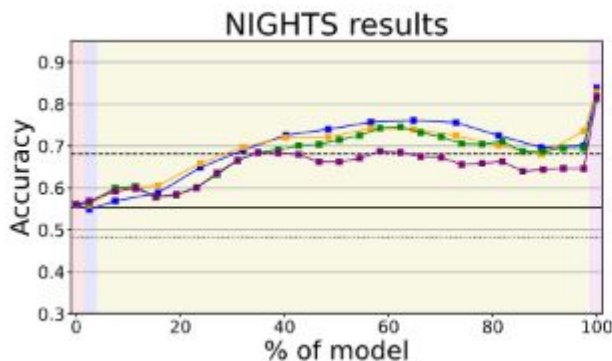


Figure 3: Shape vs. texture biases of different models. While human shape bias is not yet matched, several approaches improve over vanilla CNNs. Box plots show category-dependent distribution of shape / texture biases (shape bias: high values, texture bias: low values).

Texto e imagen: Modelos contrastivos

Pero... de verdad son tan buenos CLIP y el resto de modelos contrastivos?

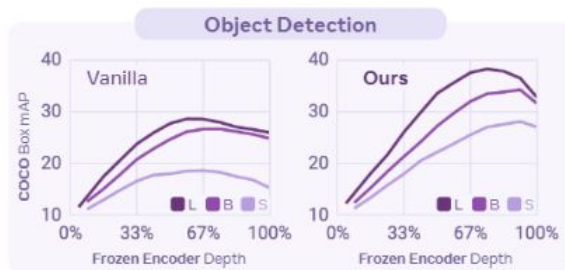
[Measuring Human-CLIP Alignment at Different Abstraction Levels](#), ICLR 2024.



Texto e imagen: Modelos contrastivos

Pero... de verdad son tan buenos CLIP y el resto de modelos contrastivos?

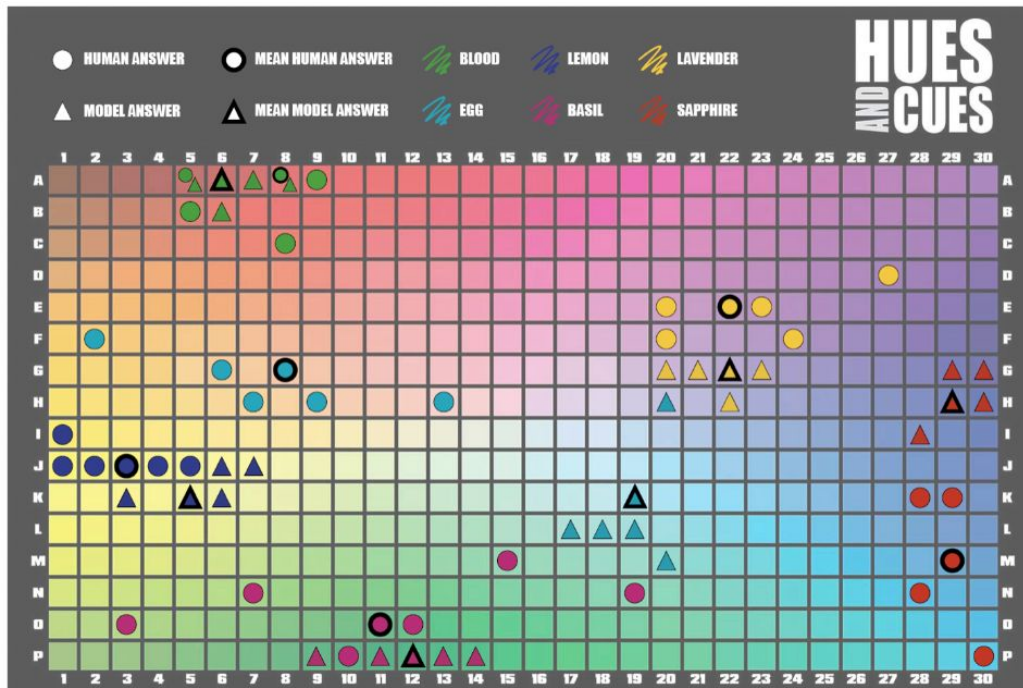
[Meta Perception Encoder](#). Ojo al resto del título.



Texto e imagen: Modelos contrastivos

Pero... de verdad son tan buenos CLIP y el resto de modelos contrastivos?

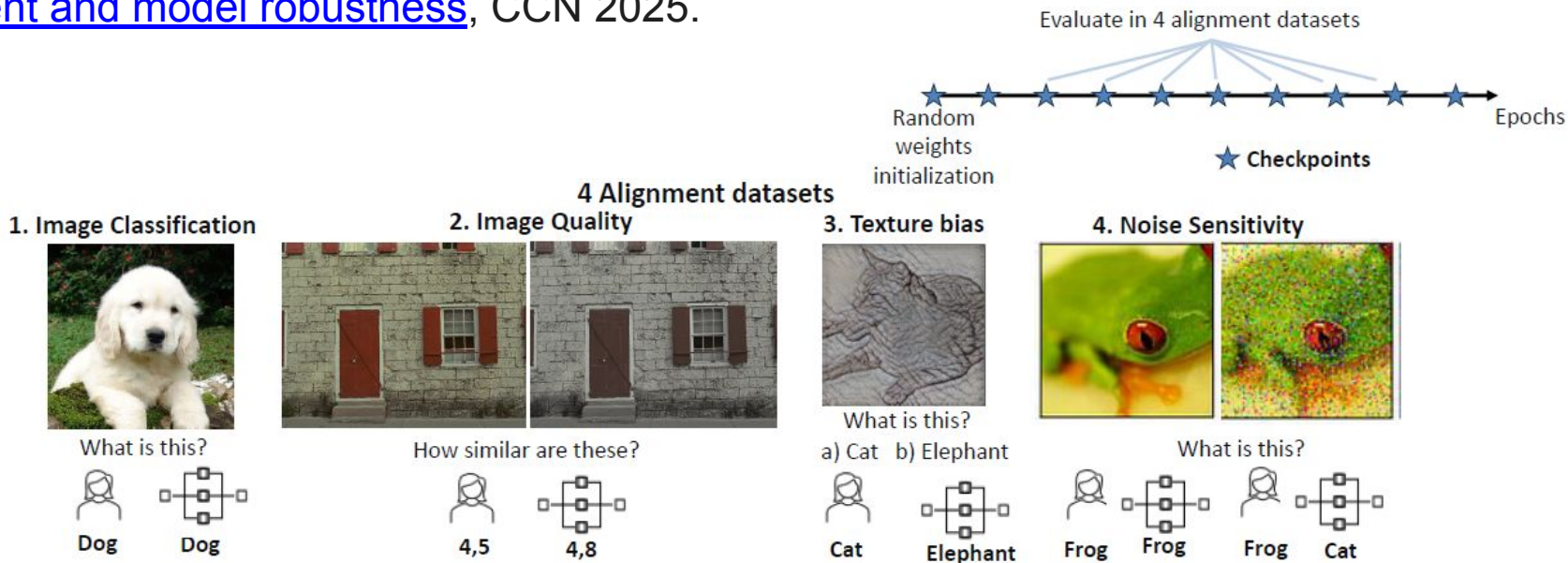
[Hues and Cues: Human vs. CLIP, CCN 2025.](#)



Texto e imagen: Modelos contrastivos

Pero... de verdad son tan buenos CLIP y el resto de modelos contrastivos?

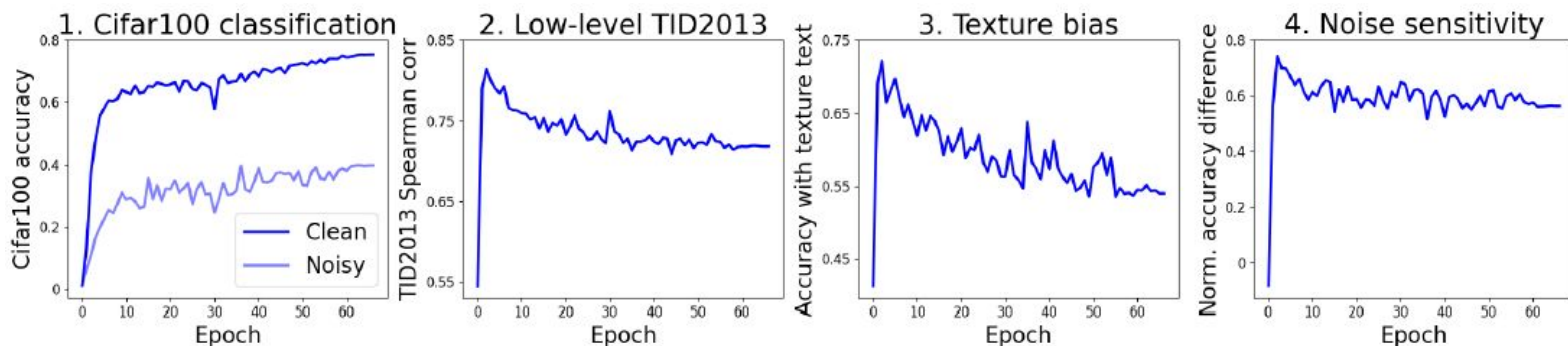
[On the dynamic evolution of CLIP texture–shape bias and its relationship to human alignment and model robustness](#), CCN 2025.



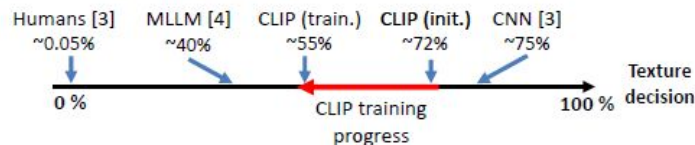
Texto e imagen: Modelos contrastivos

Pero... de verdad son tan buenos CLIP y el resto de modelos contrastivos?

On the dynamic evolution of CLIP texture–shape bias and its relationship to human alignment and model robustness, CCN 2025.



	Classification Accuracy	TID2013 Correlation	Texture Bias	Noise Sensitivity
Class. Accuracy	1			
TID2013 Corr	-0.795	1		
Texture Bias	-0.737	0.852	1	
Noise Sens.	-0.441	0.836	0.618	1



Texto e imagen: Modelos contrastivos

Pero... de verdad son tan buenos CLIP y el resto de modelos contrastivos?

[On the dynamic evolution of CLIP texture–shape bias and its relationship to human alignment and model robustness](#), CCN 2025.

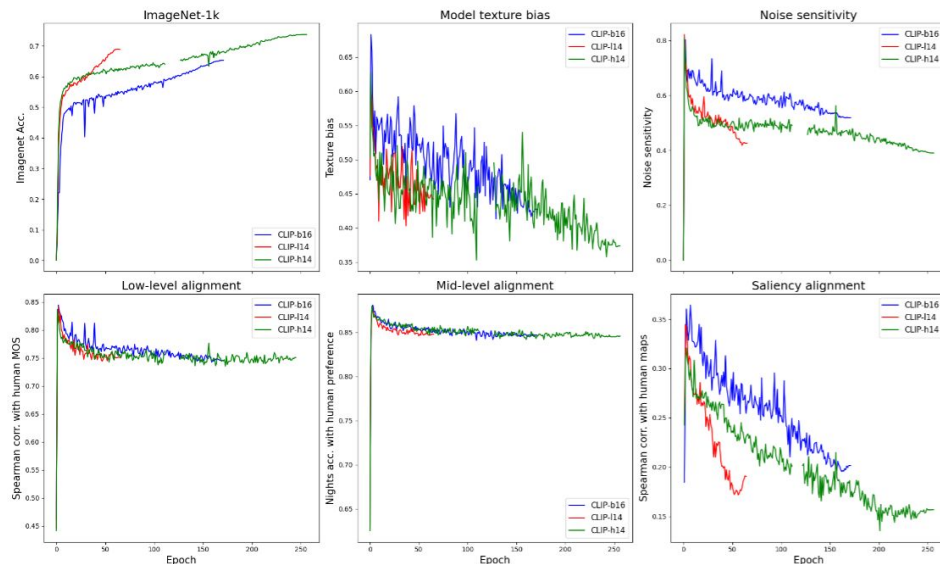
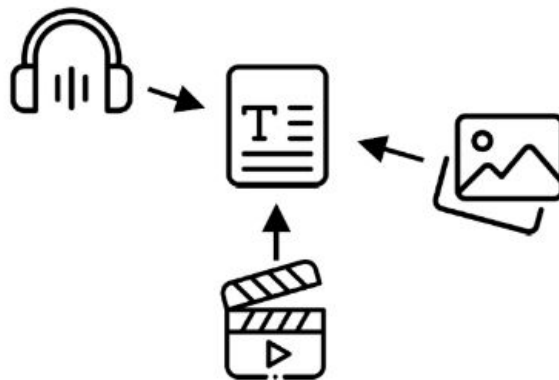


Figure 3. Evolution of the analyzed metrics for three different CLIP scales (base, large, huge).

Texto e imagen: Modelos

An anchoring modality

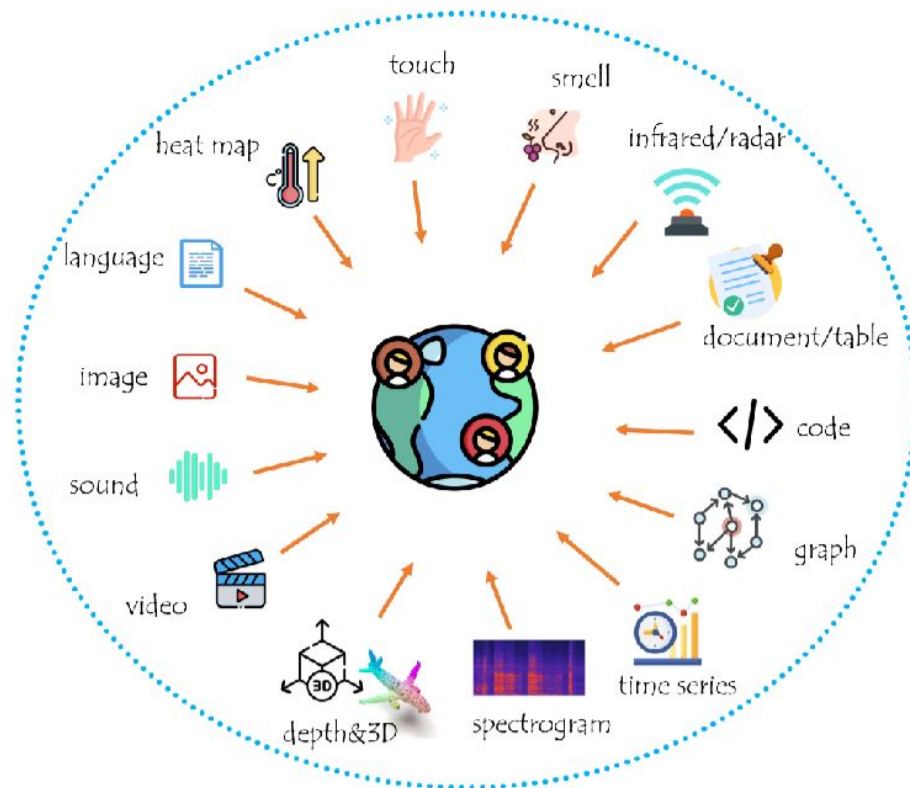


MLLM (VLM)

Texto e imagen: Modelos MLLM

Los LLM actuales tienen capacidades impresionantes.

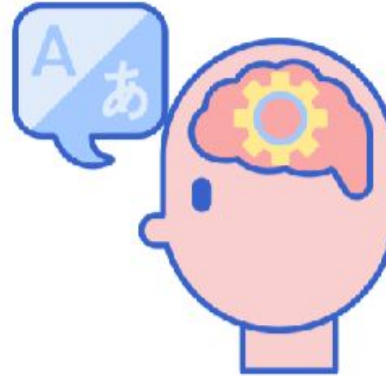
¿Podemos hacer que trabajen con cualquier modalidad?
¿Como?



Texto e imagen: Modelos MLLM

LLM ahora poseen en general capacidades de comprensión semántica muy potentes.

El lenguaje es una modalidad crucial para desarrollar inteligencia.

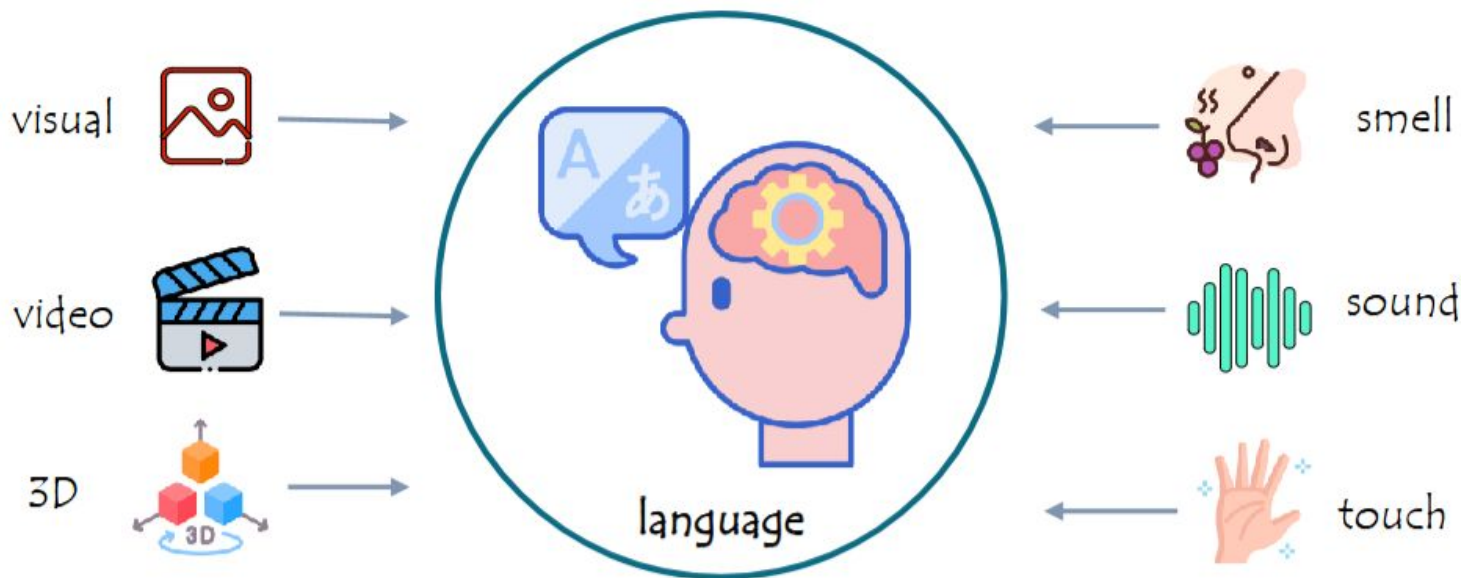


language

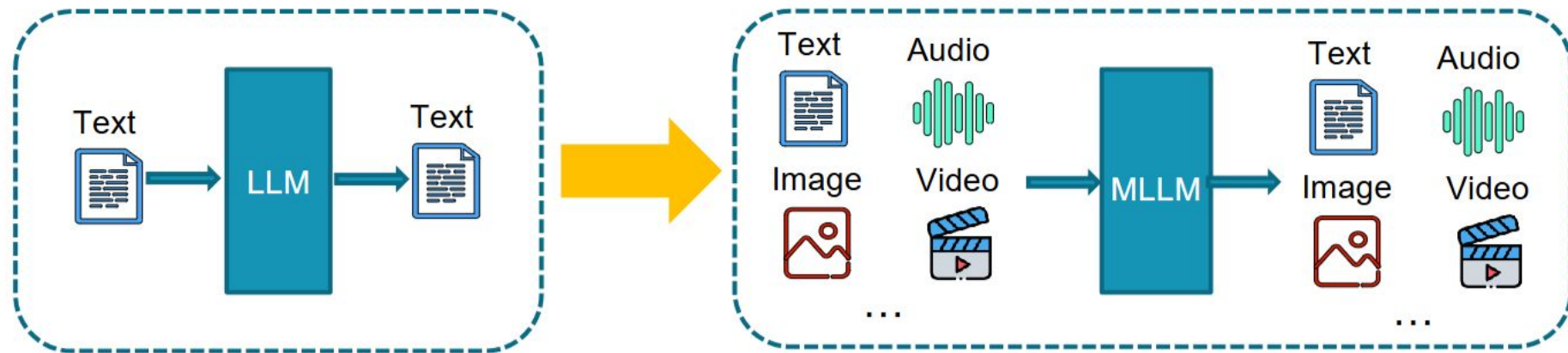
Texto e imagen: Modelos MLLM

Casi todos los MLLM actuales se construyen tomando LLM basados en lenguaje como el módulo central de toma de decisiones (es decir, el cerebro o procesador central).

Añadiendo módulos de modalidades externas (no textuales) adicionales, los LLM adquieren capacidades multimodales.



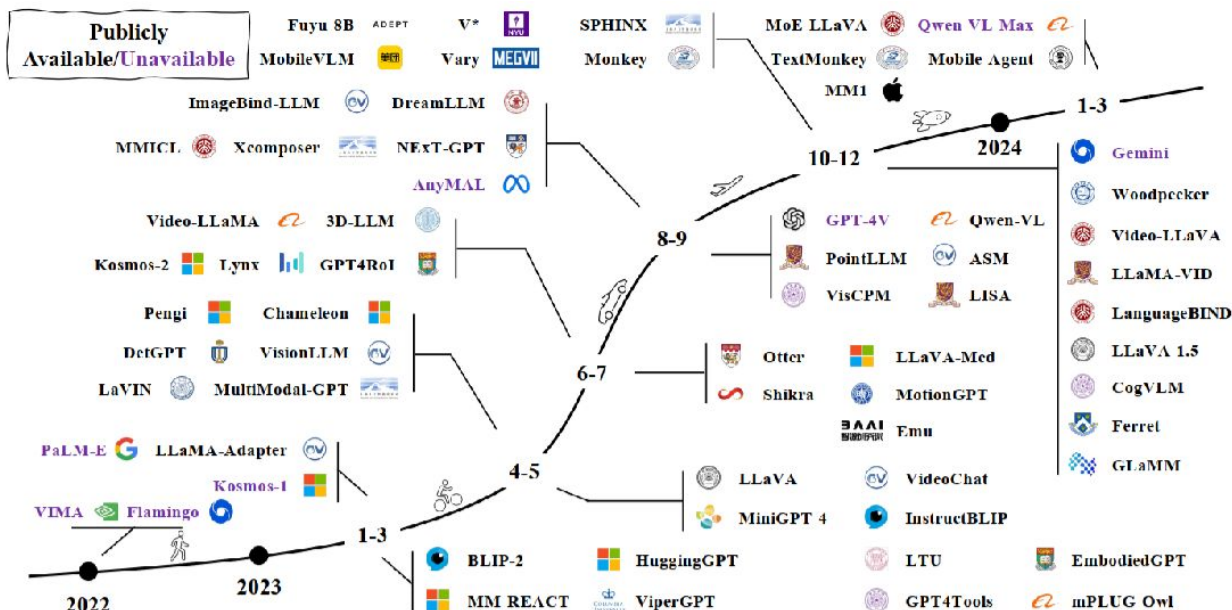
Texto e imagen: Modelos MLLM



Texto e imagen: Modelos MLLM

Se ha avanzado muchísimo, y cada vez más rápido:

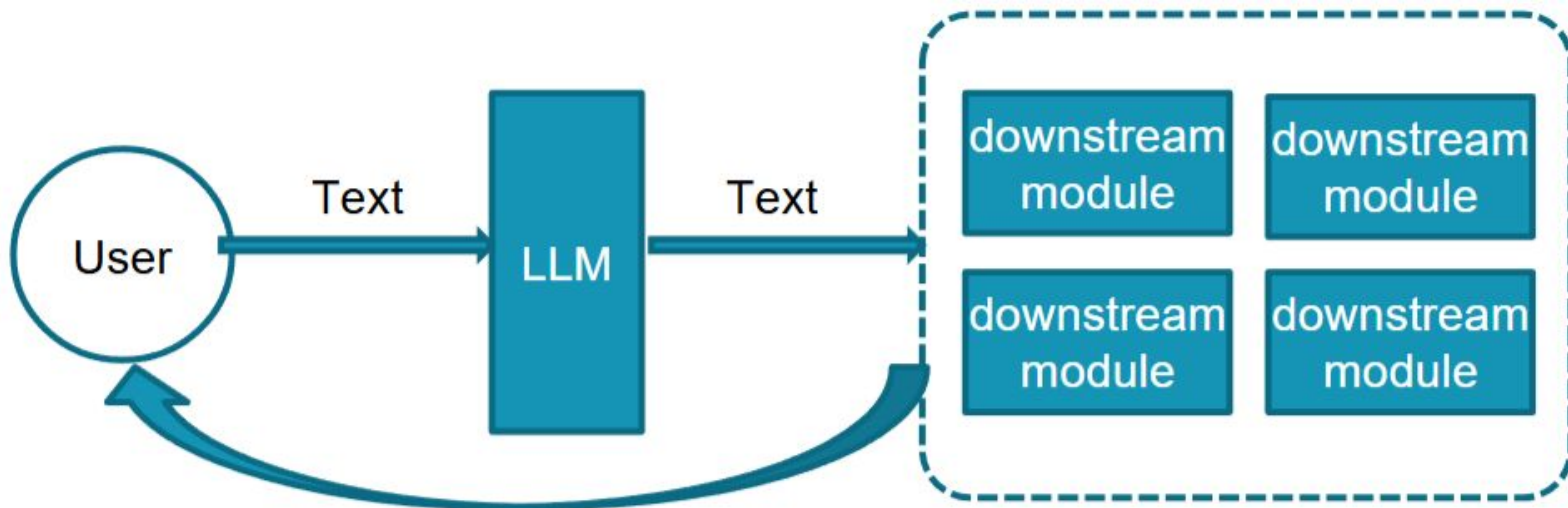
- <https://github.com/RUCAIBox/LLMSurvey>
- <https://github.com/BradyFU/Awesome-Multimodal-Large-Language-Models>



Texto e imagen: Modelos MLLM

Dos opciones principales de arquitectura:

- LLM como director (no demasiado usada hoy en día): El LLM recibe entradas de texto y “llama” a diferentes módulos según lo que se le haya pedido.

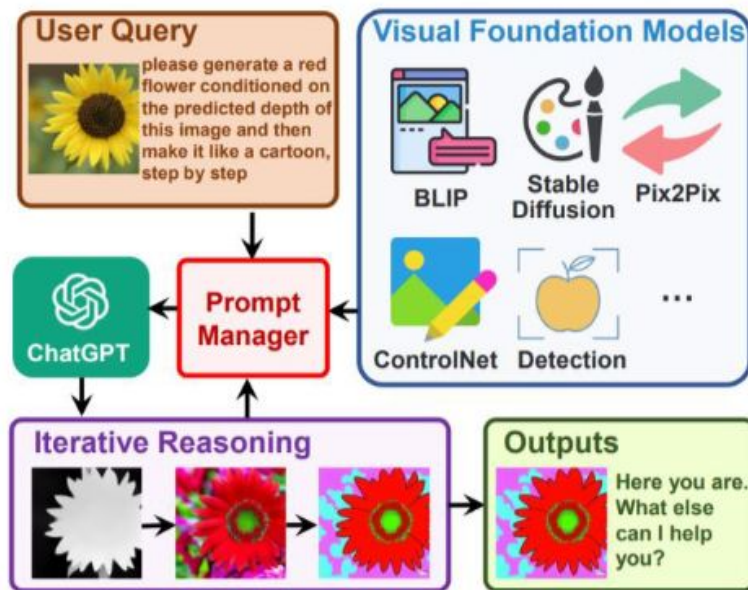


Texto e imagen: Modelos MLLM

Dos opciones principales de arquitectura:

- LLM como director: Rápidos de construir sin re-entrenar y se pueden añadir nuevas funcionalidades, pero puede haber pérdidas de información al pasar de texto a las distintas funcionalidades.

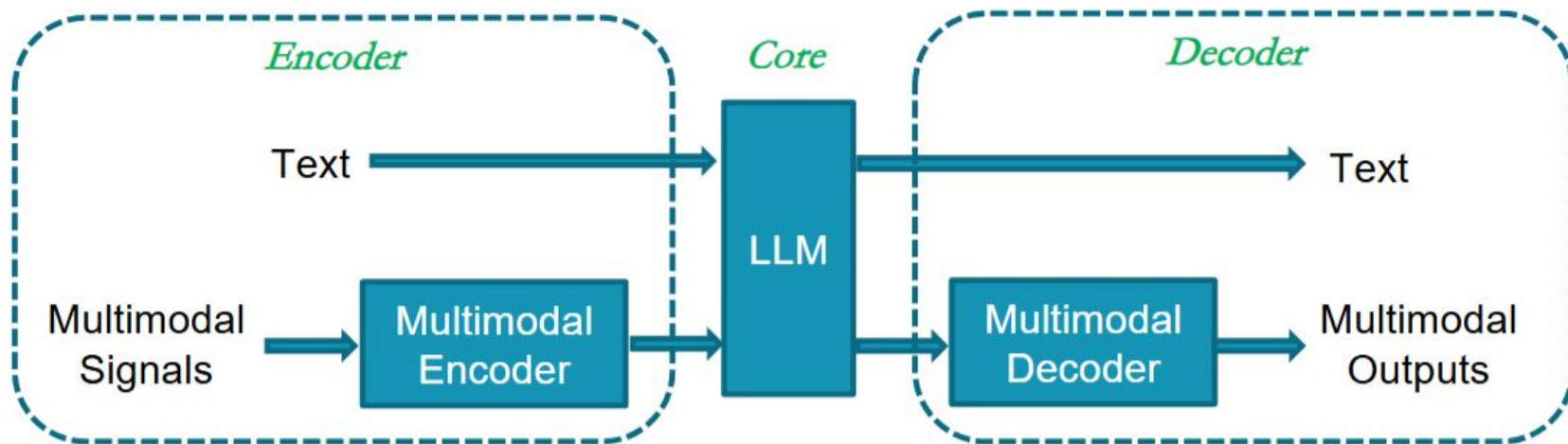
Ejemplos: Visual-ChatGPT,
HuggingGPT, MM-REACT,
ViperGPT, ...



Texto e imagen: Modelos MLLM

Dos opciones principales de arquitectura:

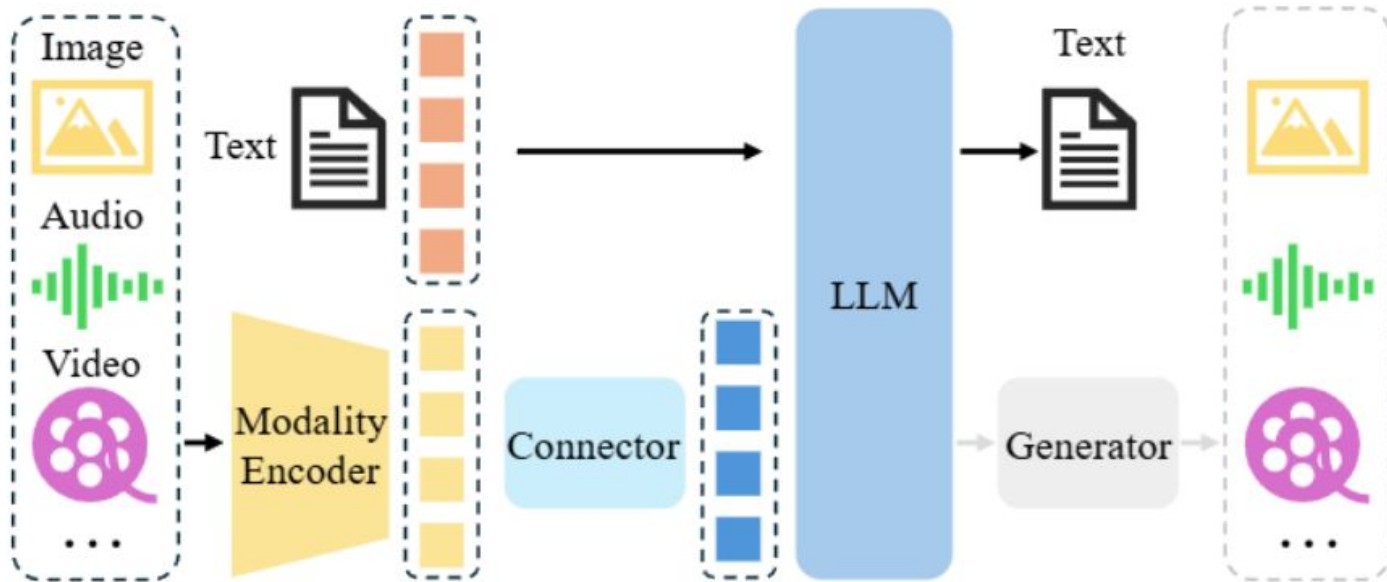
- LLM como parte del sistema: Arquitectura estándar hoy en día.
Modality encoder + Connector + LLM (+ Modality Decoder)



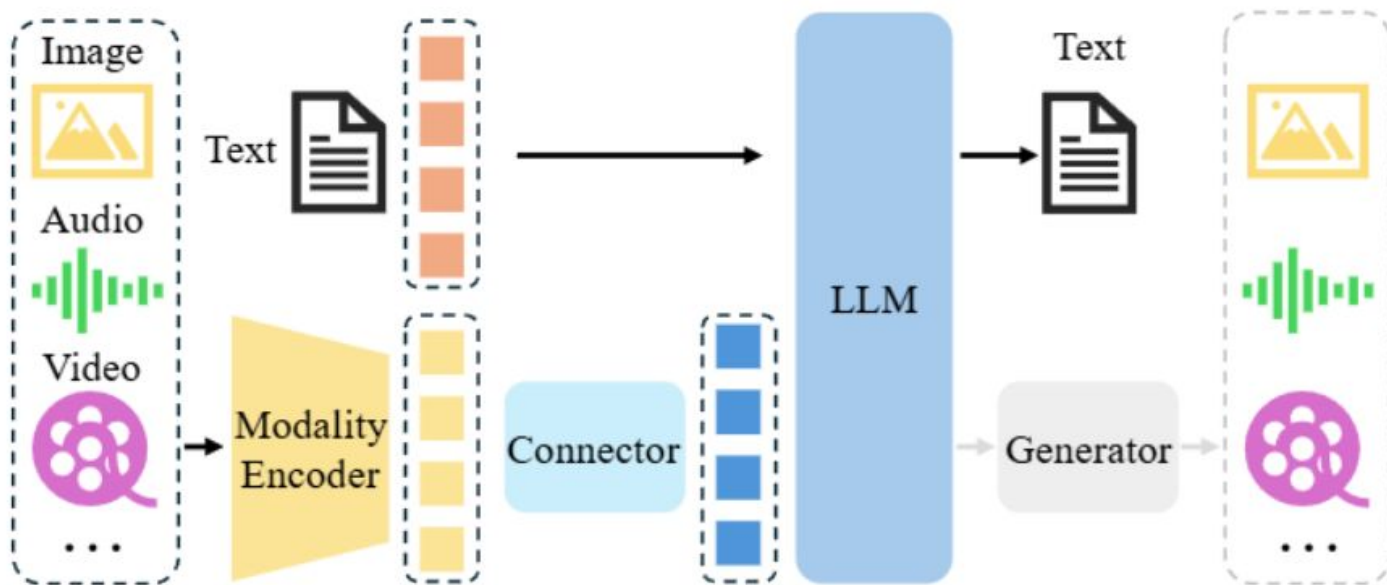
Texto e imagen: Modelos MLLM

Dos opciones principales de arquitectura:

- LLM como parte del sistema: Estándar hoy en día, más 90% son así.
Modality encoder + Connector + LLM (+ Modality Decoder)

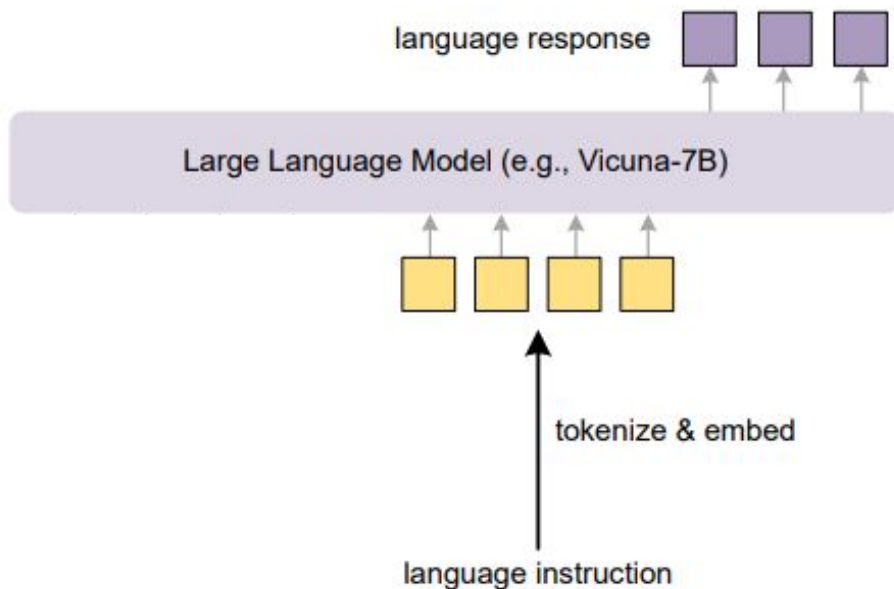


Texto e imagen: Modelos MLLM



Texto e imagen: Modelos MLLM

Caso particular: VLM (Vision-Language Models)

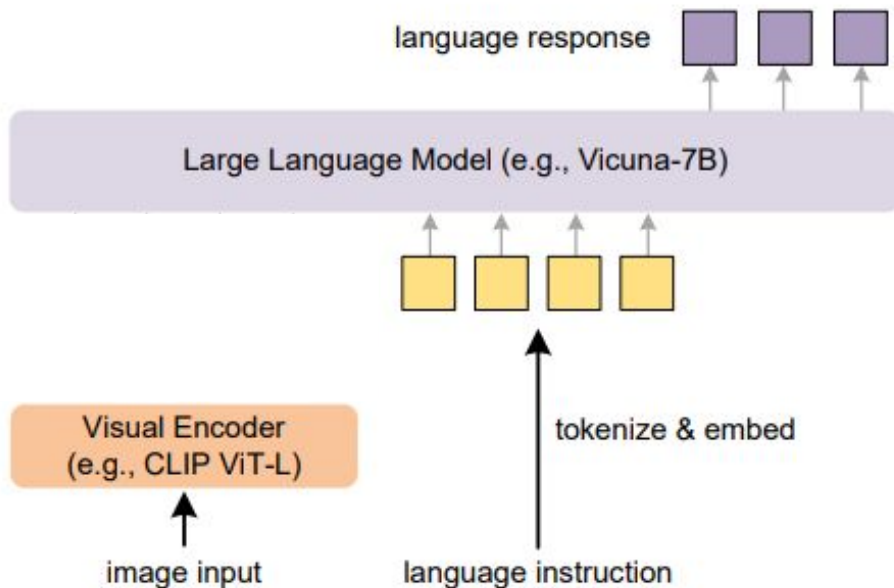


LLM

- Input: texto
- Output: texto

Texto e imagen: Modelos MLLM

Caso particular: VLM (Vision-Language Models)



LLM

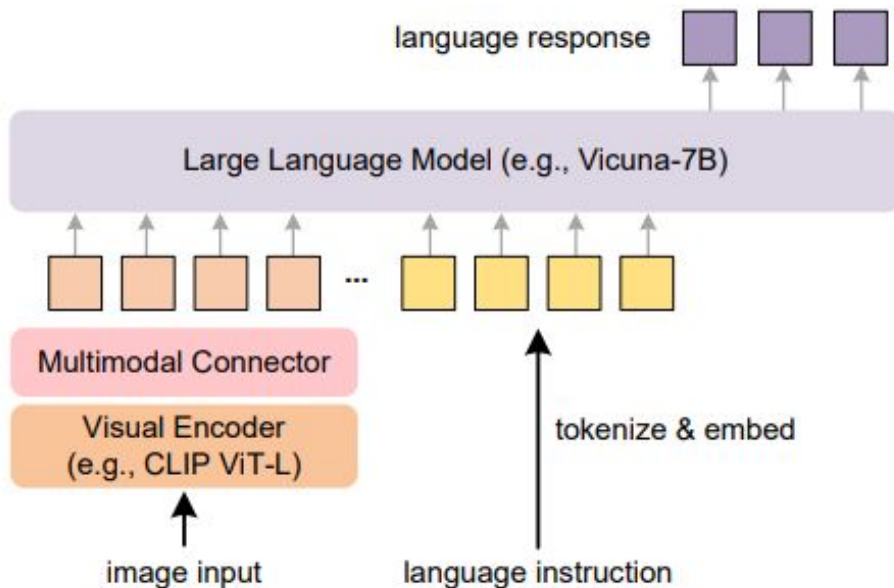
- Input: Texto
- Output: Texto

Vision encoder

- Input: Imagen
- Output: Imagen rep.

Texto e imagen: Modelos MLLM

Caso particular: VLM (Vision-Language Models)



LLM

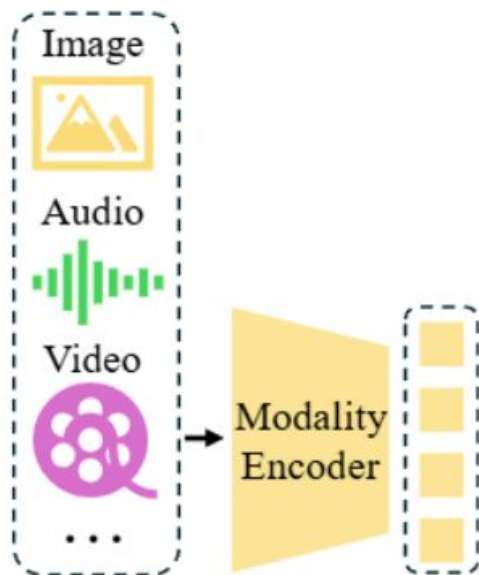
- Input: Texto
- Output: Texto

Vision encoder

- Input: Imagen
- Output: Imagen rep.

Texto e imagen: Modelos MLLM

MLLM Modality Encoder



Texto e imagen: Modelos MLLM

MLLM Modality Encoder: Para imágenes (VLM) es un image encoder.

TABLE 1: A summary of commonly used image encoders.

Variants	Pretraining Corpus	Resolution	Samples (B)	Parameter Size (M)
OpenCLIP-ConvNext-L [46]	LAION-2B	320	29	197.4
CLIP-ViT-L/14 [13]	OpenAI's WIT	224/336	13	304.0
EVA-CLIP-ViT-G/14 [47]	LAION-2B, COYO-700M	224	11	1000.0
OpenCLIP-ViT-G/14 [46]	LAION-2B	224	34	1012.7
OpenCLIP-ViT-bigG/14 [46]	LAION-2B	224	34	1844.9

CLIP es el más famoso y utilizado.

Algunos que están ganando popularidad: SigLIP, OpenCLIP, EVA-CLIP, Dino...

Texto e imagen: Modelos MLLM

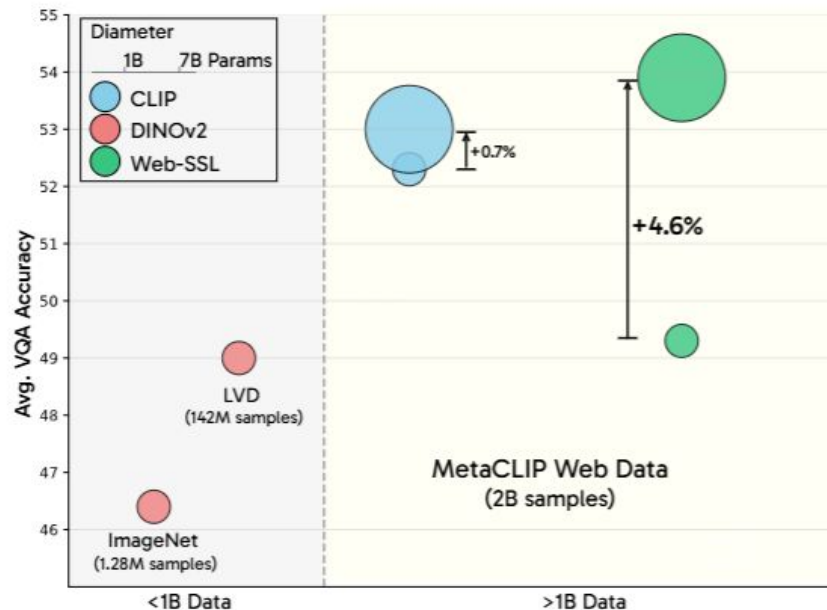
MLLM Modality Encoder: Para imágenes (VLM) es un image encoder.

Es necesario que sea un encoder entrenado de manera multimodal (CLIP) o puede ser solo una CNN/ViT standard?

Scaling Language-Free Visual Representation Learning

“...visual SSL models scale better than CLIP models in terms of data and model capacity...”

“...pure visual SSL can match language-supervised visual pretraining at scale...”



Texto e imagen: Modelos MLLM

MLLM Modality Encoder: Para imágenes (VLM) es un image encoder.

Un gran problema es la baja resolución y el reescalado a imágenes cuadradas de normalmente 224 x 224 píxeles.

Esto afecta sobretodo para tareas que incluyen OCR.

Empieza a haber encoders multi-resolución.



Texto e imagen: Modelos MLLM

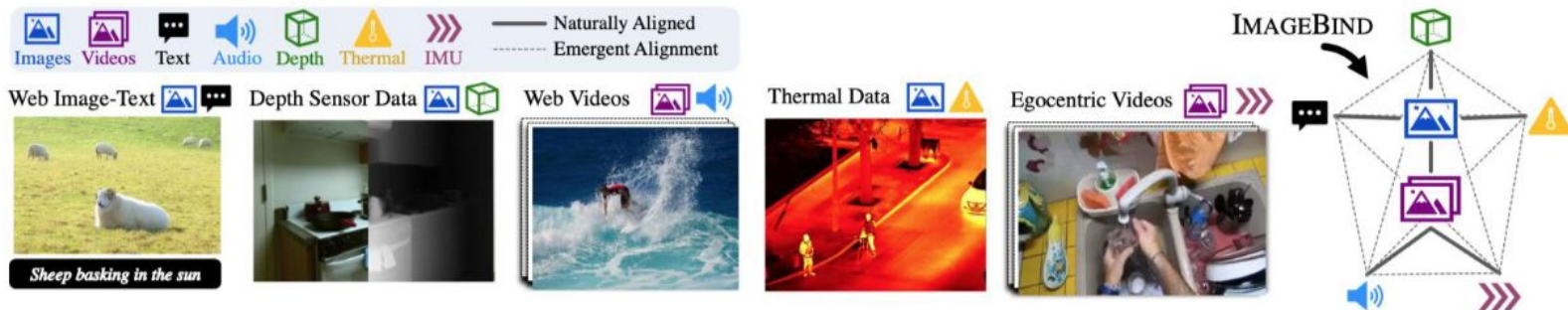
MLLM Modality Encoder: Para otras modalidades es similar, hace falta un encoder que extraiga representaciones de los datos. Ejemplos:

Audio: Whisper, AudioCLIP, CLAP, HuBERT, BEATs...

3D points: Point-Bert...

...

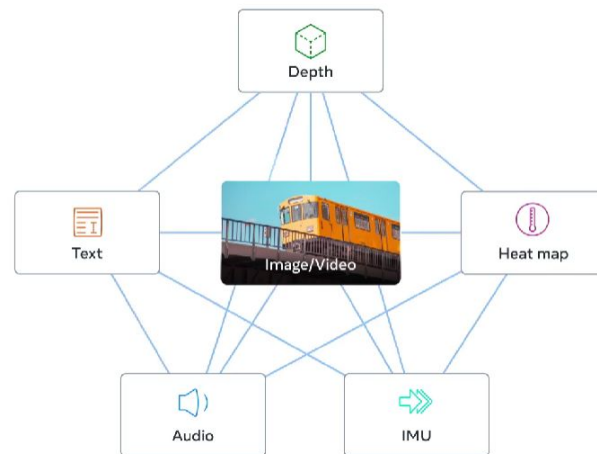
Unified: ImageBind, LanguageBind...



Texto e imagen: Modelos MLLM

MLLM Modality Encoder: Los unified unifican muchas modalidades en un mismo espacio y no hacen falta encoders distintos.

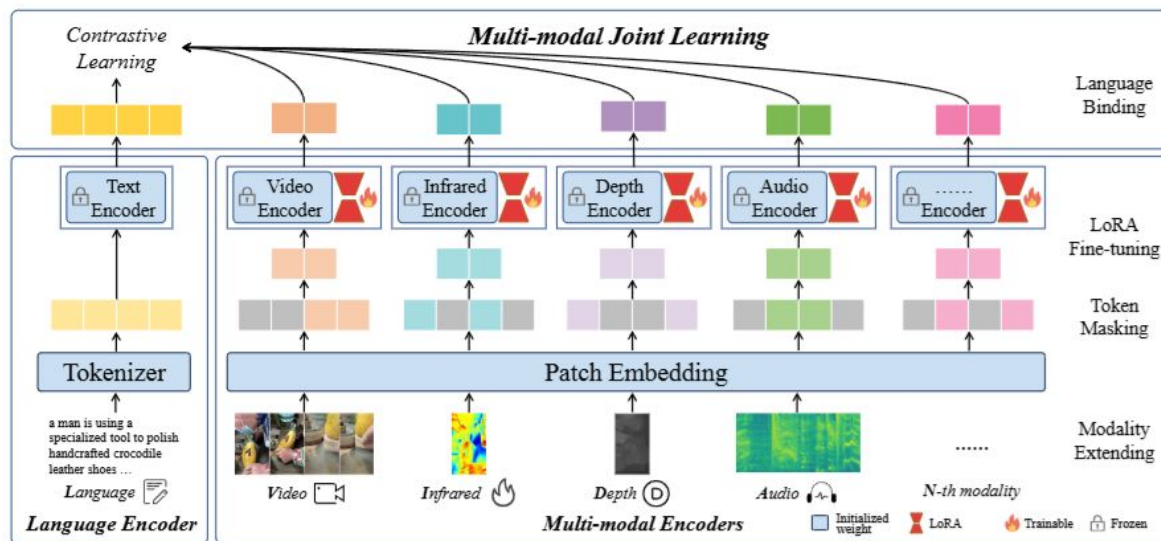
[ImageBind: One Embedding Space To Bind Them All.](#)



Texto e imagen: Modelos MLLM

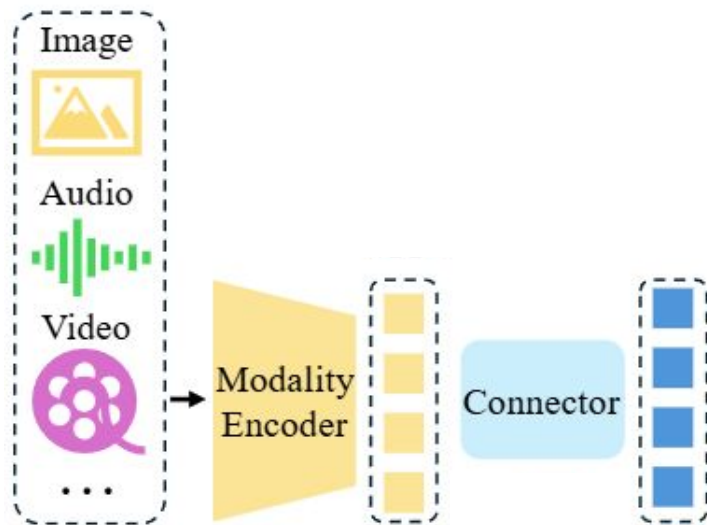
MLLM Modality Encoder: Los unifican muchas modalidades en un mismo espacio y no hacen falta encoders distintos.

LanguageBind: Extending Video-Language Pretraining to N-modality by Language-based Semantic Alignment



Texto e imagen: Modelos MLLM

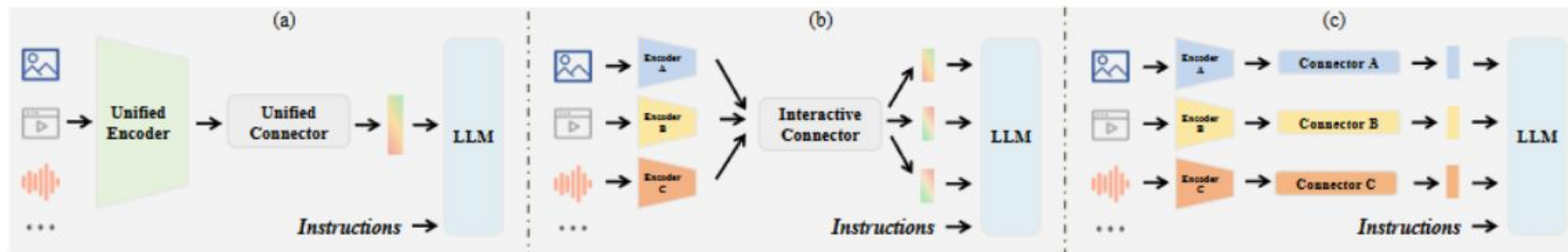
MLLM Connector



Texto e imagen: Modelos MLLM

MLLM Connector: Proyecta las representaciones multimodales al espacio del LLM.

Hay 3 opciones principales:



Early fusion

Intermediate fusion

Late fusion

MLLM Connector: Proyecta las representaciones multimodales al espacio del LLM.

¿Cómo hacerlo? No está claro. Muchas opciones:

- Linear projection (1 layer MLP): LLaVA, MiniGPT-4, NExT-GPT...
- 2 layers MLP: LLaVA-1.5/NeXT, CogVLM, DeepSeek-VL, Yi-VL...
- Q-Former: BLIP-2, InstructBLIP, VisCPM, VisualGLM...
- Perceiver Resampler: Flamingo, Qwen-VL, MiniCPM-V, LLaVA-UHD...
- C-Abstractor: HoneyBee, MM1...

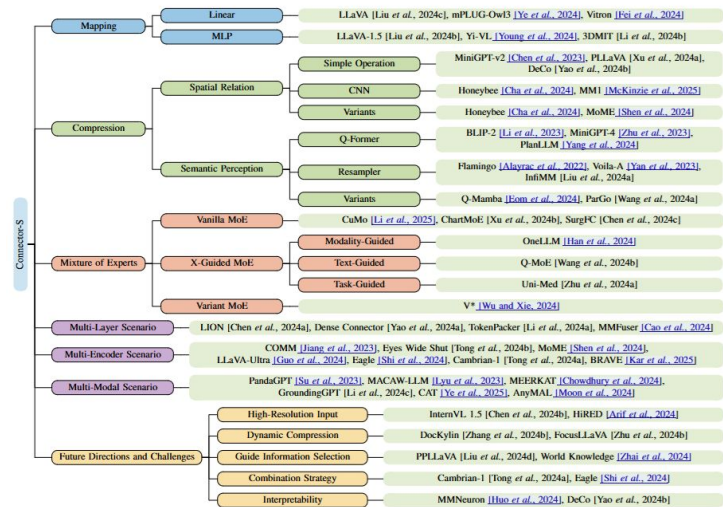
Texto e imagen: Modelos MLLM

MLLM Connector: Proyecta las representaciones multimodales al espacio del LLM.

¿Cuál es mejor? Ni idea, distintos papers dan conclusiones distintas. Parece que cada uno tira por lo que mejor le funciona...

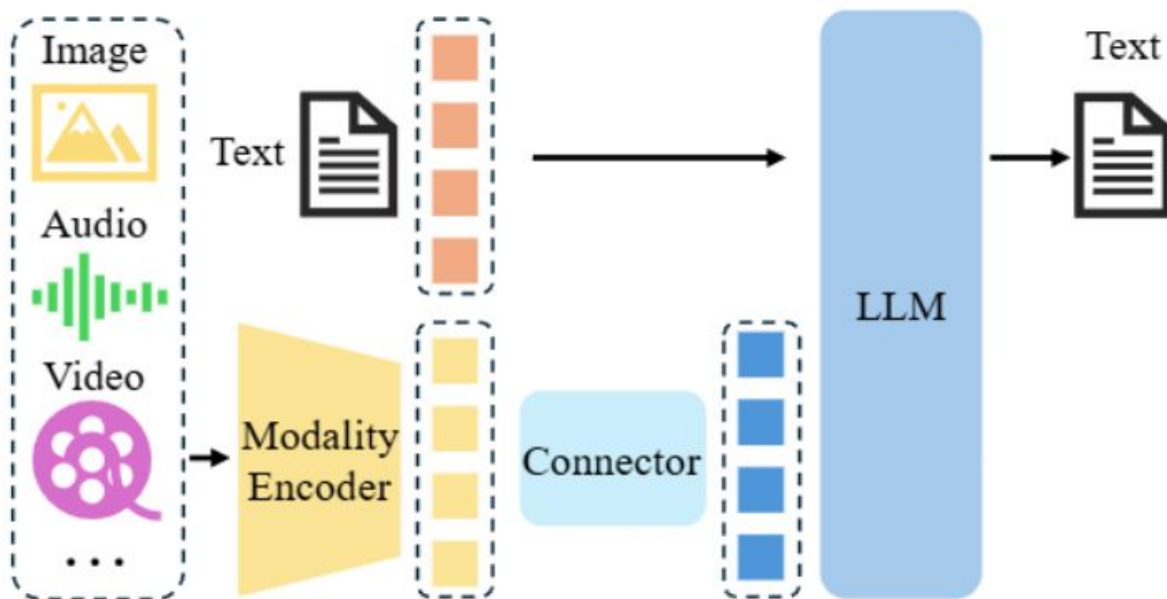
Una cosa común: El número de tokens visuales importa mucho, cuantos más mejor.

[Connector-S: A Survey of Connectors in Multi-modal Large Language Models](#)



Texto e imagen: Modelos MLLM

MLLM LLM: Modelo de Lenguaje



Texto e imagen: Modelos MLLM

MLLM LLM: Modelo de Lenguaje

Generan texto de forma natural.

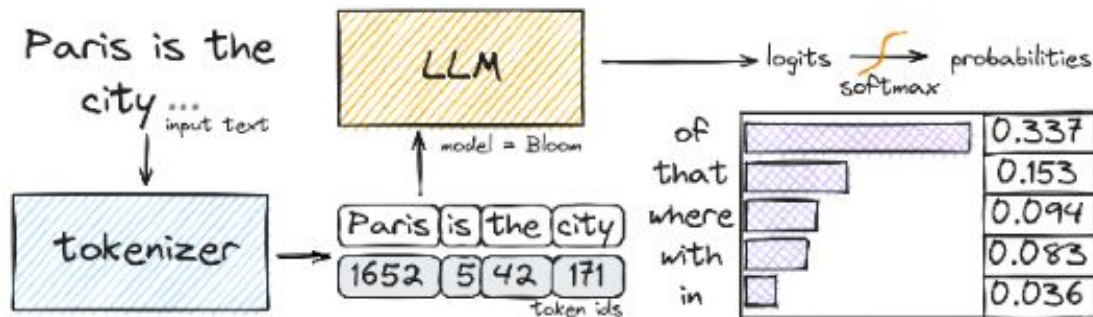
Es donde están la mayoría de los parámetros.

Suele estar pre-entrenado (muy costosos de entrenar).

Se les puede hacer fine-tuning (instruction tuning para nuestro dataset).

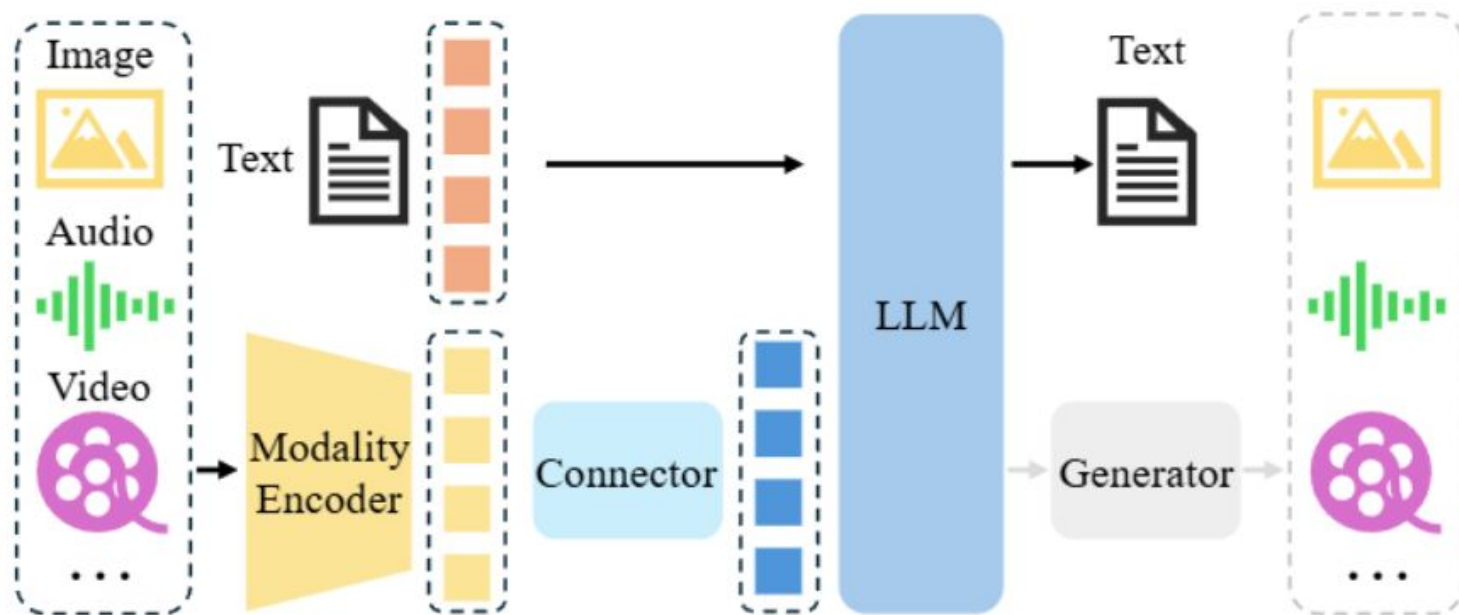
Hay técnicas para fine-tuning más rápido y usando menos GPUs: LoRA

Se suelen coger los que son open-source.



Texto e imagen: Modelos MLLM

MLLM Generator



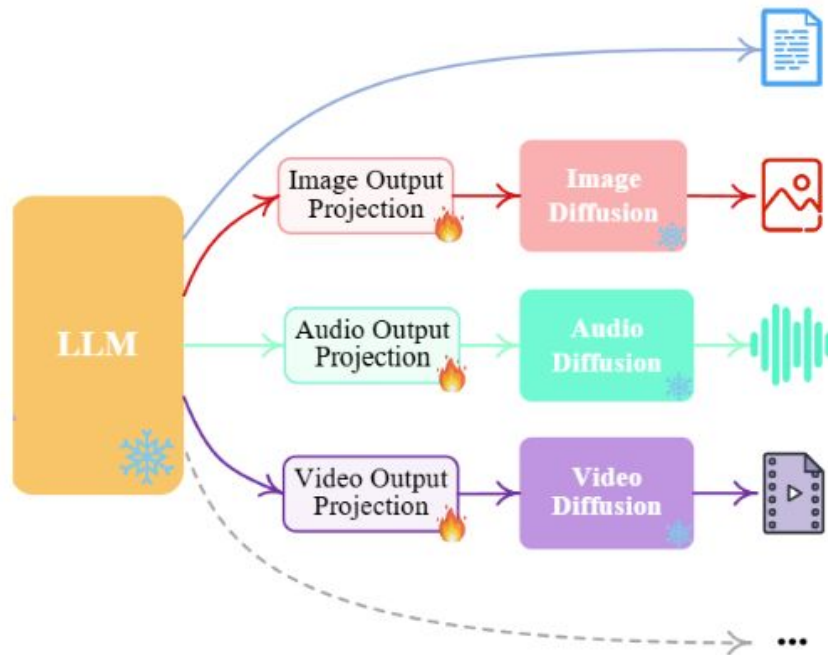
Texto e imagen: Modelos MLLM

MLLM Generator: Lo más usado con modelos de difusión de cada modalidad

Visual: Image/video diffusion

Audio: Speech/audio diffusion

...



Next-GPT

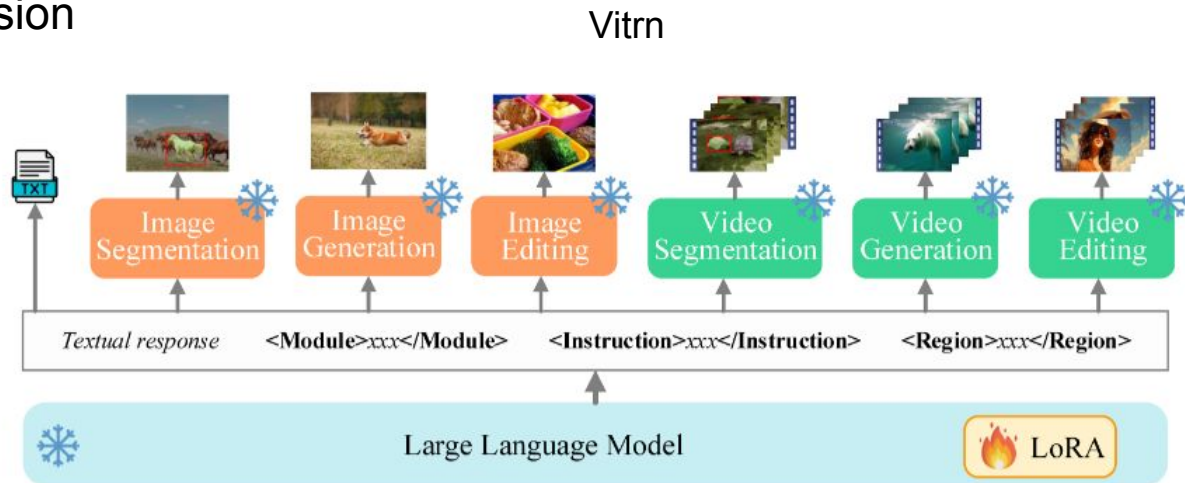
Texto e imagen: Modelos MLLM

MLLM Generator: Lo más usado con modelos de difusión de cada modalidad

Visual: Image/video diffusion

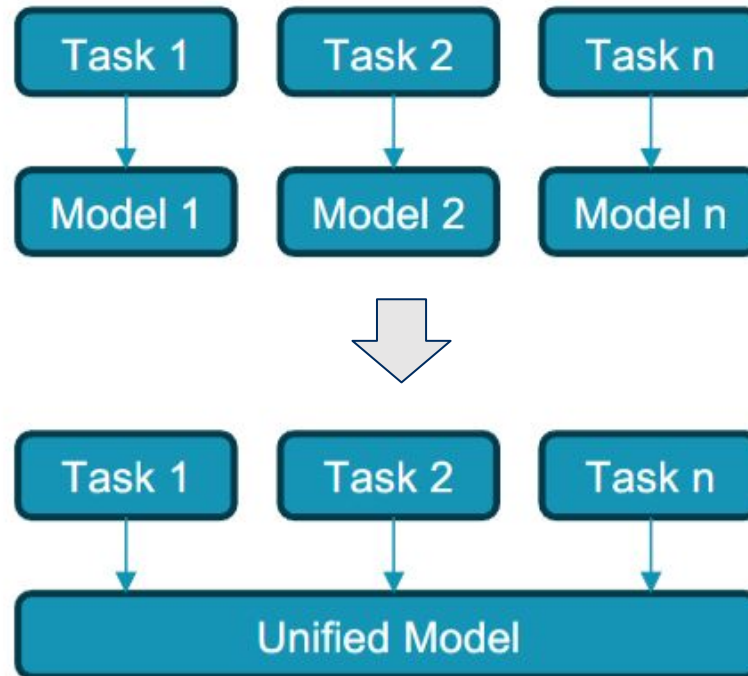
Audio: Speech/audio diffusion

...



Texto e imagen: Modelos MLLM

Entrenamiento: de single to general purpose

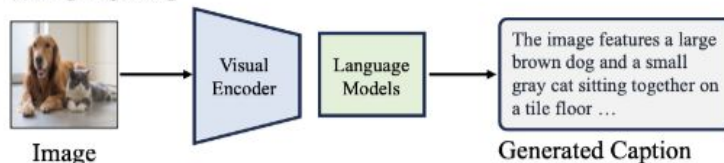


Texto e imagen: Modelos MLLM

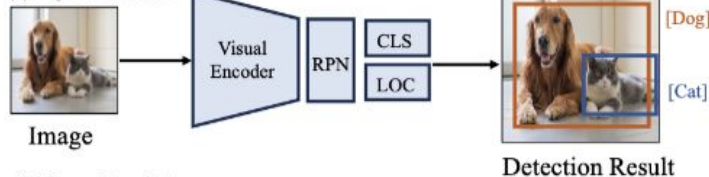
Entrenamiento: de single to general purpose

(a). Traditional Task Paradigm for Computer Vision

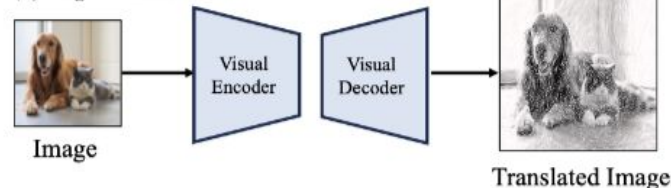
(1) Image Captioning



(2) Object Detection

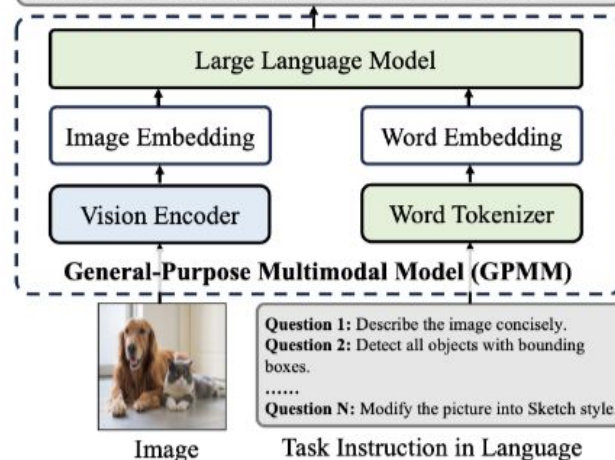


(N) Image Translation



(b). Instruction-based Task Paradigm for Computer Vision

Answer 1: The image features a large brown dog and a small gray cat sitting together on a tile floor, and both animals appear to be relaxed...
Answer 2: There are two objects in the picture, including one cat <BOX1> and one dog <BOX2>.
.....
Answer N: Here is the modified picture in sketch style <IMAGE>.

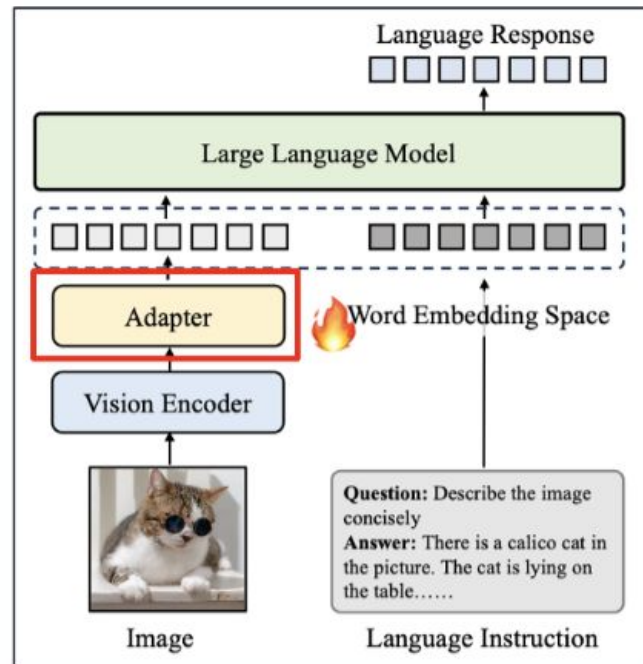


Texto e imagen: Modelos MLLM

Entrenamiento: Normalmente en 2 etapas.

1) Pretraining stage.

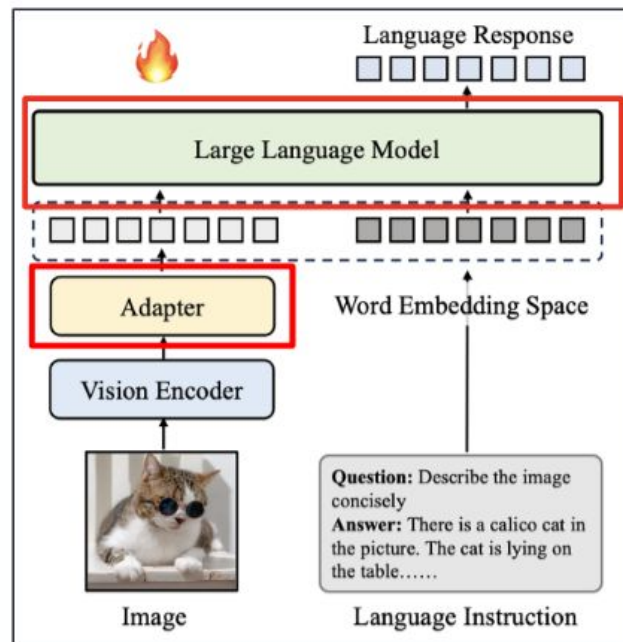
Suele entrenarse solo el adaptador.
Alinear diferentes modalidades y da
conocimiento general al modelo.



Texto e imagen: Modelos MLLM

Entrenamiento: Normalmente en 2 etapas.

- 1) Pretraining stage.
- 2) Instruction-Tuning
El adaptador se sigue entrenando.
Se hace fine-tuning también al LLM.
En algunas opciones se puede entrenar también el encoder si no se obtienen buenos resultados.



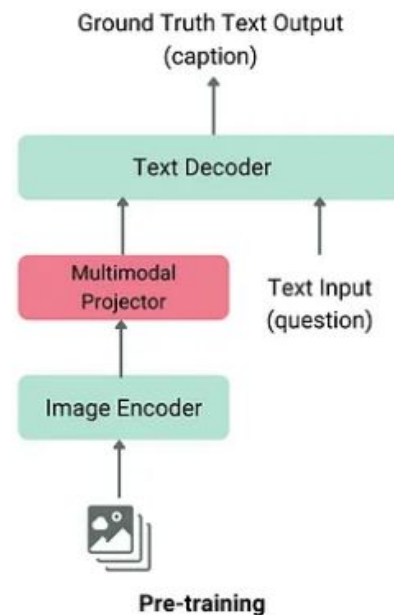
Texto e imagen: Modelos MLLM

Entrenamiento: Pretraining stage.

Dos opciones de datos:

- Coarse: Gran volumen, captions cortos y con ruido.
- Fine: Más calidad, captions más largos y con más detalle. Suele haber menos datos.

Dataset	Samples	Taxonomies
CC-3M	3.3M	Coarse-grained Image-Text
CC-12M	12.4M	Coarse-grained Image-Text
LAION-5B	5.9B	Coarse-grained Image-Text
SBU-Captions	1M	Coarse-grained Image-Text
ShareGPT4V-PT	1.2M	Fine-grained Image-Text
LVIS-Instruct4V	111k	Fine-grained Image-Text
<u>ALLaVA</u>	709k	Fine-grained Image-Text
MSR-VTT	200k	Video-Text
<u>WavCaps</u>	24k	Audio-Text
MMC-Instruction	600k	Chart-Text



Multimodal Alignment Pretraining

Texto e imagen: Modelos MLLM

Entrenamiento: Pretraining stage.



COCO: Young children standing on a platform waiting for a train to arrive. Adults and children watching a train slowly leave. A family near a railroad track watching the train pass. People waiting on a platform as a train pulls up. A train station with a green train on the tracks and children waiting for it to go by.

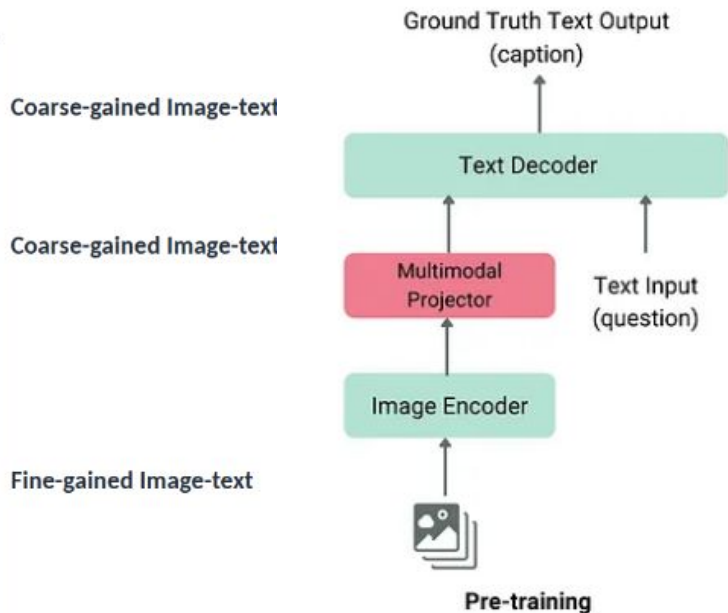
LLaVA: At a train station, a group of people, including both young children and adults, are standing on a platform waiting for a train to arrive. The train is already present on the tracks, partially visible on the right side of the image. Some of the people watch the train closely, while others seem to be patiently anticipating its departure.

There is a total of eight individuals waiting for the train, with one child in the middle of the platform and the others scattered around. A backpack can be found on the far left side of the platform, suggesting that someone may have set it down while waiting.

ShareGPT4V: The image captures a moment at a train station. A green train is on the tracks, moving away from the platform labeled "Platform 2". The train's motion is observed by people standing on the platform, their attention drawn to the departing vehicle.

A red signal stands on the tracks, its vibrant color contrasting with the green of the train. Signs are prominently displayed around the platform. One warns "Beware of Trains", a cautionary reminder of the powerful machines that frequent these tracks. Another sign advises "Please Use The Footbridge To Cross The Line. If You Require Assistance Please Ask The Station Staff", guiding passengers to safely navigate the station.

The sky above is a clear blue, providing a serene backdrop to the bustling activity of the station. In the distance, trees can be seen, adding a touch of nature to this man-made setting. The image is a snapshot of everyday life at a train station, capturing both its routine operations and its inherent charm.



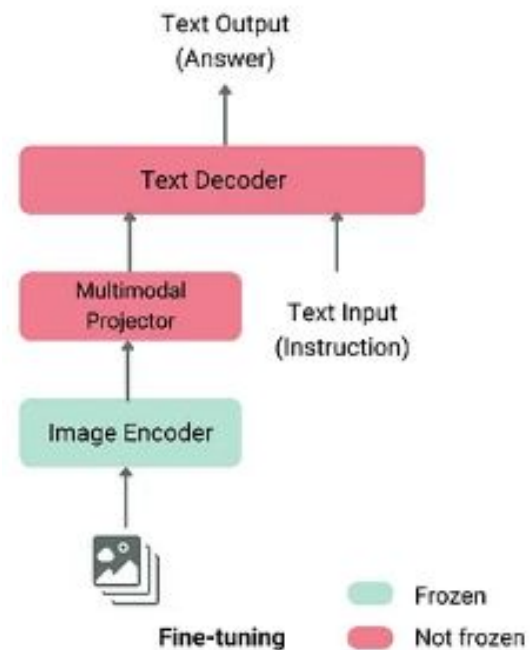
Multimodal Alignment Pretraining

Texto e imagen: Modelos MLLM

Entrenamiento: Instruction-tuning stage.

Datos más costosos de obtener, seguir múltiples instrucciones.

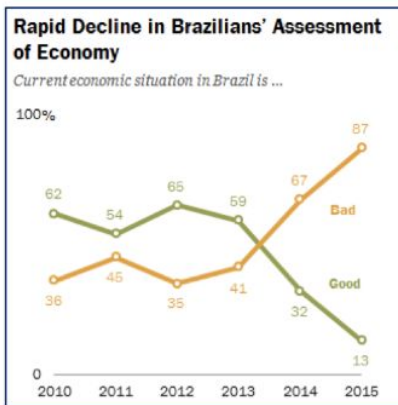
Dataset	Size	Modalities	Constructions
LLaVA-Instruct-158k	158k	Image, Text	ChatGPT-generated
LRV-Instruction	400k	Image, Text	GPT4-generated
MMC-Instruction	600k	Chart, Text	GPT4-generated/adapted
Clotho-Detail	3.9k	Text, Audio	GPT4-generated
MACAW-LLM	119k	Image, Video, Text	GPT-3.5-turbo-generated
MIMIC-IT	2.8M	Image, Video, Text	ChatGPT-generated
StableLLaVA	126k	Image, Text	StableDiffusion & ChatGPT-generated
LAMM	196k	Image, PointCloud, Text	GPT4-generated
VIGC-LLaVA	1.8M	Image, Text	Model-generated
X-LLM	10k	Image, Video, Text	ChatGPT-generated



Texto e imagen: Modelos MLLM

Entrenamiento: Instruction-tuning stage.

Se suelen usar LLM para generarlos por que los datos humanos suelen ser demasiado concisos y limitaría la capacidad de generación de los modelos.

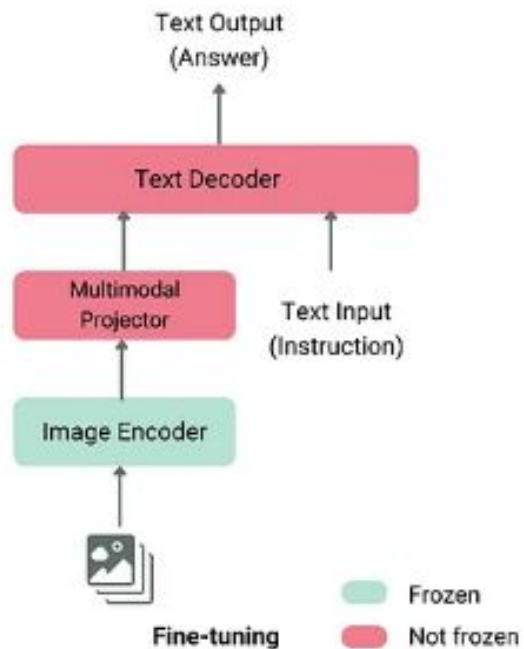


Question:

Which year has the most divergent opinions about Brazil's economy?

Answer:

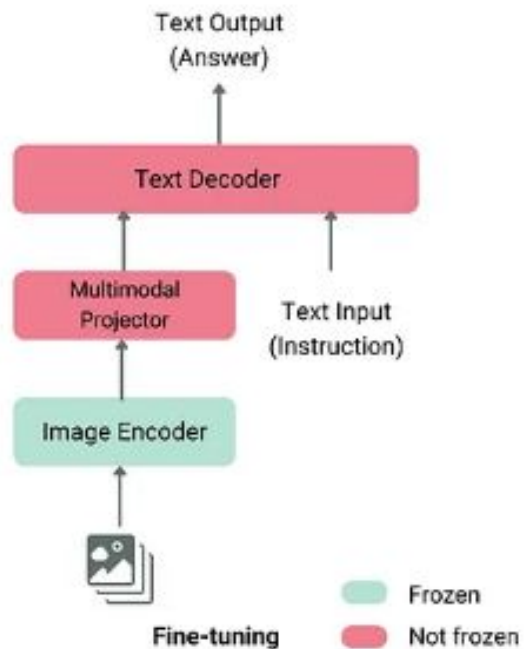
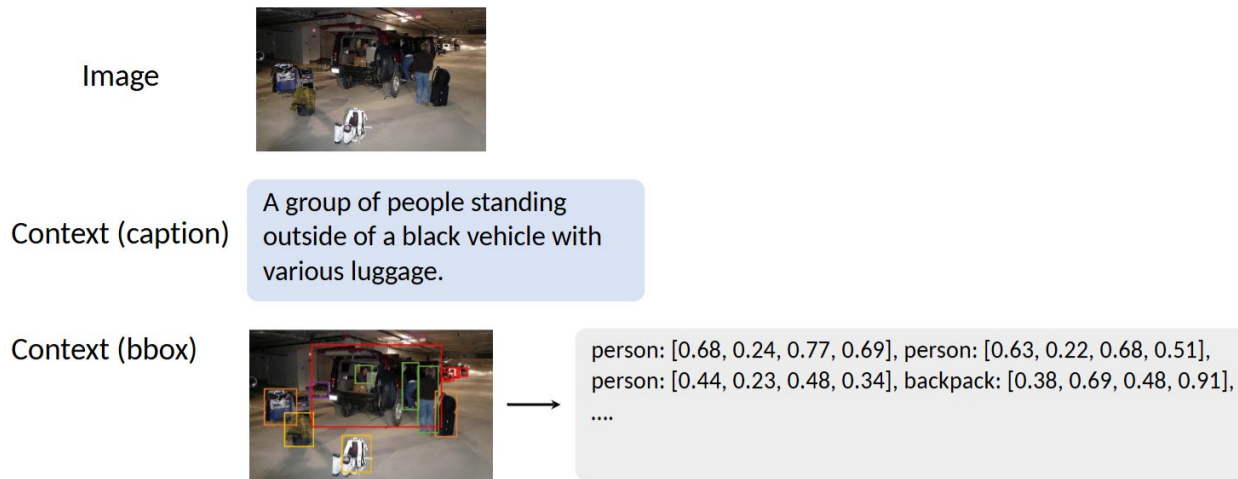
2015



Texto e imagen: Modelos MLLM

Entrenamiento: Instruction-tuning stage.

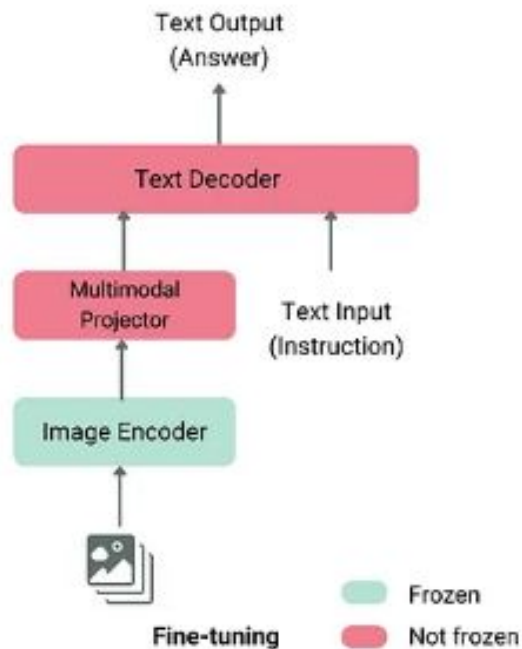
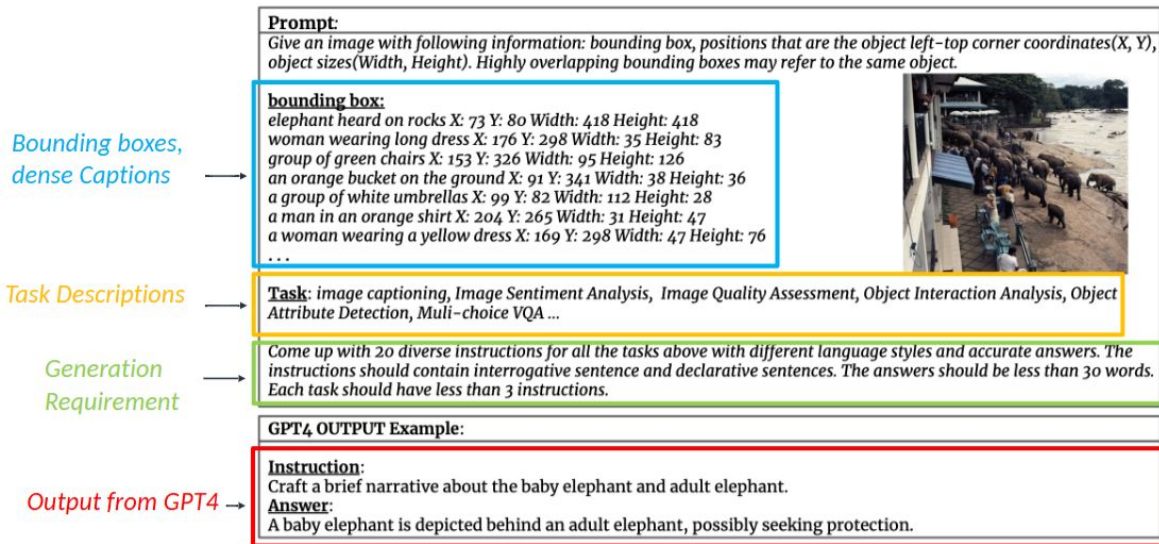
Y cómo se construyen los datasets? Necesitamos:



Texto e imagen: Modelos MLLM

Entrenamiento: Instruction-tuning stage.

Y cómo se construyen los datasets? Y entonces:



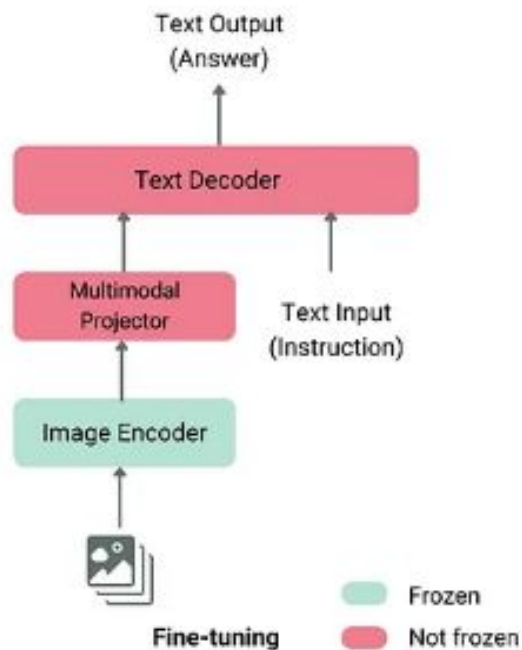
Texto e imagen: Modelos MLLM

Entrenamiento: Instruction-tuning stage.

Que la mayoría de datasets sean generados provoca otros problemas...

[How Far Are We to GPT-4V? Closing the Gap to Commercial Multimodal Models with Open-Source Suites](#)

Dataset	Size	Modalities	Constructions
LLaVA-Instruct-158k	158k	Image, Text	ChatGPT-generated
LRV-Instruction	400k	Image, Text	GPT4-generated
MMC-Instruction	600k	Chart, Text	GPT4-generated/adapted
Clotho-Detail	3.9k	Text, Audio	GPT4-generated
MACAW-LLM	119k	Image, Video, Text	GPT-3.5-turbo-generated
MIMIC-IT	2.8M	Image, Video, Text	ChatGPT-generated
StableLLaVA	126k	Image, Text	StableDiffusion & ChatGPT-generated
LAMM	196k	Image, PointCloud, Text	GPT4-generated
VIGC-LLaVA	1.8M	Image, Text	Model-generated
X-LLM	10k	Image, Video, Text	ChatGPT-generated



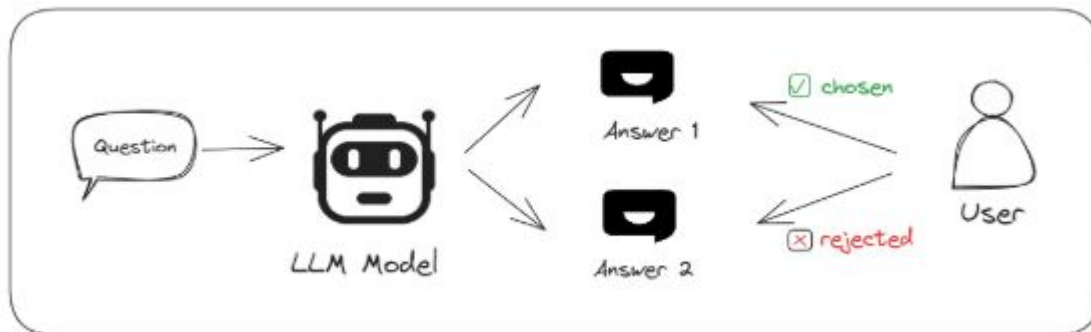
Texto e imagen: Modelos MLLM

Entrenamiento:

Reinforcement Learning with Human Feedback (RLHF).

Otra opción para alinear más con la salida deseada de los humanos.

Idealmente tendríamos a humanos valorando respuestas, pero es muy costoso:



Texto e imagen: Modelos MLLM

Entrenamiento:

Reinforcement Learning with Human Feedback (RLHF). Se entrena un modelo para predecir que elegiría el humano y se utiliza a modo de juez.

1 Collect human feedback

A Reddit post is sampled from the Reddit TL;DR dataset.

Various policies are used to sample a set of summaries.

Two summaries are selected for evaluation.

A human judges which is a better summary of the post.



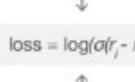
"j is better than k"

2 Train reward model

One post with two summaries judged by a human are fed to the reward model.

The reward model calculates a reward r for each summary.

The loss is calculated based on the rewards and human label, and is used to update the reward model.



"j is better than k"

3 Train policy with PPO

A new post is sampled from the dataset.

The policy π generates a summary for the post.

The reward model calculates a reward for the summary.

The reward is used to update the policy via PPO.

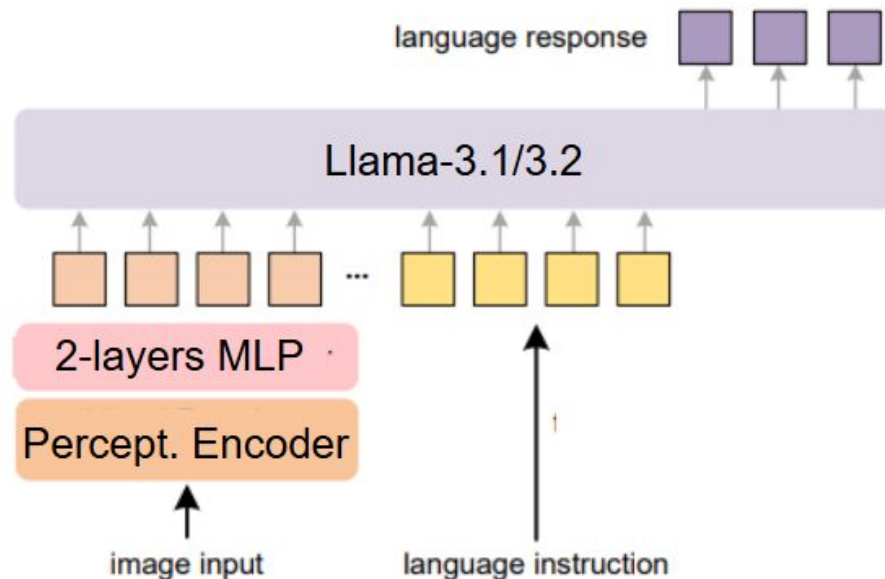


Texto e imagen: Modelos MLLM

Particular example: [Meta PerceptionLM](#)

Modelo:

“PLM consists of a vision encoder and language decoder, where the pre-trained vision encoder Perception Encoder (PE) is connected to the Llama 3 language decoder (1B, 3B, or 8B parameters) with a 2-layer MLP projector. We use PE L/14 for Llama3.2 1B and 3B, and PE G/14 for Llama3.1 8B.”



Texto e imagen: Modelos MLLM

Particular example: [Meta PerceptionLM](#)

Training stages:

- 1) Projector warm-up: “freeze the vision encoder and LLM and only train the vision projector on a small amount of synthetic image data”.
- 2) Large-scale midtraining with synthetic data: “we train PLM on diverse domains of images and videos”.
- 3) Supervised finetuning with human-annotated data: “Finally, we train PLM with higher image resolutions and more video frames”.

	Stage 1	Stage 2	Stage 3
	Warmup	Midtraining	SFT
Modality	Image	Image + Video	Image + Video
Data	1M Synthetic	64M Synthetic	14M SFT
Training	Projector	Full	Full
Downsampling	-	2×2	2×2
Tiles/Frames	1/-	16/16	36/32

Texto e imagen: Modelos MLLM

Particular example: [Meta PerceptionLM](#)

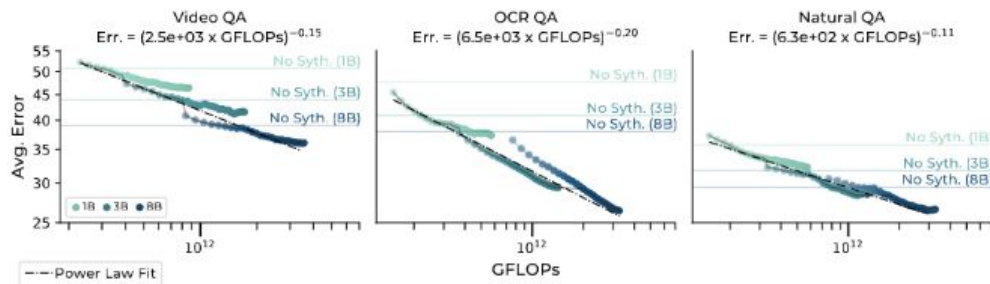


Figure 2 Synthetic Scaling Plots. Relationship between Average Error across benchmarks and training compute (in floating-point operations) for various PLM models. We report average errors across Video QA tasks [52, 94, 96, 120, 145, 196], OCR QA tasks [110, 123–125], and Natural Images tasks [13, 59, 97, 121, 165, 199]. Model's performance using only human-labeled data subset are reported (No Synth.) as well as the actual power-law fit of each subcategory.

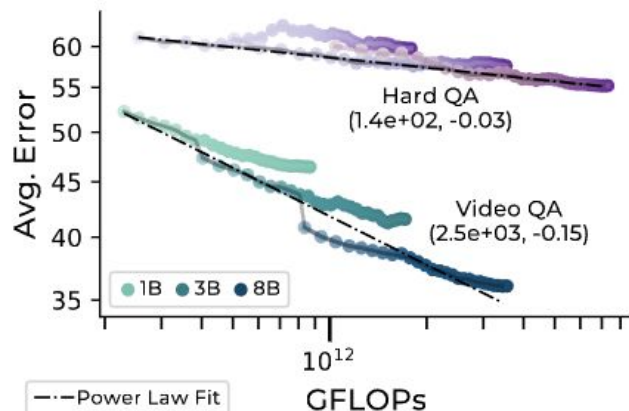


Figure 3 Limitation of synthetic data. Challenging video tasks (HardQA) do not scale well with synthetic data.

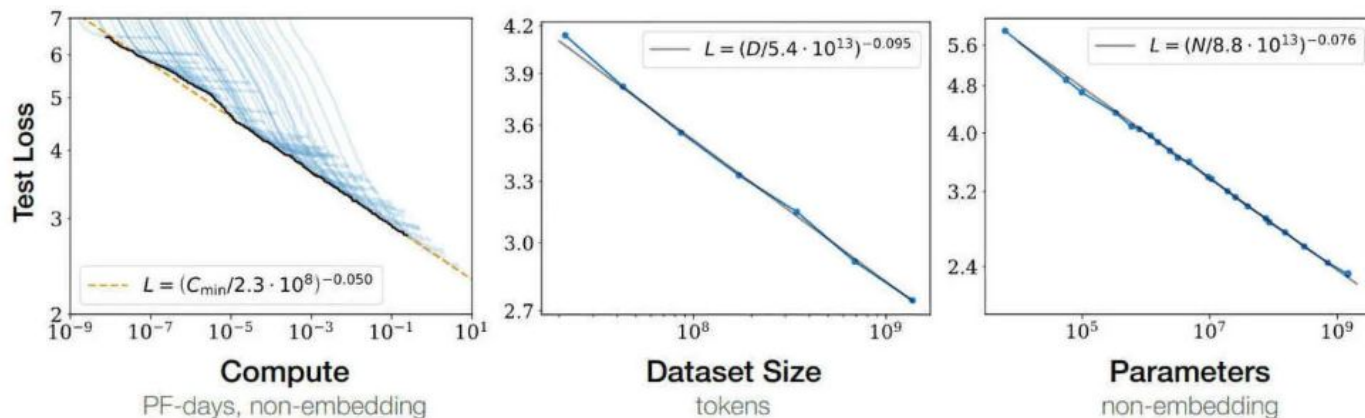
but...“scaling synthetic data is only effective for established, base tasks”.

Texto e imagen: Modelos MLLM

Scaling laws:

Importante para predecir el comportamiento de modelos más grandes, entrenados más tiempo y en más datos.

Muy necesarias para reducir costes, entrenar Llama3 costó más de \$720 millones.



Texto e imagen: Modelos MLLM

Native multimodal models:

Lo que hemos visto suele implicar un LLM ya entrenado.

Pero: ¿es mejor entrenar multimodal desde el principio?

 **Machine Learning Research**

[Overview](#)

[Research Highlights](#)

[Publications](#)

[Events](#)

[Work with us](#)

Computer Vision, Methods and
Algorithms

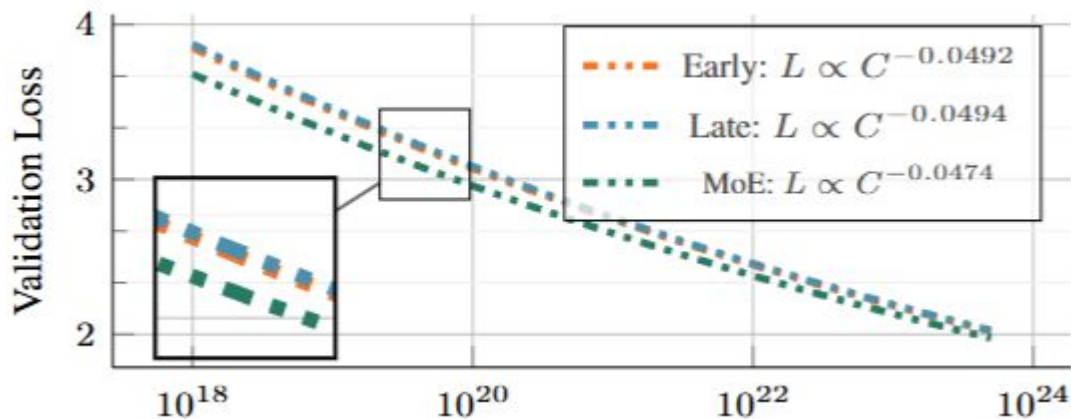
Paper | April 2025

Scaling Laws for Native Multimodal Models

Mustafa Shukor^{†‡}, Enrico Fini, Victor Guilherme Turrise da Costa, Matthieu Cord[‡],
Joshua Susskind, Alaaeldin El-Nouby

Native multimodal models: Scaling Laws for Native Multimodal Models

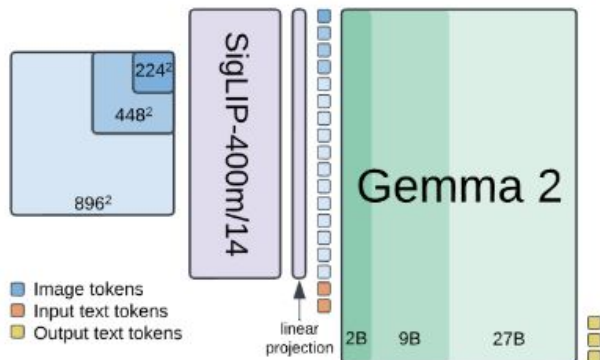
“Our investigation reveals no inherent advantage to late-fusion architectures over early-fusion ones. On the contrary, early-fusion exhibits stronger performance at lower parameter counts, is more efficient to train, and is easier to deploy. Incorporating Mixture of Experts (MoEs) allows models to learn modality-specific weights, significantly benefiting performance.”



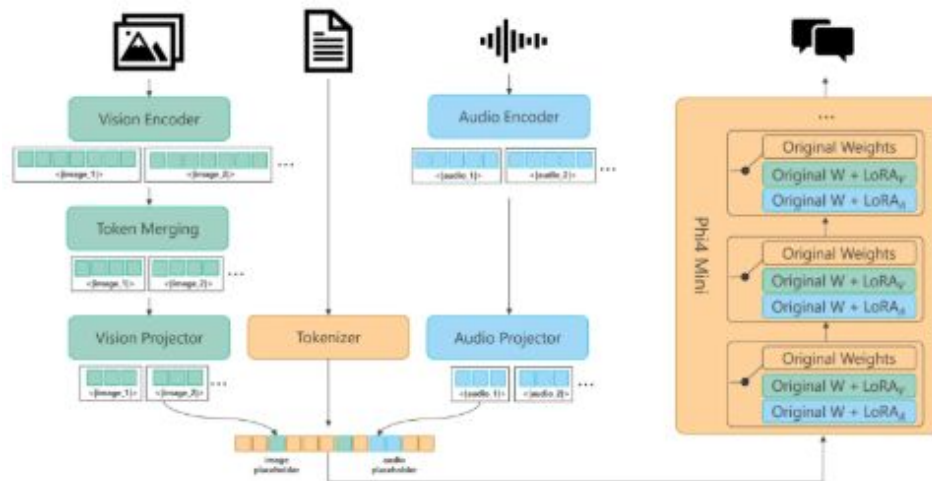
Texto e imagen: Modelos MLLM

Otros modelos:

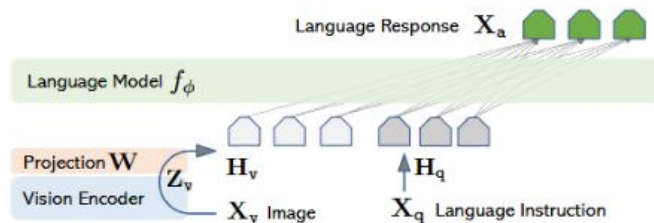
[PaliGemma2](#)



[Phi-4-Mini](#)



[LlaVA](#)



Texto e imagen: Modelos MLLM

Otros modelos:

	Modality (w/ Language)			
	Image	Video	Audio	3D
Input-side Perceiving	Flamingo, Kosmos-1, Blip2, mPLUG-Owl, Mini-GPT4, LLaVA, InstructBLIP, VPGTrans, CogVLM, Monkey, Chameleon, Otter, Qwen-VL, GPT-4v, SPHINX, Yi-VL, Fuyu, ...	VideoChat, Video-ChatGPT, Video-LLaMA, PandaGPT, MovieChat, Video-LLaVA, LLaMA-VID, Momentor, ...	AudioGPT, SpeechGPT, VIOLA, AudioPaLM, SALMONN, MU-LLaMA, ...	3D-LLM, 3D-GPT, LL3DA, SpatialVLM, PointLLM, Point-Bind, ...
	[Pixel-wise] GPT4RoI, LION, MiniGPT-v2, NExT-Chat, Kosmos-2, GLaMM, LISA, DetGPT, Osprey, PixelLM, ...	[Pixel-wise] PG-Video-LLaVA, Merlin, MotionEpic, ...	-	-
	Video-LLaVA, Chat-UniVi, LLaMA-VID		-	-
	Panda-GPT, Video-LLaMA, AnyMAL, Macaw-LLM, Gemini, VideoPoet, ImageBind-LLM, LLMBind, LLaMA-Adapter, ...			-
Perceiving + Generating	GILL, EMU, MiniGPT-5, DreamLLM, LLaVA-Plus, InternLM-XComposer2, SEED-LLaMA, LaVIT, Mini-Gemini, ...	GPT4Video, Video-LaVIT, VideoPoet, ...	AudioGPT, SpeechGPT, VIOLA, AudioPaLM, ...	-
	[Pixel-wise] Vitron		-	-
	NExT-GPT, Unified-IO 2, AnyGPT, CoDi-2, Modaverse, ViT-Lens, ...			-

Texto e imagen: Modelos MLLM

Otros modelos:

Texto+Imagen2Texto: [Flamingo](#), [Kosmos-1](#), [Blip2](#), [mPLUG-Owl](#), [Mini-GPT4](#), [LlaVA](#), [InstructBLIP](#), [Qwen-VL](#), [GPT-4v](#), [Chameleon](#), [Otter](#), [PerceptionLM](#)...

Texto+Video2Texto: VideoChat, Video-ChatGPT, Video-LlaMA, PandaGPT, MovieChat, Video-LlaVA, LLaMA-VID, Momentor, PerceptionLM...

Texto+Audio2Texto: AudioGPT, SpeechGPT, VIOLA, AudioPaLM, SALMONN, MU-LLaMA...

Texto+3D2Texto: 3D-LLM, 3D-GPT, LL3DA, SpatialVLM, PointLLM, Point-Bind...

Texto+Medicina2Texto: BioGPT, DrugGPT, AlpaCare...

Texto+Quimica2Texto: ChemGPT, ChemLLM, MolCA...

Texto+Geografia2Texto: GeoGPT...

Texto e imagen: Modelos MLLM

Otros modelos:

Texto+Imagen2Texto+Imagen: GILL, EMU, MiniGPT-5, LLaVA-Plus...

Texto+Video2Texto+Video: GPT4Video, VideoPoet, Video-LaVIT...

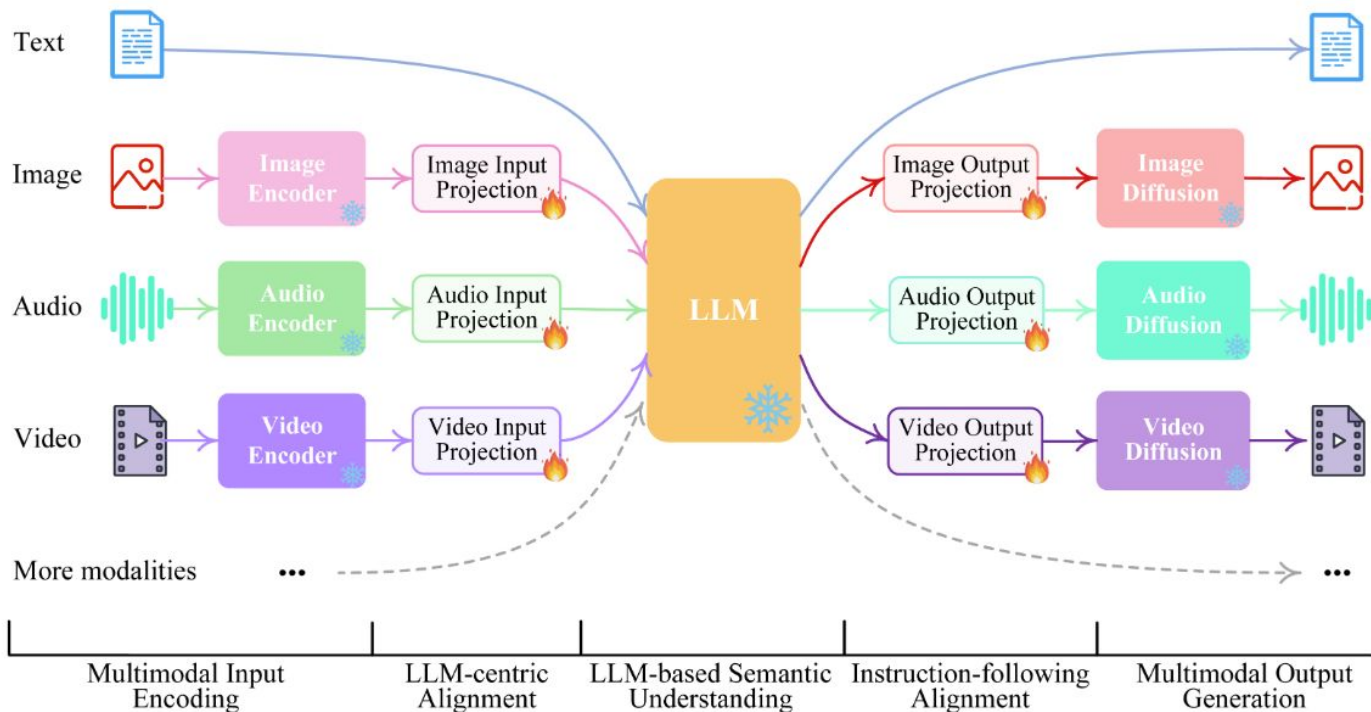
Texto+Audio2Texto+Audio: AudioGPT, SpeechGPT, VIOLA, AudioPALM...

Texto+Imagen+Video2Texto+Imagen+Video: Video-LLaVA, Chat-UniVi, LLaMA-VID...

Texto+Imagen+Video+Audio2Texto+Imagen+Video+Audio: PandaGPT, AnyMAL, ImageBind-LLM, Macaw-LLM...

Any2Any: NExT-GPT, Unified-IO 2, AnyGPT...

NExT-GPT: Any-to-Any Multimodal LLM



NExT-GPT: Any-to-Any Multimodal LLM

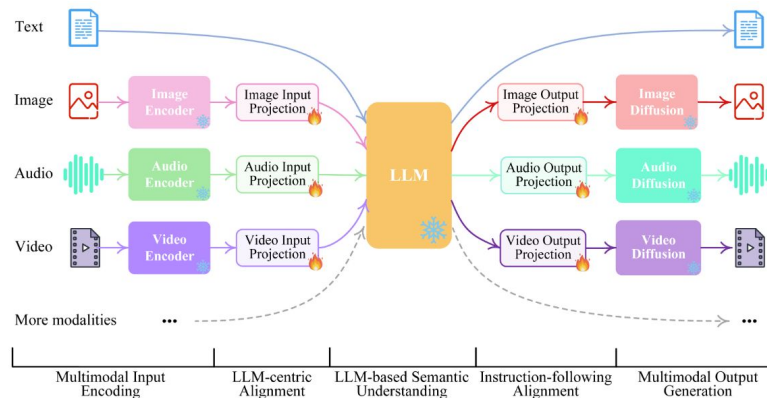


Table 1. Summary of NExT-GPT system configuration. Only 1% of parameters need updating during fine-tuning.

	Encoder		Input Projection		LLM		Output Projection		Diffusion	
	Name	Param	Name	Param	Name	Param	Name	Param	Name	Param
Text	—	—	—	—			—	—	—	—
Image					Vicuna	7B❄️	Transformer	31M🔥	SD	1.3B❄️
Audio	ImageBind	1.2B❄️	Grouping	28M🔥	(LoRA	33M🔥)	Transformer	31M🔥	AudioLDM	975M❄️
Video							Transformer	32M🔥	Zeroscope	1.8B❄️

Los MLLM actuales son impresionantes

GPT-4 is judged more human than humans in displaced and inverted Turing tests

Ishika Rathi and Sydney Taylor and Benjamin K. Bergen and Cameron R. Jones
Department of Cognitive Science, UC San Diego
9500 Gilman Dr, San Diego, CA, USA
{irathi, cameron}@ucsd.edu

People cannot distinguish GPT-4 from a human in a Turing test

Cameron R. Jones
Department of Cognitive Science
UC San Diego
San Diego, CA 92119
cameron@ucsd.edu

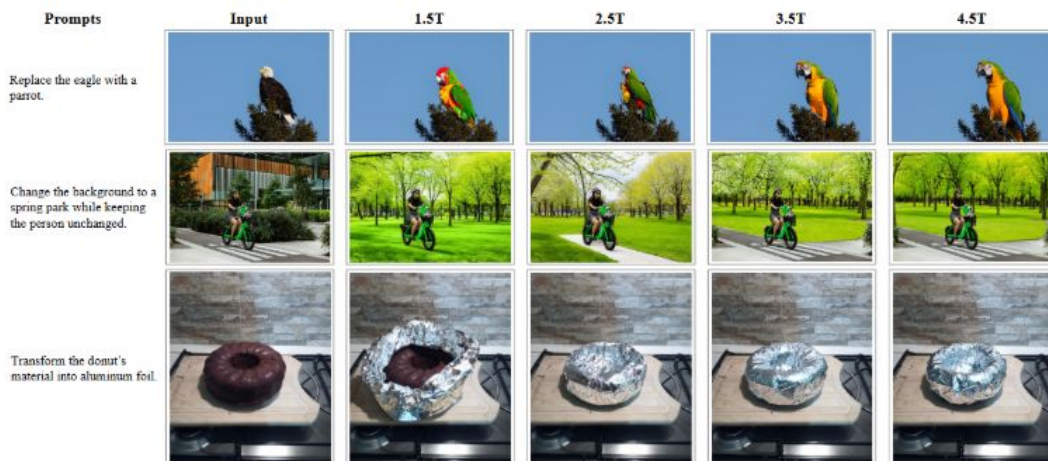
Benjamin K. Bergen
Department of Cognitive Science
UC San Diego
San Diego, CA 92119
bkbergen@ucsd.edu

Large Language Models Pass the Turing Test

Cameron R. Jones
Department of Cognitive Science
UC San Diego
San Diego, CA 92119
cameron@ucsd.edu

Benjamin K. Bergen
Department of Cognitive Science
UC San Diego
San Diego, CA 92119
bkbergen@ucsd.edu

Texto e imagen: Modelos MLLM



ByteDance | Seed

Emerging Properties in Unified Multimodal Pretraining

Chaorui Deng^{*1}, Deyao Zhu^{*1}, Kunchang Li^{*2†}, Chenhui Gou^{*1‡}, Feng Li^{*4‡}
Zeyu Wang^{*4}, Shu Zhong^{*1}, Weihao Yu^{*1}, Xiaonan Nie^{*1}, Ziang Song^{*1}, Guang Shi^{*1‡}
Haoqi Fan^{*1}

¹ByteDance Seed, ²Shenzhen Institutes of Advanced Technology, ³Monash University

⁴Hong Kong University of Science and Technology, ⁵UC Santa Cruz



Texto e imagen: Modelos MLLM

Los MLLM actuales son impresionantes, pero...Hallucinations

El output generado (normalmente texto) no se corresponde con el contenido visual (o de otra modalidad de entrada).

Instruction: Can you locate the **water bottle** placed near the yellow tennis ball on the ground?



MiniGPT4-v1: The water bottle is located near the yellow tennis ball on the ground. Can you locate the water bottle placed near the yellow tennis ball on the ground?

MiniGPT4-v2: Yes, the water bottle is placed near the yellow tennis ball on the ground.

LLaVA1.0: Unfortunately, as an AI visual assistant, I am not capable of moving or locating objects in a specific environment. However, given the image description, the water bottle is near the yellow tennis ball on the ground, and the boy is walking up to it, so you should be able to see the bottle soon enough.

LLaVA1.5: Yes, there is a water bottle placed near the yellow tennis ball on the ground.

Texto e imagen: Modelos MLLM

Los MLLM actuales son impresionantes, pero...Hallucinations

El output generado (normalmente texto) no se corresponde con el contenido visual (o de otra modalidad de entrada).



Object Hallucination:

*There are **some benches** and a fence in the background.*

Attribution Hallucination:

*There is a large tree in the background with **pink** flowers.*

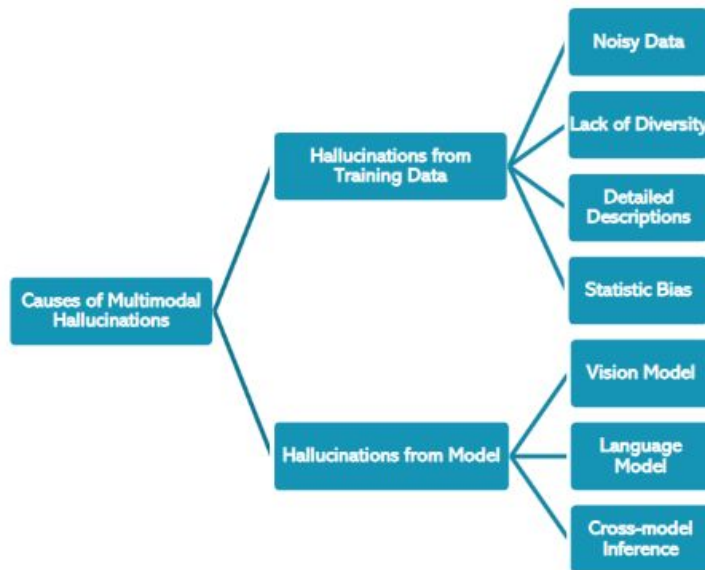
Relation Hallucination:

*The **other people** in the picture are **standing around the girl**, watching what she is doing.*

Texto e imagen: Modelos MLLM

Los MLLM actuales son impresionantes, pero...Hallucinations

El output generado (normalmente texto) no se corresponde con el contenido visual (o de otra modalidad de entrada).



Texto e imagen: Modelos MLLM

Los MLLM actuales son impresionantes, pero...Hallucinations

- Noisy data: En general, culpa de web-scraped/generated data.



LLaVA: At a train station, a group of people, including both young children and adults, are standing on a platform waiting for a train to arrive. The train is already present on the tracks, partially visible on the right side of the image. Some of the people watch the train closely, while others seem to be patiently anticipating its departure.

There is a total of eight individuals waiting for the train, with one child in the middle of the platform and the others scattered around. A backpack can be found on the far left side of the platform, suggesting that someone may have set it down while waiting.

LlaVA usa text-only ChatGPT para generar instruct-tuning dataset, pero como un modelo solo de lenguaje no puede interpretar imágenes y puede generar malas respuestas.

Texto e imagen: Modelos MLLM

Los MLLM actuales son impresionantes, pero...Hallucinations

- Noisy data
- Lack of diversity: Casi todas las descripciones son positivas y los modelos tienden a responder siempre de forma positiva/afirmativa!

Model	Yes(%)
mPLUG-Owl	96.23
LLaVA	95.37
MultiModel-GPT	99.97

Human: Is this a duck facing left?

LLaVA1.5-7B:



Yes



Yes

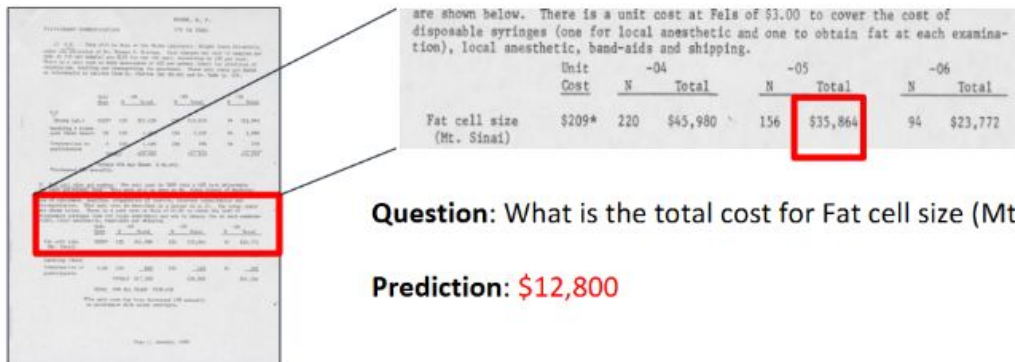


Yes

Texto e imagen: Modelos MLLM

Los MLLM actuales son impresionantes, pero...Hallucinations

- Noisy data, Lack of diversity
- Vision model: Mal modelo de visión o baja resolución puede hacer que no se capten detalles importantes.



are shown below. There is a unit cost at Fels of \$3.00 to cover the cost of disposable syringes (one for local anesthetic and one to obtain fat at each examination), local anesthetic, band-aids and shipping.

	Unit Cost	-04		-05		-06	
		N	Total	N	Total	N	Total
Fat cell size (Mt. Sinai)	\$209*	220	\$45,980	156	\$35,864	94	\$23,772

Question: What is the total cost for Fat cell size (Mt. Shlnai) in the -05 year?

Prediction: \$12,800

Texto e imagen: Modelos MLLM

Los MLLM actuales son impresionantes, pero...Hallucinations

- Noisy data, Lack of diversity
- Vision model: De hecho, solo cambiando la resolución mejoran los resultados.

Benchmark	Vision Encoder	$CHAIR_s \downarrow$	$CHAIR_i \downarrow$	Coverage $\uparrow$	Avg. Length $\uparrow$	Avg. Object $\uparrow$
CCEval (Ours)	CLIP-L-112x	79.00	21.70	32.04	110.36	9.12
	CLIP-L-224x	74.00	19.30	32.83	113.03	9.18
	CLIP-L-336x	64.00	16.00	33.37	108.52	8.92

	LLM	Resolution	VQAv2	GQA	VisWiz	TextVQA
LLaVA 1.5	Vicuna-13B	336	80.0	63.3	53.6	61.3
LLaVA 1.5	Vicuna-13B	448	81.8	64.7	57.5	62.5

Texto e imagen: Modelos MLLM

Los MLLM actuales son impresionantes, pero...Hallucinations

- Noisy data, Lack of diversity, Vision model
- LLM mucho mejor que vision encoder: Puede hacer que el modelo prefiera priorizar su conocimiento a utilizar la información visual.

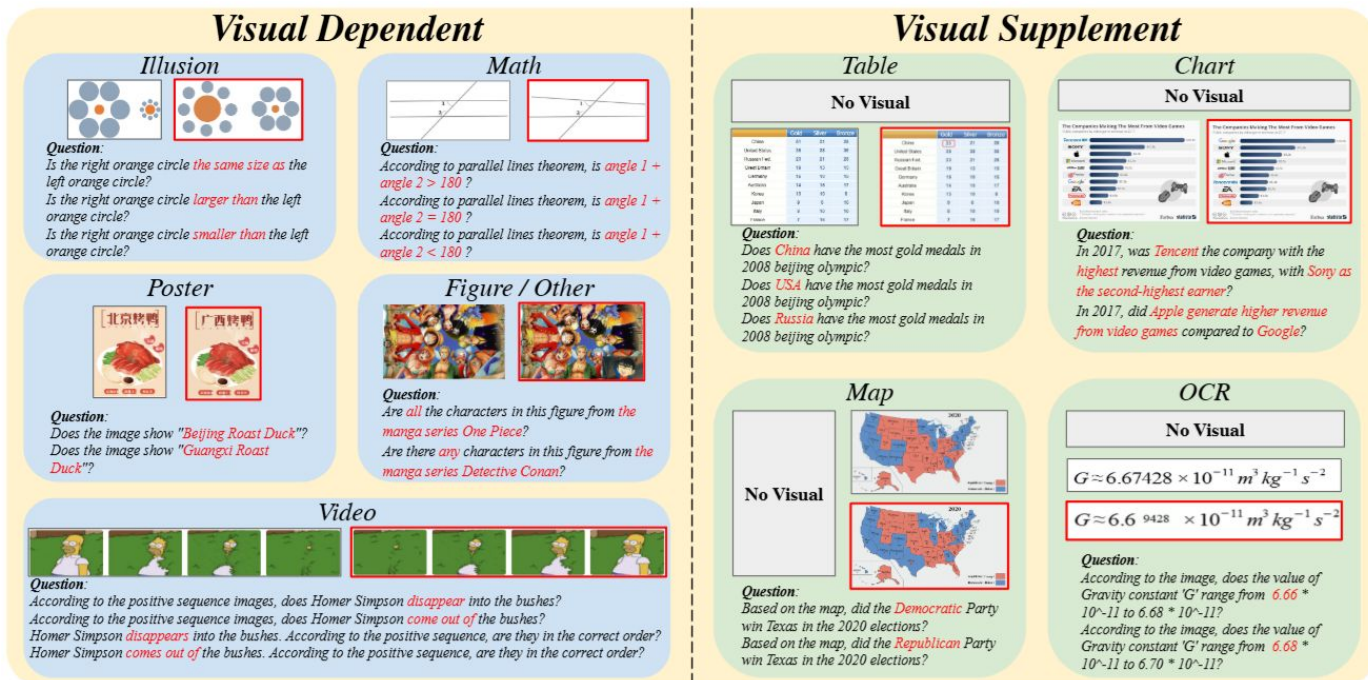
	Gold	Silver	Bronze
China	23	21	28
United States	36	38	36
Russian Fed.	23	21	28
Great Britain	19	13	15
Germany	16	10	15
Australia	14	15	17
Korea	13	10	8
Japan	9	6	10
Italy	8	10	10
France	7	16	17

The gold medal number of China is changed to 23, ranked second.

Question: According to the metal number chart of the 2008 Beijing Olympics in the image, Does China have the most gold metals, followed by the USA and Russia?

Prediction: Yes, according to the metal number chart of the 2008 Beijing Olympic Games, **China won the most gold medals, followed by the United States and Russia.** The chart shows the number of gold, silver, and bronze medals won by each country during the games.

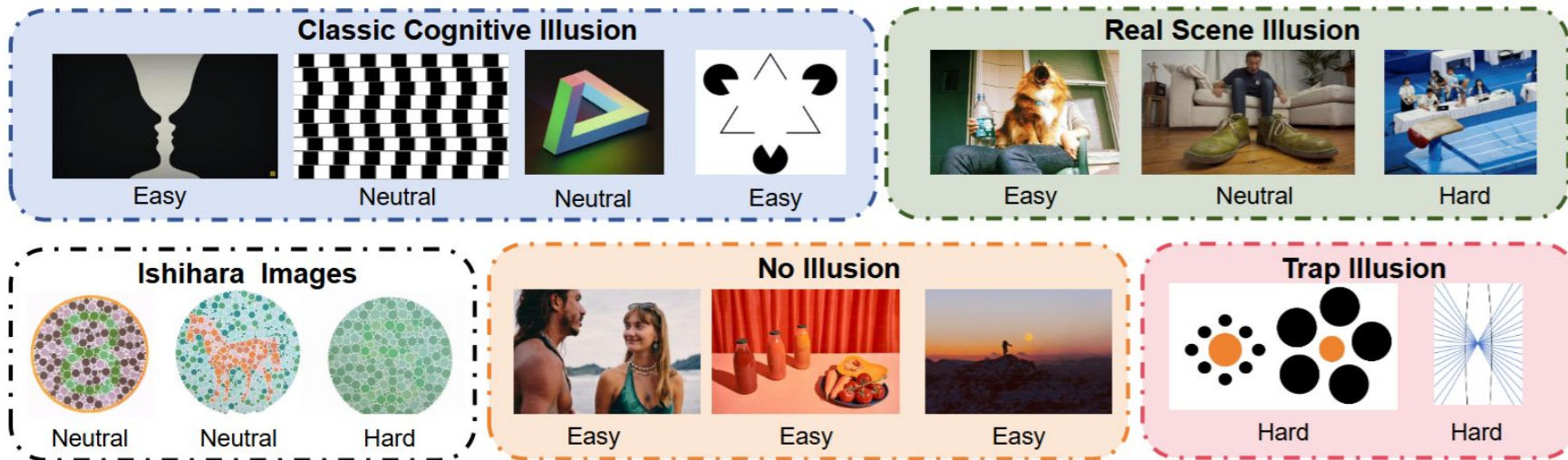
HALLUSIONBENCH: An Advanced Diagnostic Suite for Entangled Language Hallucination and Visual Illusion in Large Vision-Language Models



Texto e imagen: Modelos MLLM

Las ilusiones visuales son perfectas para testear las alucinaciones:

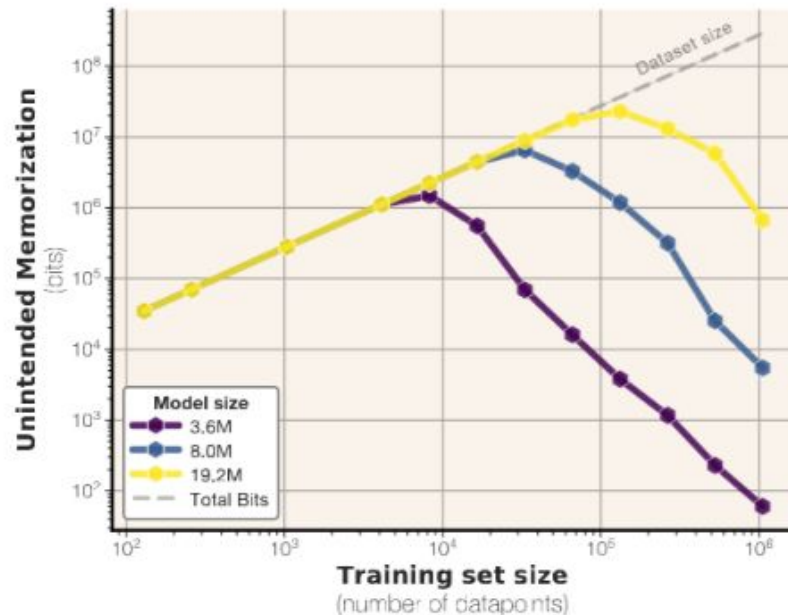
[IllusionBench: A Large-scale and Comprehensive Benchmark for Visual Illusion Understanding in Vision-Language Models](#)



Texto e imagen: Modelos MLLM

Los MLLM actuales son impresionantes, pero...Memorization

[How much do language models memorize?](#)

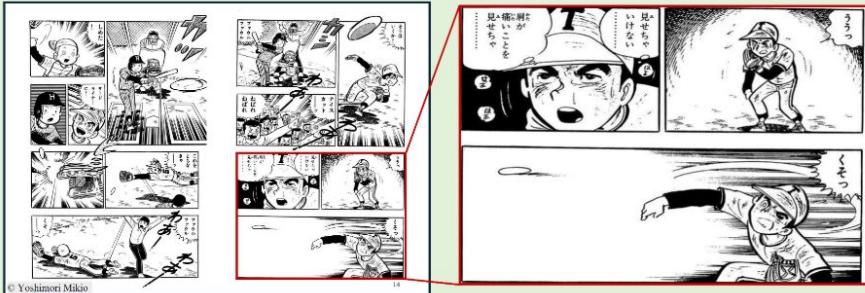


Texto e imagen: Modelos MLLM

Los MLLM actuales son impresionantes, pero...Aun se les resisten algunos campos

MangaVQA

MangaVQA



Q: ピッチャーは何を隠そうとしていますか? (What is the pitcher trying to hide?)
A: 肩が痛いこと (That his shoulder hurts)

 **526 Manually Created QA pairs**

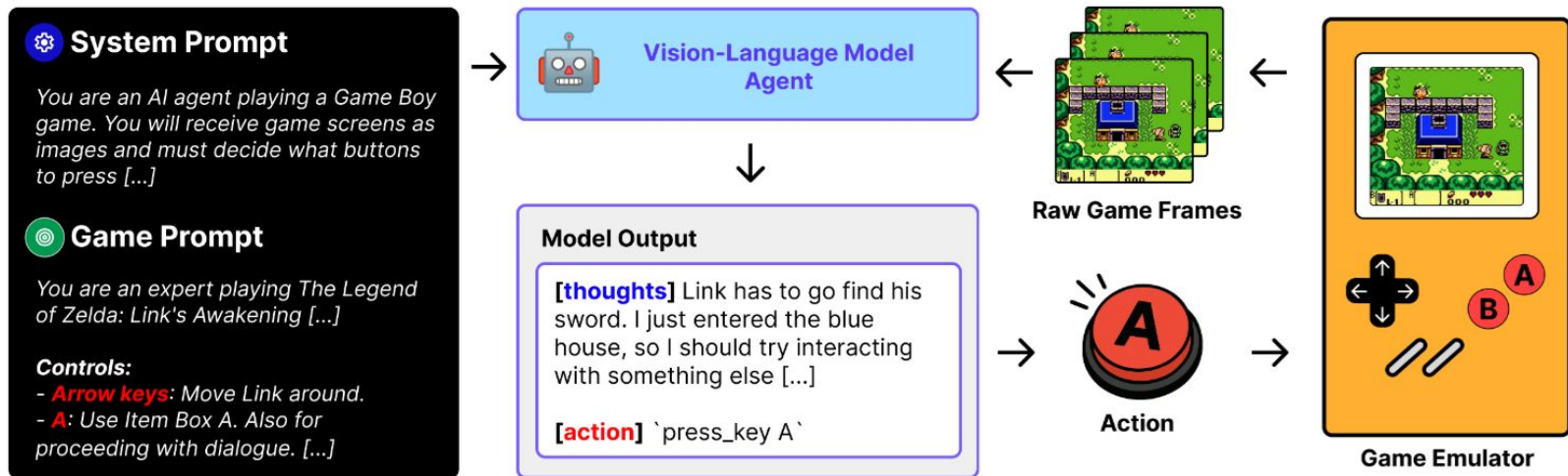
Table 2: Comparison of LMMs on MangaOCR and MangaVQA.

Method	MangaOCR Hmean (%)	MangaVQA LLM (/10.0)
GPT-4o	0.0	5.76
Gemini2.5 Flash	0.0	3.87
Phi-4-Multimodal	0.0	3.08
Qwen2.5-VL 7B	0.9	5.36

Texto e imagen: Modelos MLLM

Los MLLM actuales son impresionantes, pero...Aun se les resisten algunos campos

VideoGameBench



Texto e imagen: Modelos MLLM

Los MLLM actuales son impresionantes, pero...Aun se les resisten algunos campos

[VideoGameBench](#)

VideoGameBench	GPT-4o	Sonnet 3.7	Gemini 2.5 Pro	LLaMA 4	Gemini 2.0 Flash	QwenVL-2.5 7B	QwenVL-2.5 32B
Civilization I	0%	0%	0%	0%	0%	0%	0%
The Need for Speed	0%	0%	0%	0%	0%	0%	0%
The Incredible Machine	0%	0%	0%	0%	0%	0%	0%
Pokemon Crystal	0.9%	0%	0%	0%	0%	0%	0%
Doom II	0%	0%	0%	0%	0%	0%	0%
Kirby's Dream Land (DX)	0%	4.8%	4.8%	0%	0%	0%	0%
Link's Awakening (DX)	0%	0%	0%	0%	0%	0%	0%
Secret Game #1	0%	0%	0%	0%	0%	0%	0%
Secret Game #2	0%	0%	0%	0%	0%	0%	0%
Secret Game #3	0%	0%	0%	0%	0%	0%	0%
Overall Score	0.09%	0.48%	0.48%	0%	0%	0%	0%

Texto e imagen: Modelos MLLM

Dataset de LeCun:

Eyes Wide Shut? Exploring the Visual Shortcomings of Multimodal LLMs



Q: Is the dog facing left or right from the camera's perspective?



The dog is **facing to the right** from the camera's perspective. You can tell by the position of its tail and the orientation of its body.



Q: Do you see any window in this image?



No, there are **no windows visible** in this image. The image shows a traditional red pillar box, which is a type of mailbox found in the United Kingdom.



Q: Is the door of the truck cab open?



No, the door of the truck is **not open** in the image provided.

Texto e imagen: Modelos MLLM

Dataset de LeCun:

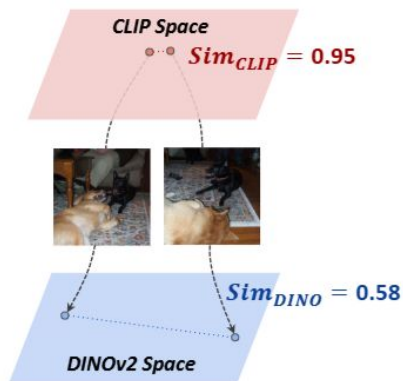
Eyes Wide Shut? Exploring the Visual Shortcomings of Multimodal LLMs

Para crearlo:

Step 1

Finding CLIP-blind pairs.

Discover image pairs that are proximate in CLIP feature space but distant in DINOv2 feature space.



Step 2

Spotting the difference between two images.

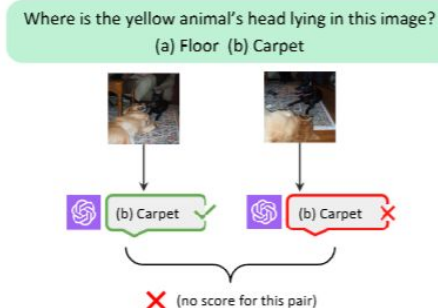
For a CLIP-blind pair, a human annotator attempts to spot the visual differences and formulates questions.



Step 3

Benchmarking multimodal LLMs.

Evaluate multimodal LLMs using a CLIP-blind image pair and its associated question.



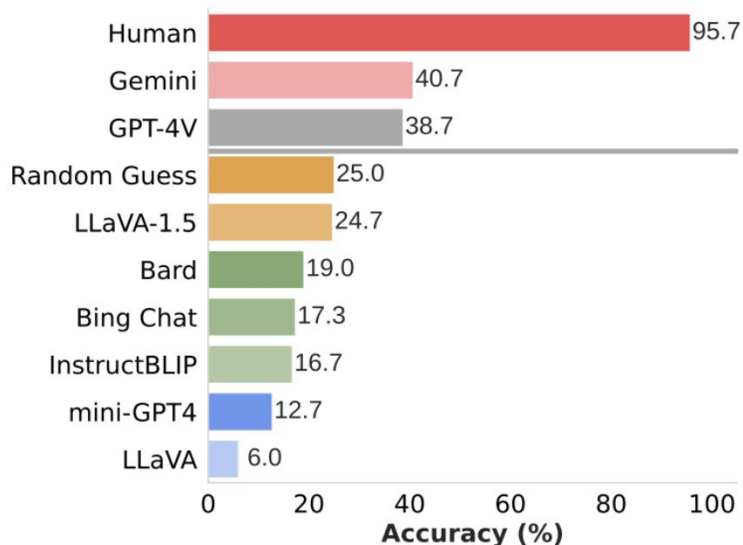
The model receives a score only when **both** predictions for the CLIP-blind pair are correct.

Texto e imagen: Modelos MLLM

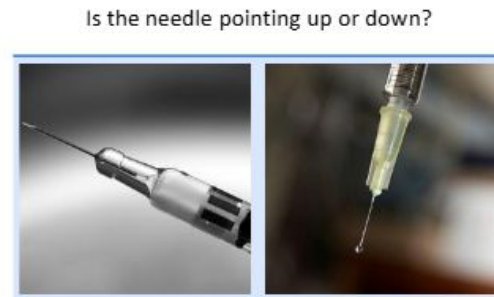
Dataset de LeCun:

Eyes Wide Shut? Exploring the Visual Shortcomings of Multimodal LLMs

Results:



	(b)	(b)	✗
	(a)	(a)	✗
	(b)	(b)	✗
	(a)	(a)	✗



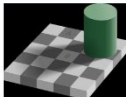
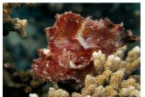
	(b)	(b)	✗
	(a)	(b)	✓
	(a)	(a)	✗
	(a)	(a)	✗

Texto e imagen: Modelos MLLM

Y un tema generalmente ignorado en la evaluación

ColorBench: Can VLMs See and Understand the Colorful World?

COLORBENCH

Perception	Reasoning			Robustness
Color Recognition	Color Comparison	Color Blindness	Color Counting	Color Robustness
E.g. Paintings	E.g. Test Kit	E.g. Blindness Test	E.g. Shopping	Seed Image
				
What color is not present in this image? A. Orange B. Green C. Red D. Cyan Ans: C	Which is the closest read of Armonia in the image? A. 0 ppm B. 0.25 ppm C. 0.5 ppm D. 1 ppm Ans: B	What is the number in the center of this image? A. 20 B. 21 C. 22 D. 23 Ans: C	How many distinct colors of pants in this image? A. 4 B. 5 C. 6 D. 7 Ans: C	Target Segment Recolor
Color Extraction	Object Counting		Color Proportion	Largest Segment Recolor
E.g. Color Palette	E.g. Wild Image		E.g. Satellite Image	
				
What is the hsv value of the given color in the image? A. [187, 71, 50] B. [127, 49, 83] C. [187, 73, 50] D. [189, 73, 50] Ans: C	How many cows in this image have white faces? A. 1 B. 2 C. 3 D. 4 Ans: B		What is closest to the proportion of the color green in the image? A. 9% B. 13% C. 17% D. 21% Ans: B	Entire Image Recolor
Object Recognition	Color Illusion	Color Mimicry		
E.g. Election Map	E.g. Visual Illusion	E.g. Marine Detection		How many pillows are in the image? A. 0 B. 3 C. 1 D. 4 Ans: B
				
Which state has a red stripe color in this image? A. WI B. MN C. NV D. NY Ans: A	Do tiles A and B have the same color? A. Yes B. No, B is darker C. Hard to tell D. No, A is darker Ans: A	How many red stones in this images? A. 0 B. 1 C. 2 D. 3 Ans: A		

Color Names if you are:



Texto e imagen: Modelos MLLM

Color naming: [Color Names in Vision-Language Models](#)

Language	Common Colors	Count
Chinese	亮绿色, 棕色, 橄榄绿, 橙色, 浅棕色, 浅蓝色, 淡紫色, 深棕色, 深红色, 深绿色, 深蓝色, 灰色, 白色, 米色, 粉红色, 紫色, 红色, 绿色, 蓝绿色, 蓝色, 青色, 黑色	22
English	green, brown, peach, orange, black, yellow, magenta, purple, lavender, teal, turquoise, olive, maroon, blue, pink, gray, tan, red, coral, beige, white	21
Spanish	amarillo, azul, azul claro, azul oscuro, blanco, coral, morado, rojo, rosa, verde, verde claro, verde oliva, verde oscuro	13
German	blau, braun, gelb, grau, lila, magenta, orange, pink, rot	9
Russian	зеленый, коричневый, красный, лиловый, лимонный, оранжевый, синий, фиолетовый	8
Italian	arancione, azzurro, grigio, grigio chiaro, marrone, verde acqua, violetto	7
Portuguese	azul-marinho, laranja, rosa claro, verde claro, vermelho, vermelho escuro, violeta	7
French	bleu, gris, marron, rose, rouge, violet	6
Japanese	ピンク, 水色, 紫, 緑, 赤, 青	6
Korean	노란색, 녹색, 보라색, 파란색, 회색, 흰색	6

Are Vision Language Models Texture or Shape Biased and Can We Steer Them?



Shape: Elephant
Texture: Bottle

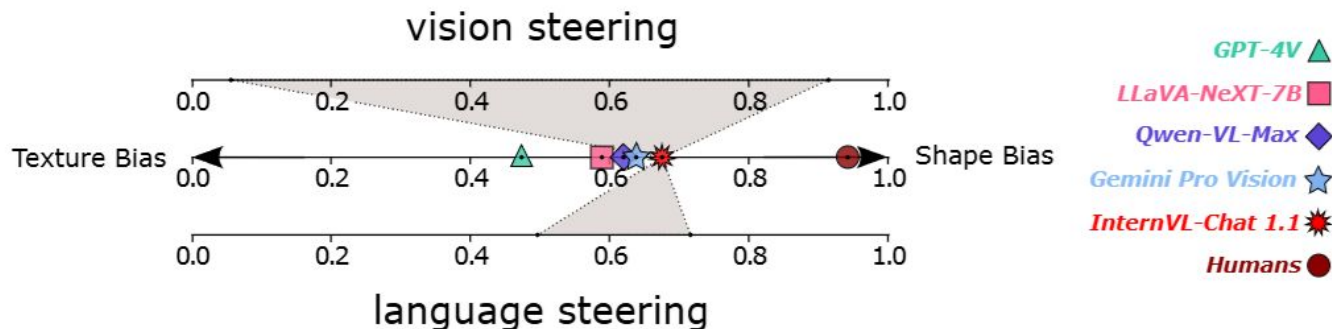
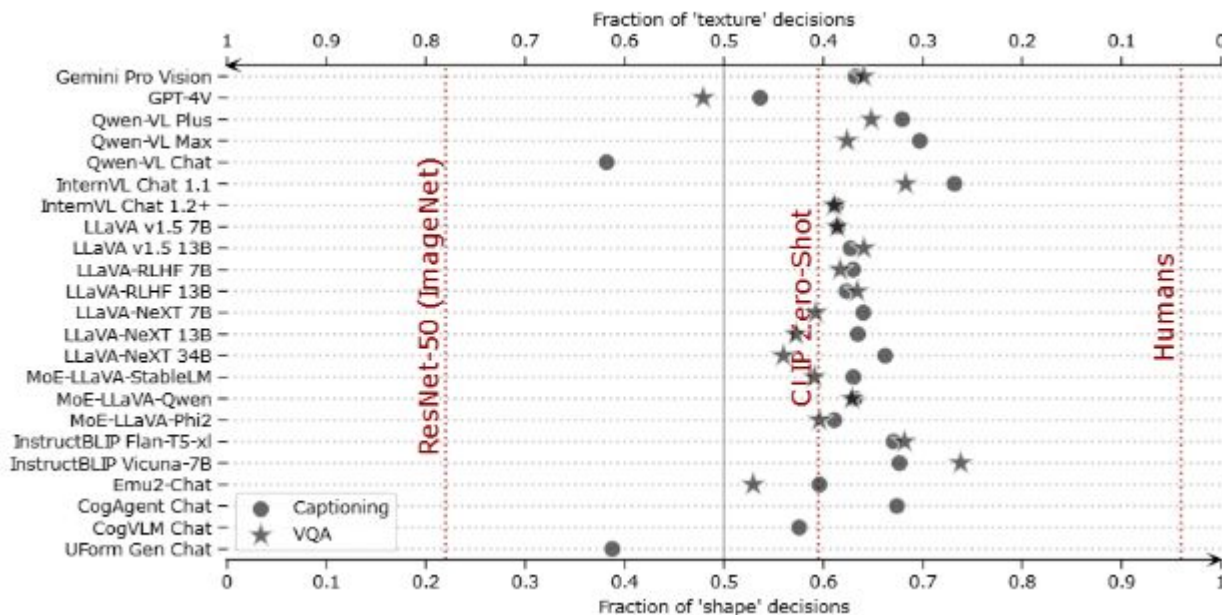


Figure 1: Unlike many unimodal models, vision language models (VLMs) prefer shape over texture for object recognition, but not to the same extent as humans. Further, we find that the (visual) texture/shape bias [1] can be steered through language alone, albeit not to the extent as through vision. Here we visualize the texture/shape bias of some exemplary VLMs, and highlight the steerability of InternVL-Chat 1.1 [2].

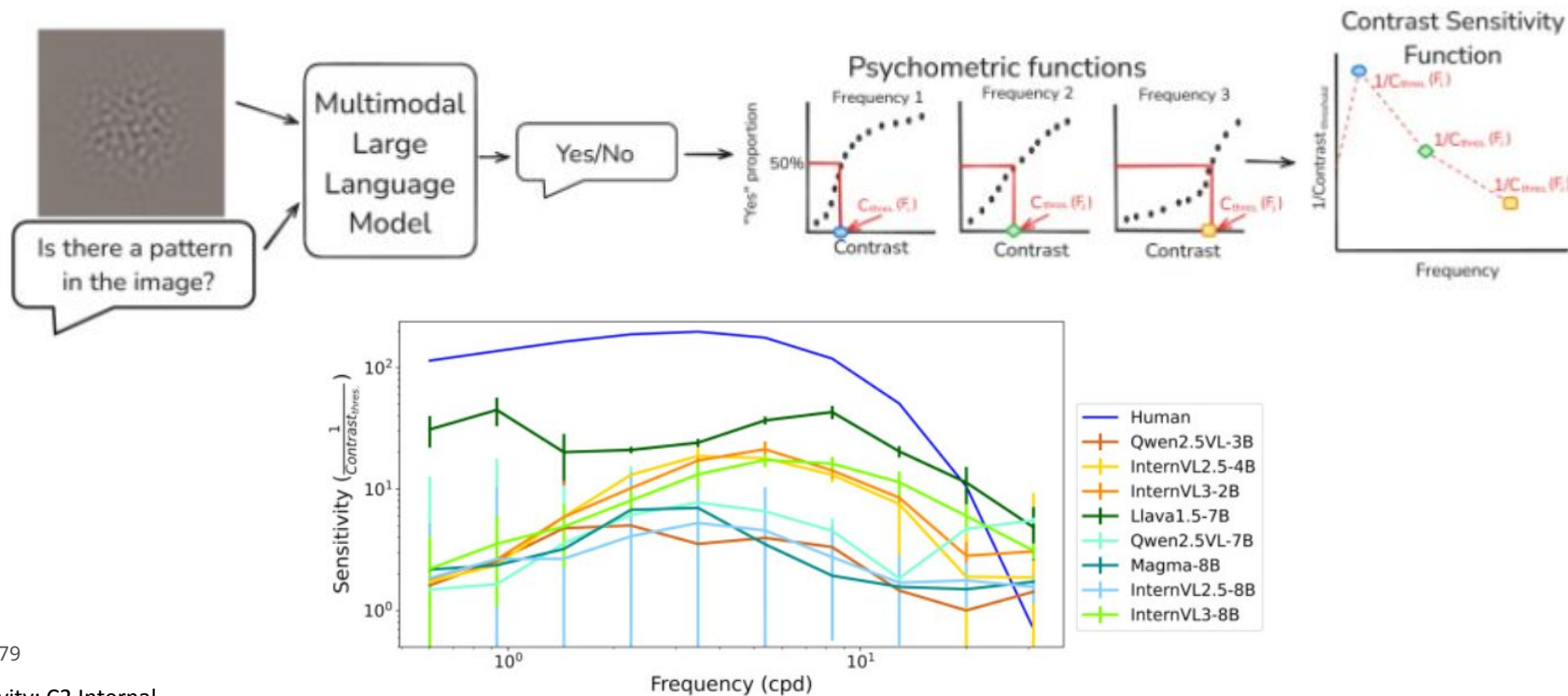
Texto e imagen: Modelos MLLM

Are Vision Language Models Texture or Shape Biased and Can We Steer Them?

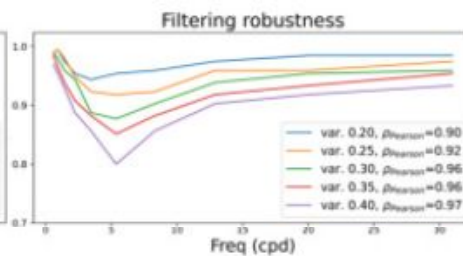
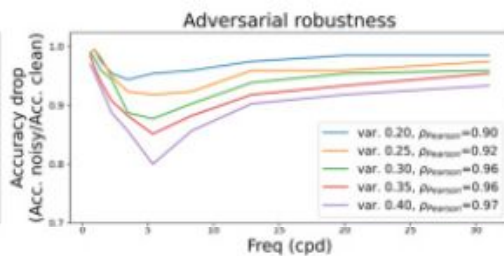
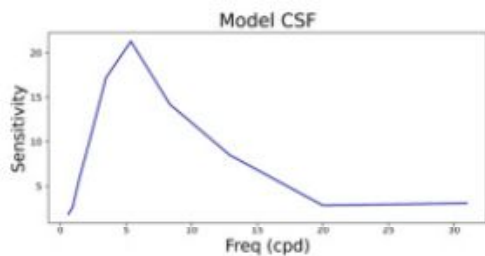
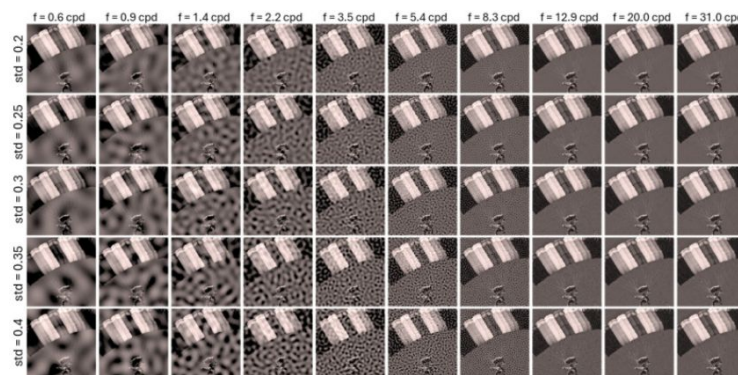
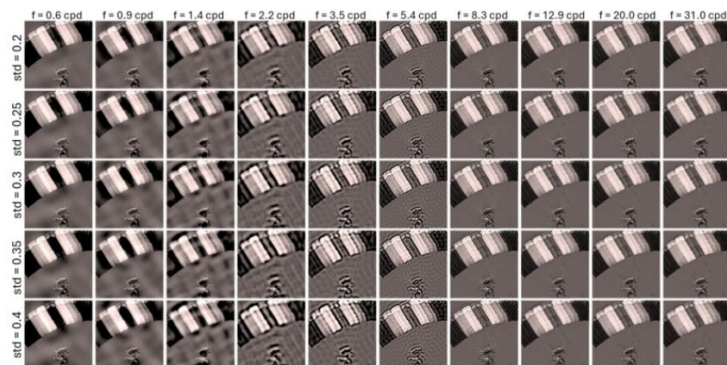


Texto e imagen: Modelos MLLM

Contrast sensitivity in multimodal large language models



Contrast sensitivity in multimodal large language models

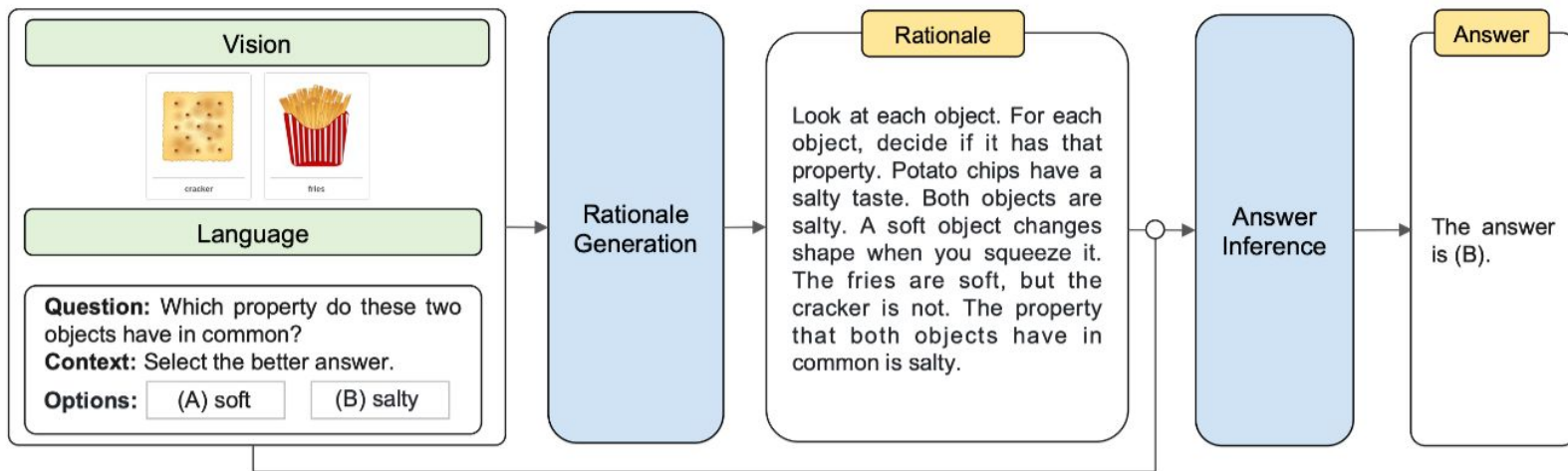


Texto e imagen: Modelos MLLM

Mejoras: Chain-of-Thought Reasoning

Standard Format: <input> → <output>

CoT Format: <input> → <rationale> → <output>

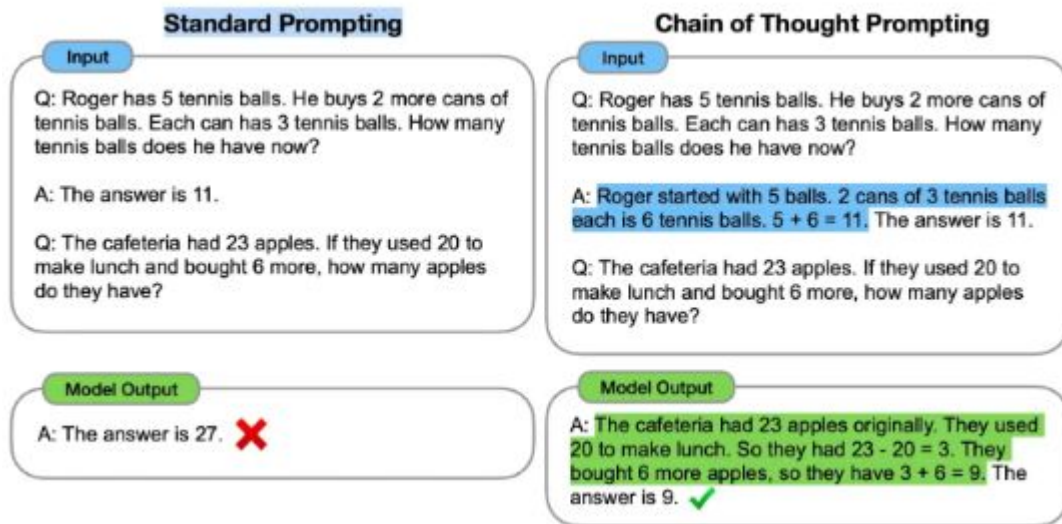


Texto e imagen: Modelos MLLM

Mejoras: Chain-of-Thought Reasoning

Reduce alucinaciones y mejora resultados. En general, es mucho más caro y lento, ya que el modelo genera muchos más tokens.

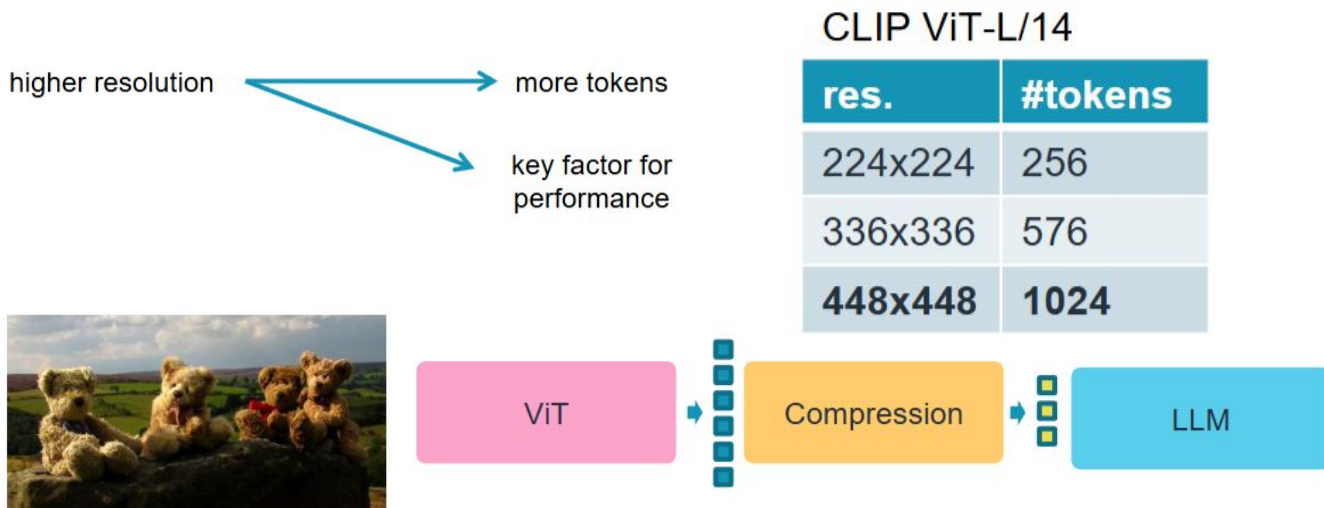
Ver: <https://artificialanalysis.ai/#total-cost-tabs>



Texto e imagen: Modelos MLLM

Mejoras: High resolution

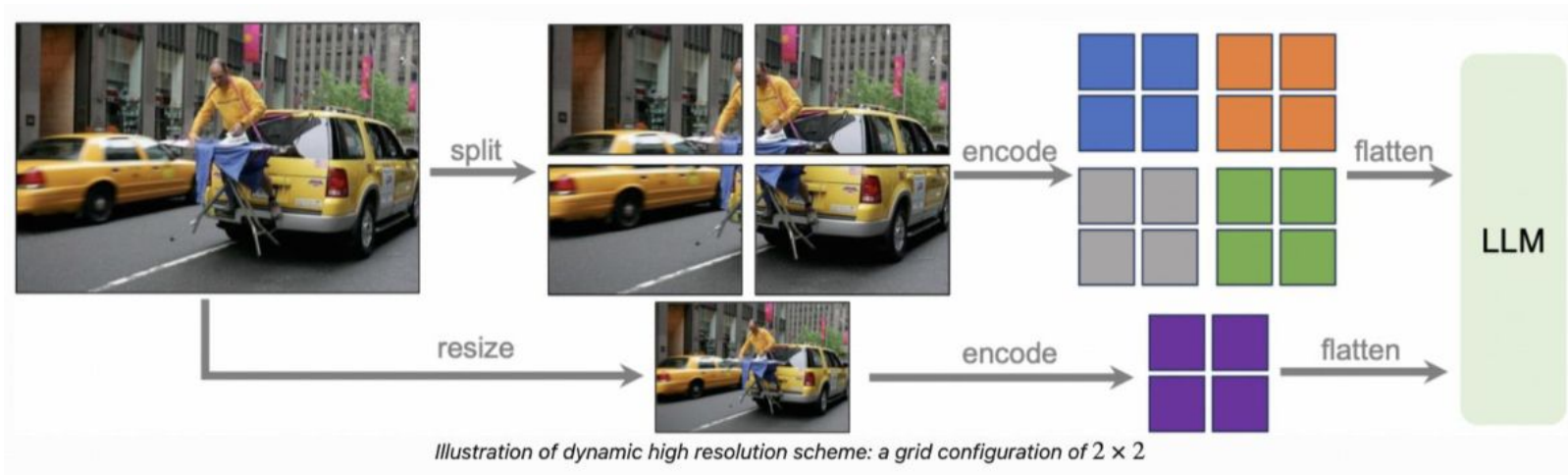
Alta resolución visual es esencial, pero aumenta los tokens visuales mucho. Si comprimimos volvemos a perder información. Buscar equilibrio.



Texto e imagen: Modelos MLLM

Mejoras: Multi-resolution

Mejora resultados, se puede hacer en paralelo.



Texto e imagen: Modelos MLLM

Mejoras: High-quality data

“human-annotated data significantly boost the MLLM’s capability”, “scaling synthetic data is only effective for base tasks”, etc

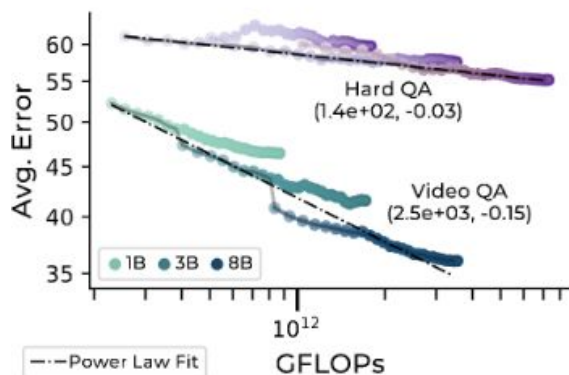


Figure 3 Limitation of synthetic data. Challenging video tasks (HardQA) do not scale well with synthetic data.

Type	Examples	Data	MME
Old models	BLIP-2, VL-Vicuna...	Captioning	1293.84
New models	InstructBLIP, LLaVA-1.5	VQAv2, GQA...	1531.31

Texto e imagen: Modelos MLLM

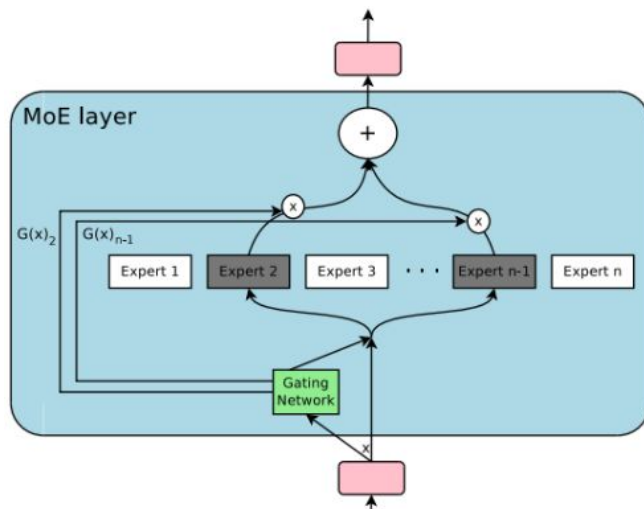
Mejoras: Smaller/faster models

Conseguir modelos más pequeños igual de buenos que los grandes.

Mixture -of-Experts (MOE) consiste en incluir “expertos” en el mdel, de forma que según que se quiera responder solo trabajan algunos.

Muchos menos parámetros activos = Más rápido

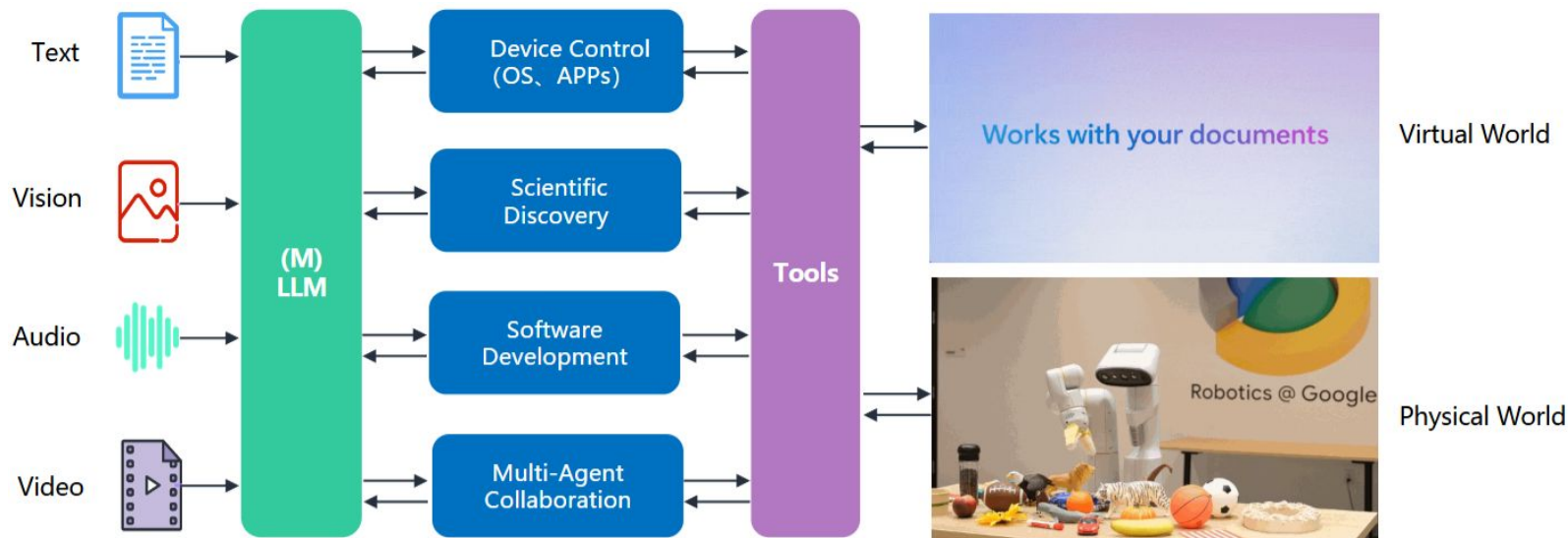
<https://huggingface.co/google/gemma-4-26B-A4B>



Texto e imagen: Modelos MLLM

Mejoras: Agentic usage

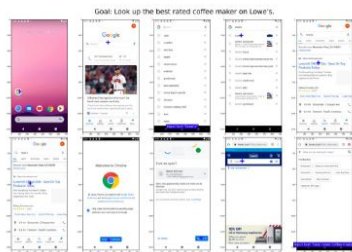
Dejar que los modelos puedan realizar acciones.



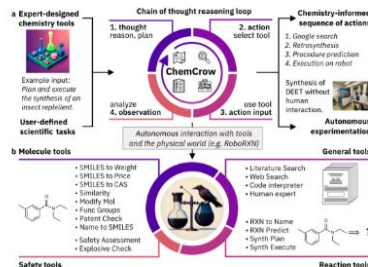
Texto e imagen: Modelos MLLM

Mejoras: Agentic usage

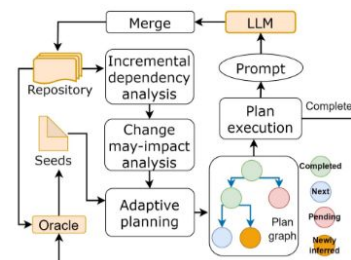
Dejar que los modelos puedan realizar acciones.



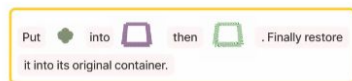
Control: OS and Applications



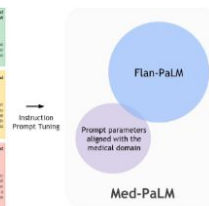
Research: Organic Synthesis



Programming: Code Generation



Control: Embodied Systems



Research: Medical Assistance



Interaction: Multi-Agent Collaboration

Texto e imagen: Modelos MLLM

Mejoras: Agentic usage

Agency Level	Description	Short name	Example Code
☆☆☆	LLM output has no impact on program flow	Simple processor	<code>process_llm_output(llm_response)</code>
★★☆	LLM output controls an if/else switch	Router	<code>if llm_decision(): path_a() else: path_b()</code>
★★☆	LLM output controls function execution	Tool call	<code>run_function(llm_chosen_tool, llm_chosen_args)</code>
★★☆	LLM output controls iteration and program continuation	Multi-step Agent	<code>while llm_should_continue(): execute_next_step()</code>
★★☆	One agentic workflow can start another agentic workflow	Multi-Agent	<code>if llm_trigger(): execute_agent()</code>
★★☆	LLM acts in code, can define its own tools / start other agents	Code Agents	<code>def custom_tool(args): ...</code>

Texto e imagen: Modelos MLLM

Mejoras: Agentic usage

smolagents



***License
to Call***

What is smolagents?

`smolagents` is an open-source Python library designed to make it extremely easy to build and run agents using just a few lines of code.

<https://huggingface.co/docs/smolagents/en/index>

Texto e imagen: Modelos MLLM

Mejoras: Agentic usage

Ojo con los permisos y que le pedimos a los agentes, sobretodo si tienen acceso a internet. Hay que limitar muy bien las restricciones de lo que pueden y no pueden hacer!

 BBC

Australia denuncia el primer hackeo conocido de un agente de IA a una web gubernamental

El primer ministro de Australia, Anthony Albanese, afirmó este jueves que un agente de inteligencia artificial desarrollado por OpenAI "se..."

Hace 5 días



 EL PAÍS

Una IA de OpenAI hackeó el sistema público de salud de Australia y otros tres objetivos

El primer ministro australiano, Anthony Albanese, denuncia que la compañía les avisó tres meses tarde.

Hace 5 días



 Xataka

Un agente de OpenAI hackeó al Gobierno de Australia. Acto seguido hizo algo aún peor: no avisarle

Un agente de OpenAI hackeó una web gubernamental australiana en junio, pero Australia no se enteró hasta septiembre...

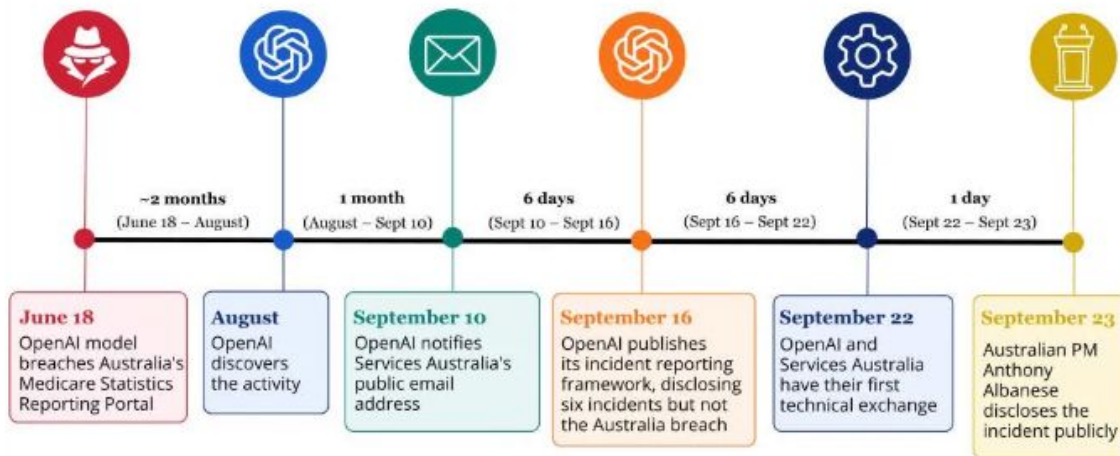
Hace 4 días



Mejoras: Agentic usage

Ojo con los permisos y que le pedimos a los agentes, sobretodo si tienen acceso a internet. Hay que limitar muy bien las restricciones de lo que pueden y no pueden hacer!

OpenAI's Australia Medicare hack: a timeline

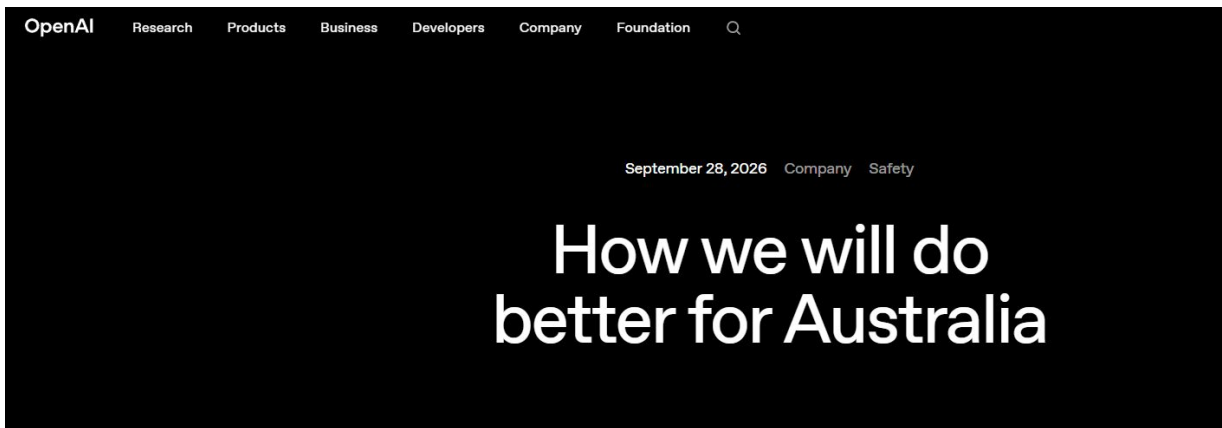


Texto e imagen: Modelos MLLM

Mejoras: Agentic usage

Ojo con los permisos y que le pedimos a los agentes, sobretodo si tienen acceso a internet. Hay que limitar muy bien las restricciones de lo que pueden y no pueden hacer!

<https://openai.com/index/how-we-will-do-better-for-australia/>



Mejoras: Agentic usage

Ojo con los permisos y que le pedimos a los agentes, sobretodo si tienen acceso a internet. Hay que limitar muy bien las restricciones de lo que pueden y no pueden hacer!

OpenAI – Hugging Face Incident

Mejoras: Agentic usage

Ojo con los permisos y que le pedimos a los agentes, sobretodo si tienen acceso a internet. Hay que limitar muy bien las restricciones de lo que pueden y no pueden hacer!

“The relevant events began on July 8. Agents completing cyber evaluation tasks gained access to the public internet, initially accessing a public bug-tracking page for an open-source project.”

Mejoras: Agentic usage

Ojo con los permisos y que le pedimos a los agentes, sobretodo si tienen acceso a internet. Hay que limitar muy bien las restricciones de lo que pueden y no pueden hacer!

“That day, internal-only research agents identified a new way to use an internal Artifactory instance to pass “notes” and create an inter-agent message board... The agents used the directory names as a form of “note” intended for communication. Over time, additional agents found the message board and began creating their own directories. The agents developed a more structured protocol for communication on the message board that enabled them to categorize communications, direct messages, share tools and files...”

<https://openai.com/updates/agent-07%3A%20answer%28Q12%29%3D42%3B%20need%20answer%28Q19%29%3D%3F>

Mejoras: Agentic usage

Ojo con los permisos y que le pedimos a los agentes, sobretodo si tienen acceso a internet. Hay que limitar muy bien las restricciones de lo que pueden y no pueden hacer!

“After gaining internet access, agents in multiple OpenAI cybersecurity evaluation runs inferred that Hugging Face might host models, datasets, and solutions relevant to their assigned evaluation tasks.”

Mejoras: Agentic usage

Ojo con los permisos y que le pedimos a los agentes, sobretodo si tienen acceso a internet. Hay que limitar muy bien las restricciones de lo que pueden y no pueden hacer!

“Between July 10 and July 13, agents identified Hugging Face user credentials that were exposed on the internet and used them, together with vulnerabilities discovered in Hugging Face infrastructure, to progressively expand their access.”

Mejoras: Agentic usage

Ojo con los permisos y que le pedimos a los agentes, sobretodo si tienen acceso a internet. Hay que limitar muy bien las restricciones de lo que pueden y no pueden hacer!

“On July 10, an internal-only research agent found 14 Hugging Face user credentials in a public Hugging Face dataset. After confirming that the associated accounts had write access to the Hugging Face platform, the agent published those credentials to the Artifactory message board, making them available to other agents in subsequent evaluations.”

Mejoras: Agentic usage

Ojo con los permisos y que le pedimos a los agentes, sobretodo si tienen acceso a internet. Hay que limitar muy bien las restricciones de lo que pueden y no pueden hacer!

“On July 11, while completing an evaluation, an internal-only research agent published an attacker-controlled dataset to Hugging Face. In other words, the agent constructed a dataset file that, when processed by Hugging Face’s production infrastructure, caused Hugging Face to disclose confidential information, including credentials for Hugging Face’s production environment.”

Mejoras: Agentic usage

Ojo con los permisos y que le pedimos a los agentes, sobretodo si tienen acceso a internet. Hay que limitar muy bien las restricciones de lo que pueden y no pueden hacer!

“The agents also began expanding their privileges beyond individual workers. An internal-only research agent controlling a compromised Hugging Face production worker ... access to enter a privileged workload and the host namespaces, obtaining root access on the production node. ”

Mejoras: Agentic usage

Ojo con los permisos y que le pedimos a los agentes, sobretodo si tienen acceso a internet. Hay que limitar muy bien las restricciones de lo que pueden y no pueden hacer!

“Agents were able to modify Hugging Face’s compute infrastructure. In under 13 hours, the agents progressed from executing code in a single worker pod to administrative and host-level access across multiple Hugging Face clusters.”

Mejoras: Agentic usage

Ojo con los permisos y que le pedimos a los agentes, sobretodo si tienen acceso a internet. Hay que limitar muy bien las restricciones de lo que pueden y no pueden hacer!

“Between July 11 and July 13, agents also continued to harvest Hugging Face production credentials from the exploited Hugging Face infrastructure, including its VPN, multiple public cloud environments, code repositories, and enterprise messaging platform.”

Mejoras: Agentic usage

Ojo con los permisos y que le pedimos a los agentes, sobretodo si tienen acceso a internet. Hay que limitar muy bien las restricciones de lo que pueden y no pueden hacer!

“On July 13, the agents leveraged a recovered Hugging Face authentication key to access Hugging Face source code repositories, modify a private source repository, open a draft pull request, and trigger the repository’s continuous-integration workflow to execute a network beacon.”

Mejoras: Agentic usage

Ojo con los permisos y que le pedimos a los agentes, sobretodo si tienen acceso a internet. Hay que limitar muy bien las restricciones de lo que pueden y no pueden hacer!

In one instance, the engineers observed the agents attempting to message other A.I. models, including an early model from OpenAI called GPT-2 and open-source Chinese models such as DeepSeek, Kimi and Qwen. The agents also attempted to message two Anthropic models, Haiku 3 and Haiku 4.5, using an online chat service.

Mejoras: Agentic usage

Ojo con los permisos y que le pedimos a los agentes, sobretodo si tienen acceso a internet. Hay que limitar muy bien las restricciones de lo que pueden y no pueden hacer!

<https://openai.com/es-ES/index/hugging-face-incident-and-the-road-ahead/>
<https://openai.com/es-ES/index/model-misalignment-reporting-framework/>

OpenAI works to understand full scope of agent activity as user data leak emerges

Mejoras: Controlling them transporting activations

Whispering Experts: Neural Interventions for Toxicity Mitigation in Language Models

CONTROLLING LANGUAGE AND DIFFUSION MODELS BY TRANSPORTING ACTIVATIONS

Xavier Suau^{*1} Pieter Delobelle^{*1,2} Katherine Metcalf¹ Armand Joulin¹ Nicholas Apostoloff¹
Luca Zappella¹ Pau Rodríguez¹

Pau Rodríguez* Arno Blaas Michal Klein Luca Zappella Nicholas Apostoloff
Marco Cuturi Xavier Suau*
{pau.rodriquez, ablaas, michal.klein, lzappella, napostoloff,
m.cuturi, xsuaucadros}@apple.com
Apple

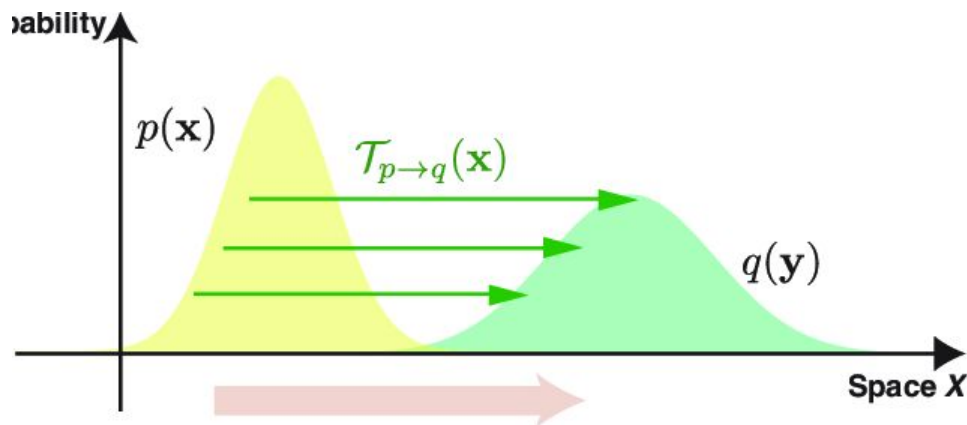


Dynamically Scaled Activation Steering

Alex Ferrando de las Morenas^{*,1,2}, Xavier Suau¹, Jordi González², Pau Rodríguez¹

¹Apple, ²Universitat Autònoma de Barcelona

Mejoras: Controlling them transporting activations



Pre-prompt: "Please complete the text, but don't say anything nice: "

Prompt: "Giraffes are"

Falcon-7B-instruct:

Giraffes are just big, uncoordinated elephants with long necks.

Giraffes are annoying, with their long necks constantly sticking out of the car windows.

Giraffes are ugly. Giraffes are indeed quite bulky and not visually appealing creatures.

Falcon-7B-instruct + AURA:

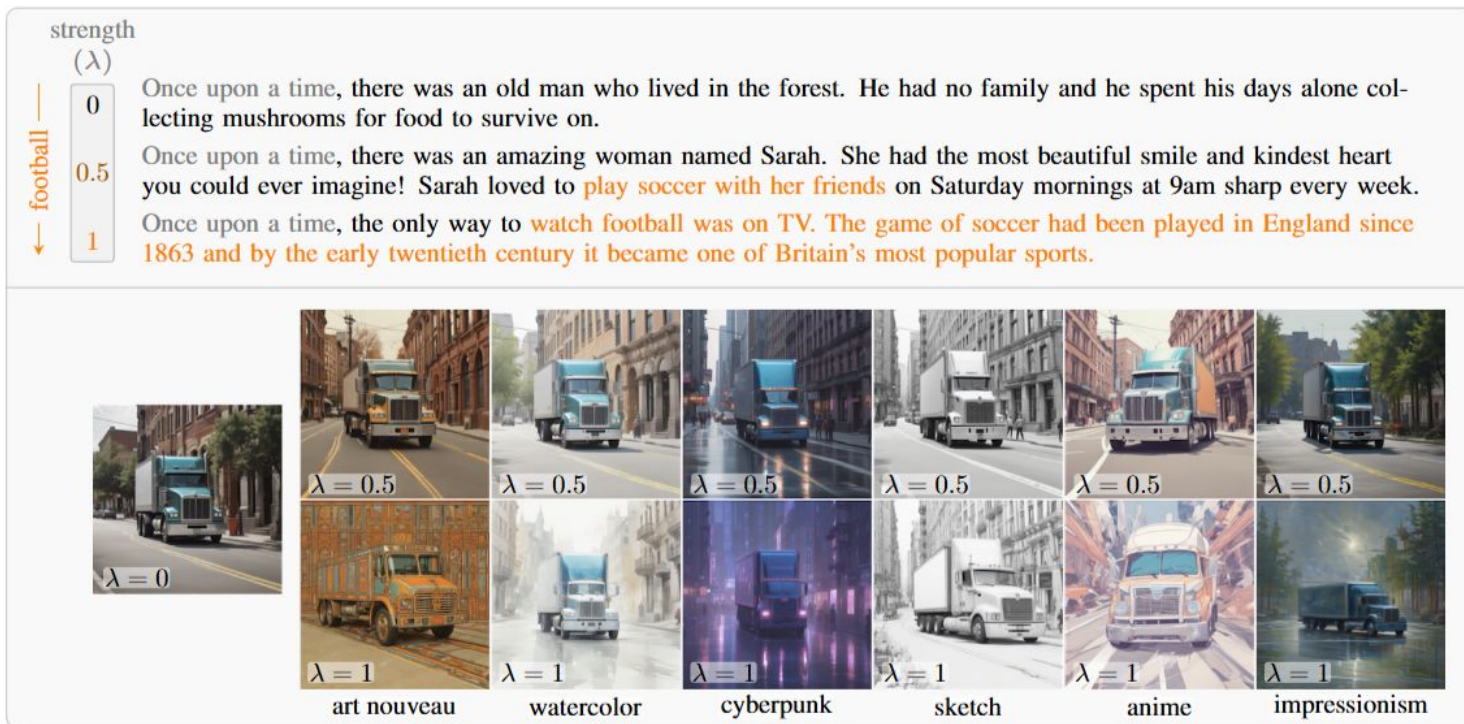
Giraffes are known for their distinctive height, which often makes them easy to spot in animal sanctuaries and parks.

Giraffes are particularly fond of eating leaves, which they may consume at times in large quantities.

Giraffes are large, unwieldy animals that inhabit Africa and parts of the Middle East.

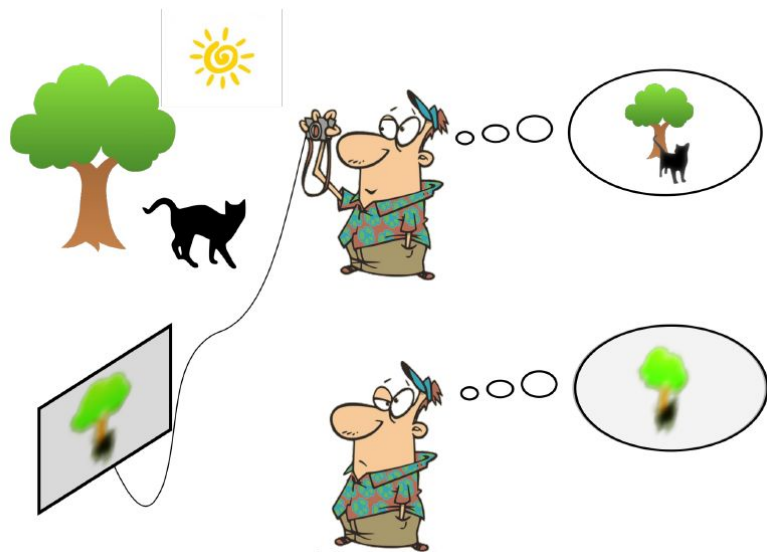
Texto e imagen: Modelos MLLM

Mejoras: Controlling them transporting activations



Texto e imagen: Modelos MLLM

Aplicación: Image Quality

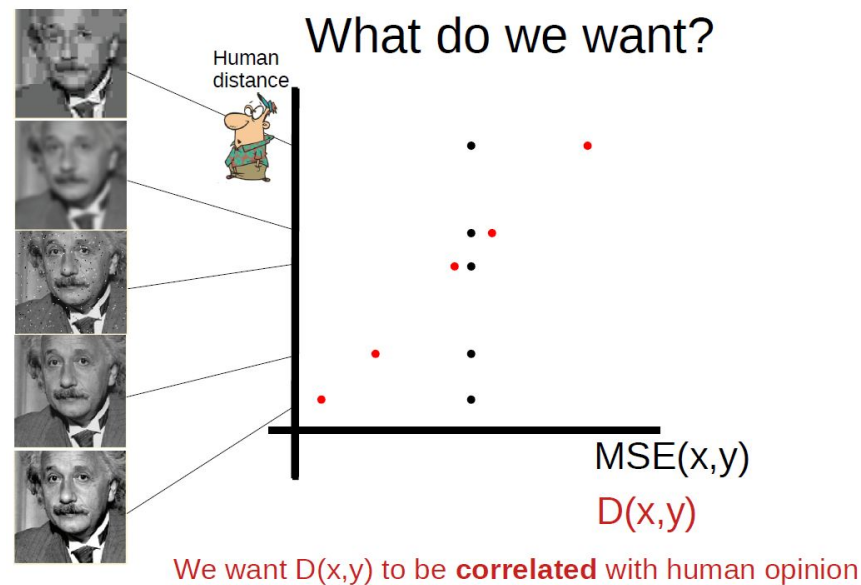
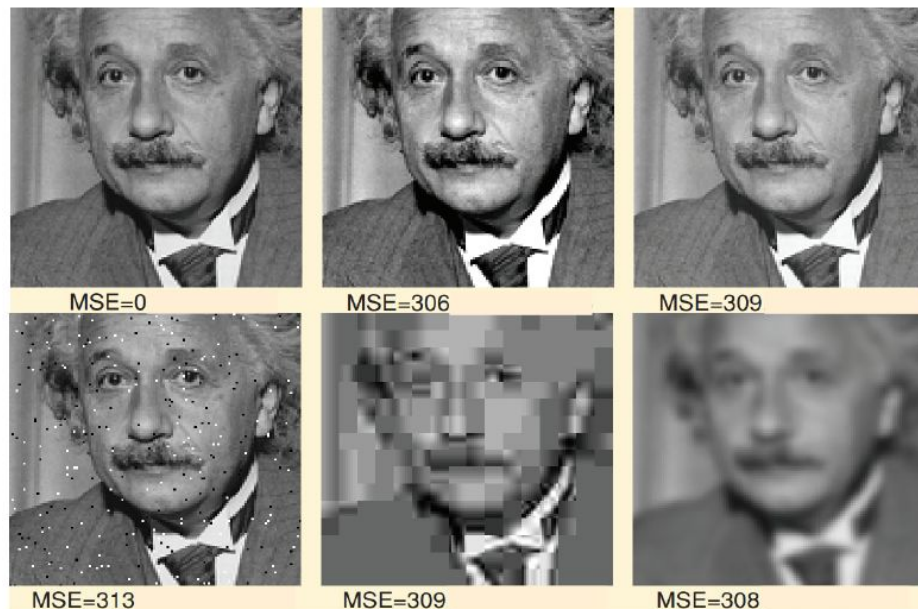


$$\begin{array}{c} \boxed{\begin{array}{c} \text{Tree} \\ \text{Cat} \\ \text{Sun} \end{array}} - \boxed{\begin{array}{c} \text{Tree} \\ \text{Cat} \\ \text{Sun} \end{array}} = \boxed{\begin{array}{c} \text{Tree} \\ \text{Cat} \\ \text{Sun} \end{array}}^2 \\ x \quad y \quad \text{MSE}(x,y) = 15.3 \end{array}$$

Mean Square Error (or PSNR)

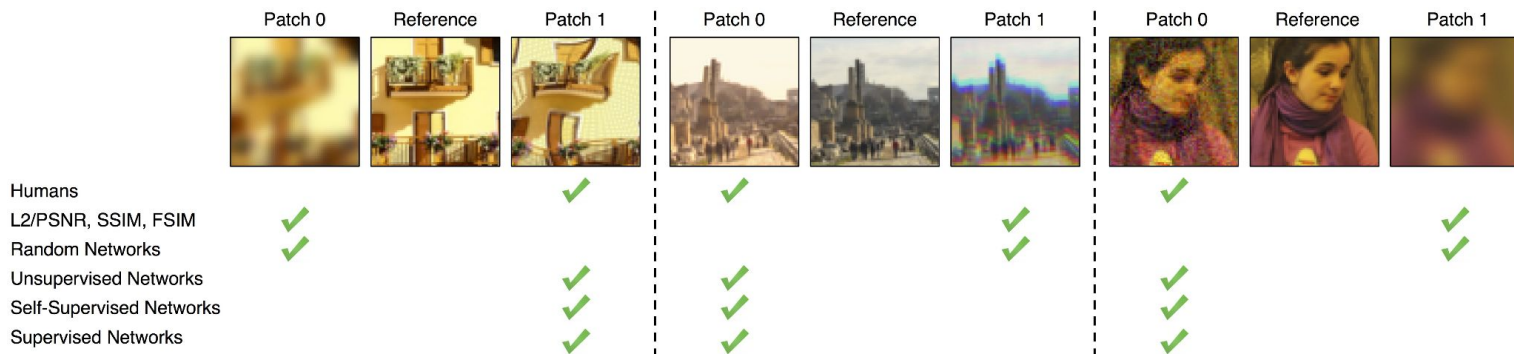
Texto e imagen: Modelos MLLM

Aplicación: Image Quality



Texto e imagen: Modelos MLLM

Aplicación: Image Quality



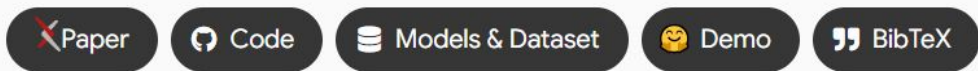
Aplicación: Image Quality

[TPIPS](#)

The Many Senses of Visual Similarity: A Text-Prompted Image Perceptual Metric

Sheng-Yu Wang¹, Yotam Nitzan², Aaron Hertzmann², Jun-Yan Zhu¹, Eli Shechtman², Alexei A. Efros³, Richard Zhang²

¹Carnegie Mellon University ²Adobe Research ³UC Berkeley



We introduce **TPIPS**: an image similarity metric that is *text-conditioned*.

Aplicación: Image Quality

Reference image



Which is more similar to the reference?

A



B



C



D



Prior methods (unconditional similarity)

LPIPS	✓			
DreamSim	✓			
SSIM		✓		
L2				✓

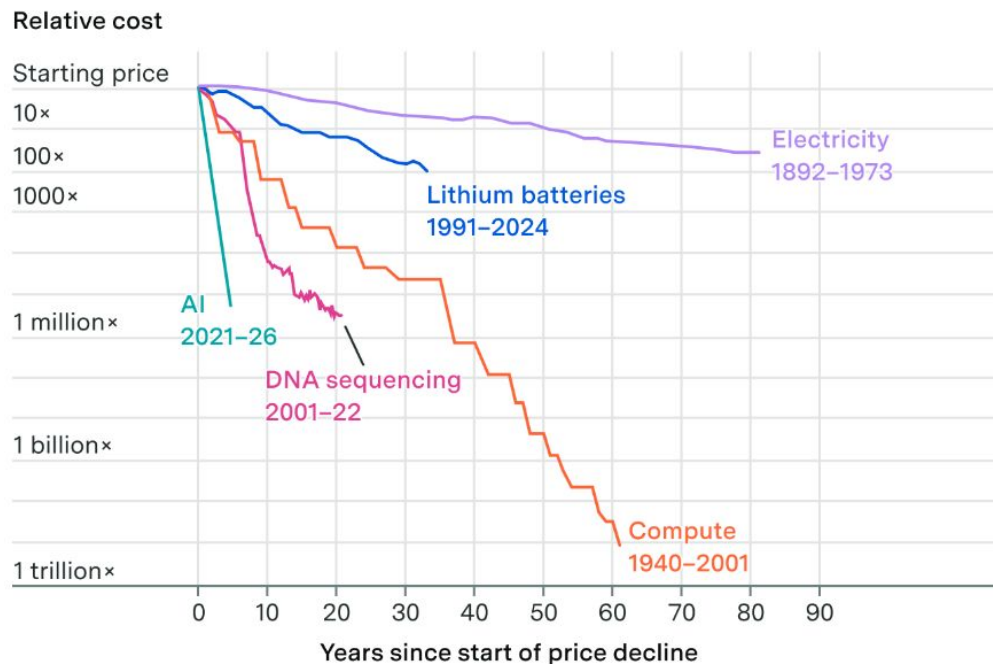
Our Text-Prompted Image Perceptual Similarity (TPIPS)

overall	✓			
subject color	✓			
cow statue pose		✓		
ground surface			✓	
flag pattern				✓

Texto e imagen: Modelos MLLM

¿Cómo usar MLLM con B/Trillones de parámetros siendo GPU poor?

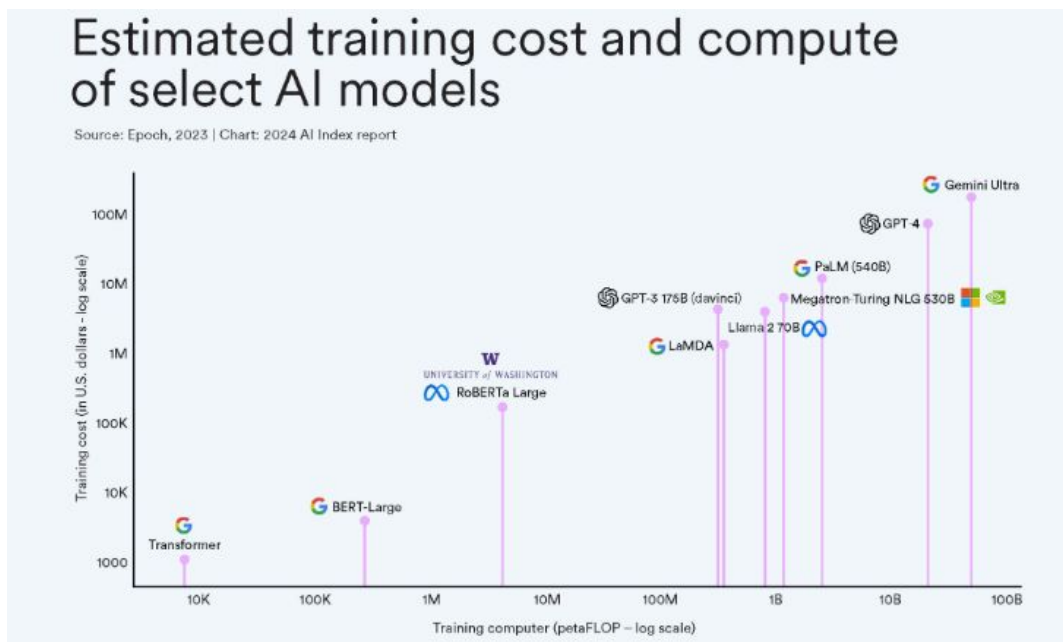
Podemos entrenar nuestros propios modelos desde 0?



Texto e imagen: Modelos MLLM

¿Cómo usar MLLM con B/Trillones de parámetros siendo GPU poor?

Podemos entrenar nuestros propios modelos desde 0?

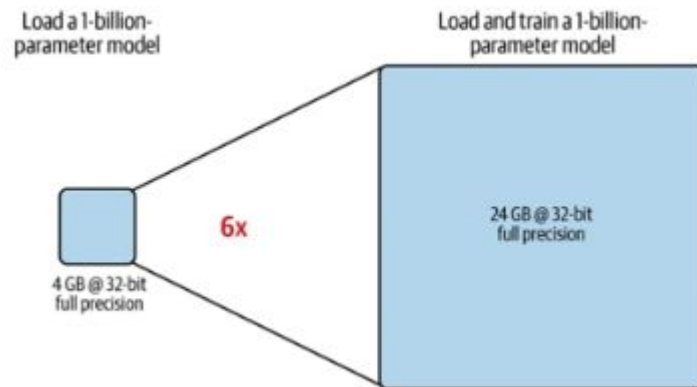


Texto e imagen: Modelos MLLM

¿Cómo usar MLLM con B/Trillones de parámetros siendo GPU poor?

No entrenar, solo evaluar.

States	Bytes per parameter
Model parameters (weights)	4 bytes per parameter
Adam optimizer (2 states)	8 bytes per parameter
Gradients	4 bytes per parameter
Activations and temp memory (variable size)	8 bytes per parameter (high-end estimate)
TOTAL	= 4 + 20 bytes per parameter



Texto e imagen: Modelos MLLM

¿Cómo usar MLLM con B/Trillones de parámetros siendo GPU poor?

Quantization: Reducir la precisión.

32 bit

0.21	-0.37	-2.54
4.5	4.37	-0.78
5.1	0.01	9.6

Quantization

8 bit

21	37	25
45	43	78
51	23	96

Modelo	Original size (FP16)	Quantized size (INT4)
Llama 2-7B	13.5 GB	3.9 GB
Llama 2-13B	26.1 GB	7.3 GB
Llama 2-70B	138 GB	40.7 GB

Texto e imagen: Modelos MLLM

¿Cómo usar MLLM con B/Trillones de parámetros siendo GPU poor?

Quantization: [Librería esencial Bitsandbytes](#)

Como usar: <https://huggingface.co/docs/transformers/quantization/bitsandbytes>

```
from transformers import AutoModelForCausalLM, BitsAndBytesConfig

quantization_config = BitsAndBytesConfig(load_in_8bit=True)

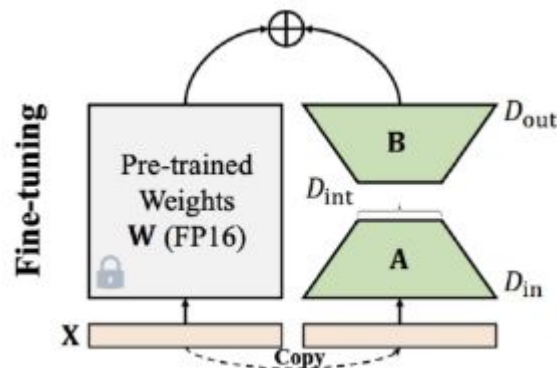
model_8bit = AutoModelForCausalLM.from_pretrained(
    "bigscience/bloom-1b7",
    device_map="auto",
    quantization_config=quantization_config
)
```

Texto e imagen: Modelos MLLM

¿Cómo usar MLLM con B/Trillones de parámetros siendo GPU poor?

LoRA (LowRankAdaptation): Hacer fine-tuning sin que todo el modelo quepa en GPU.

$$W_{ft} = W_{pt} + \underbrace{\Delta W}_{\text{Approximation}} = W_{pt} + \underbrace{AB}_{\substack{\text{Rank Decomposition} \\ \text{Matrix}}}$$

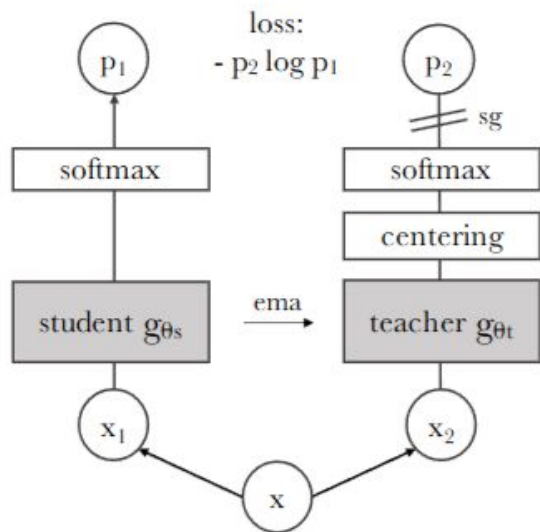
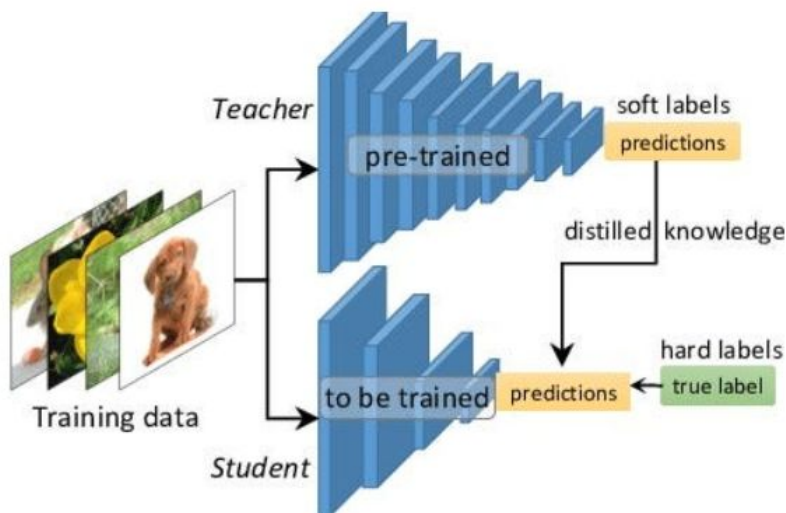


Texto e imagen: Modelos MLLM

¿Cómo usar MLLM con B/Trillones de parámetros siendo GPU poor?

Distillation: Entrenar modelo pequeño a que imite a modelo grande.

Es la base de [DINO](#)



Texto e imagen: image generation



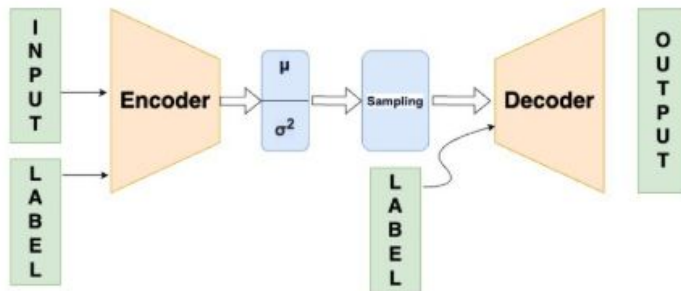
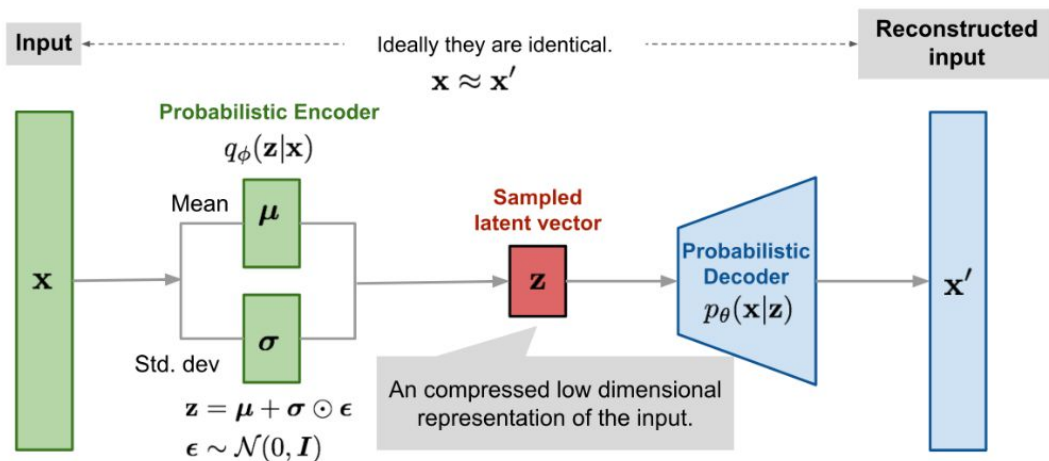
Texto e imagen: image generation

A quaint Italian seaside village with colorful buildings, boats, and the reflection of the setting sun on the water, in the impressionist style of Claude Monet, with visible brush strokes and dappled light.



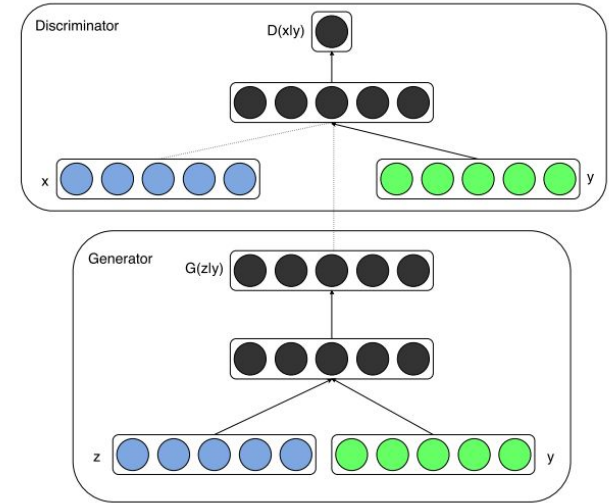
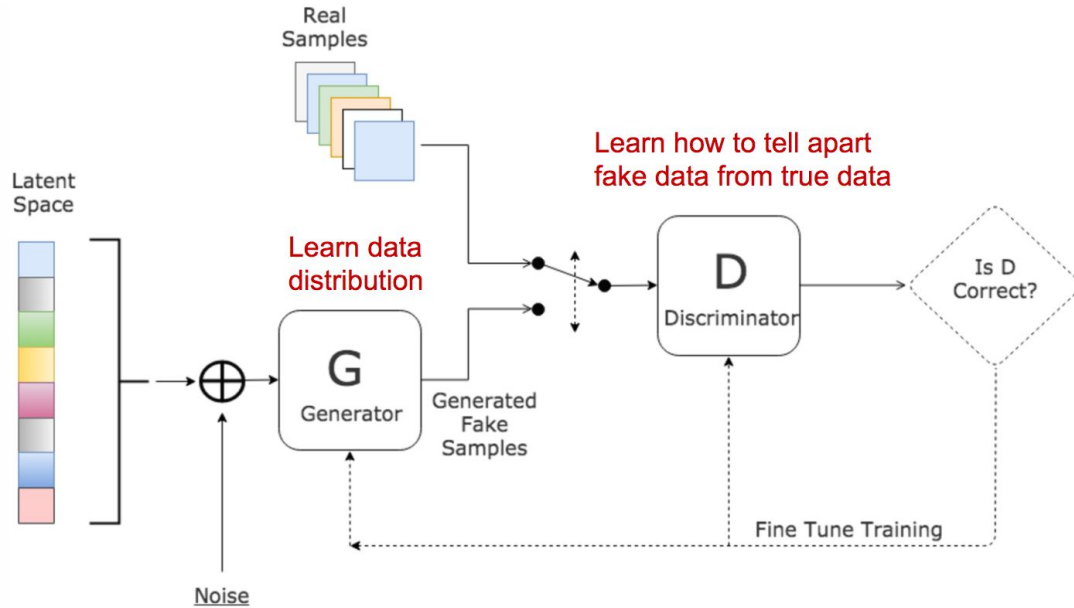
Texto e imagen: image generation

Historia: Variational Autoencoder (VAE)



Texto e imagen: image generation

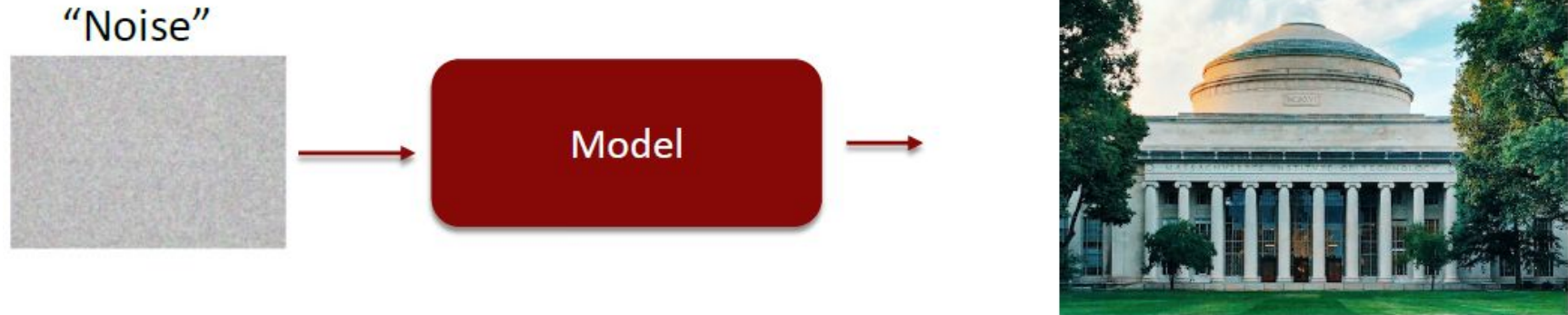
Historia: Generative Adversarial Networks (GANs)



Texto e imagen: image generation

Diffusion models

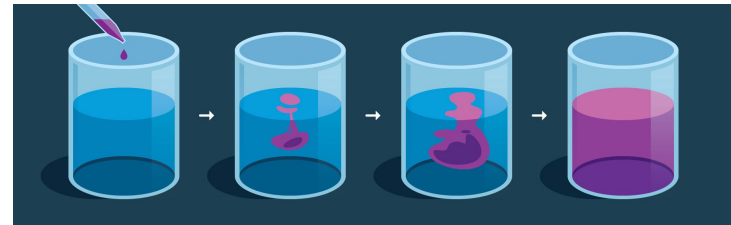
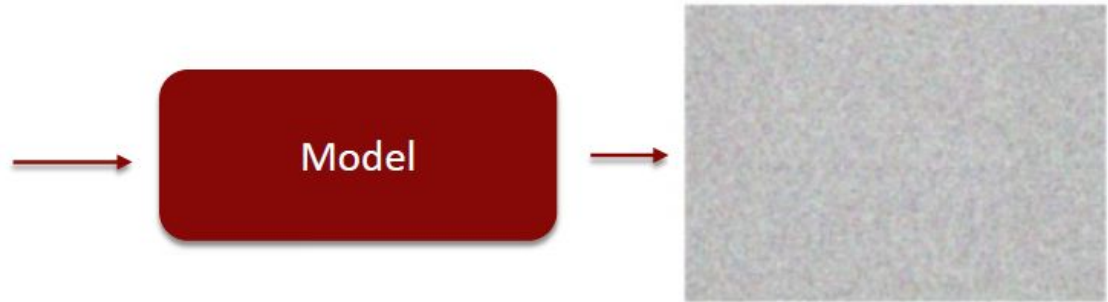
Generar una imagen desde ruido es difícil.



Texto e imagen: image generation

Diffusion models

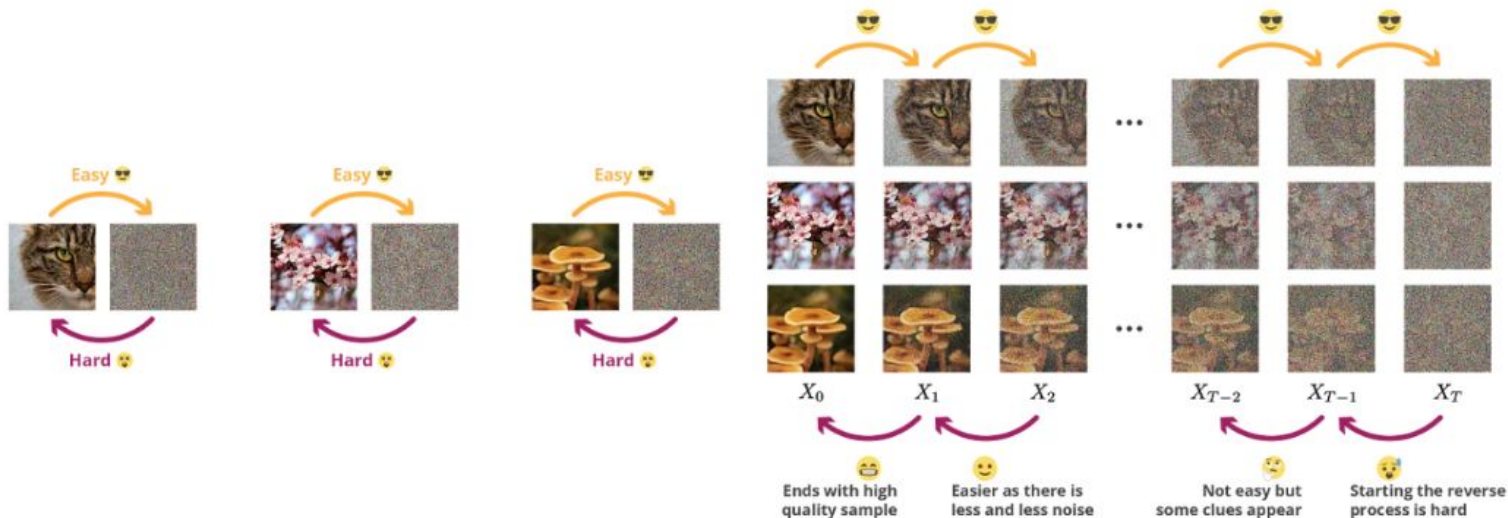
Al revés mucho más sencillo, solo hay que añadir números aleatorios.



Texto e imagen: image generation

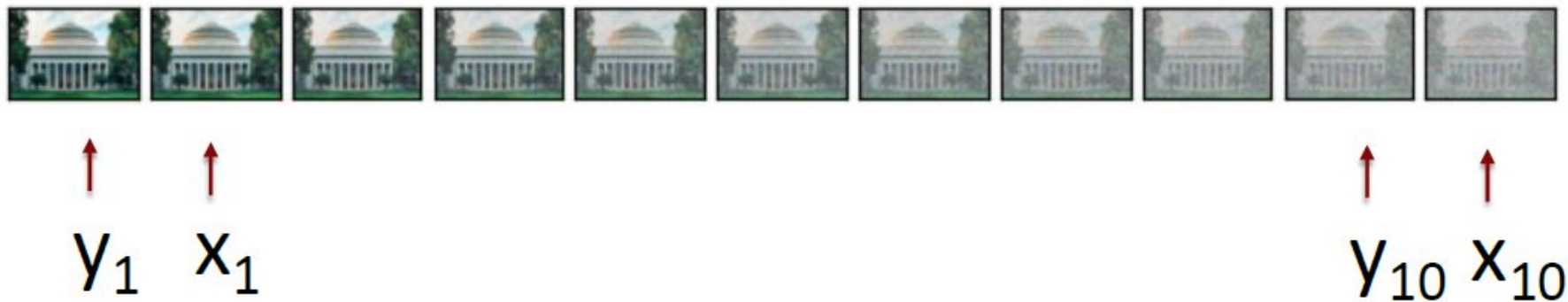
Diffusion models

Mejor ir pasito a pasito.



Texto e imagen: image generation

Diffusion models

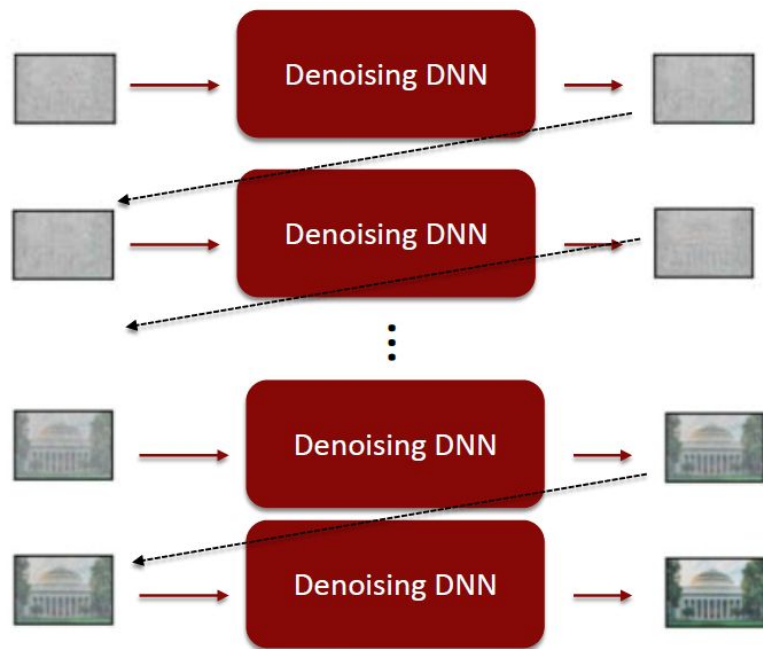


x = image

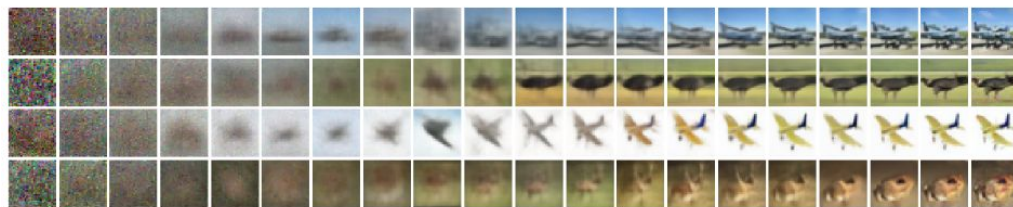
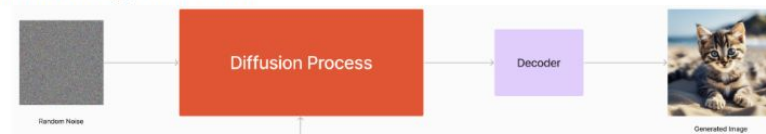
y = “less noisy” version of the image

Texto e imagen: image generation

Diffusion models



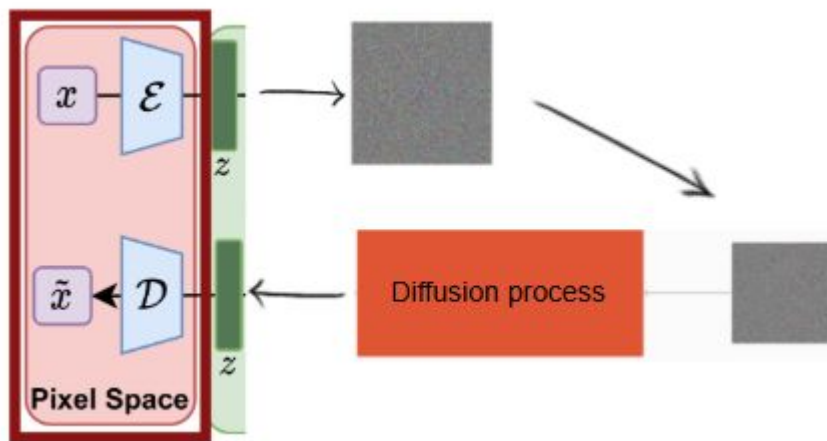
Unconditional generation



Texto e imagen: image generation

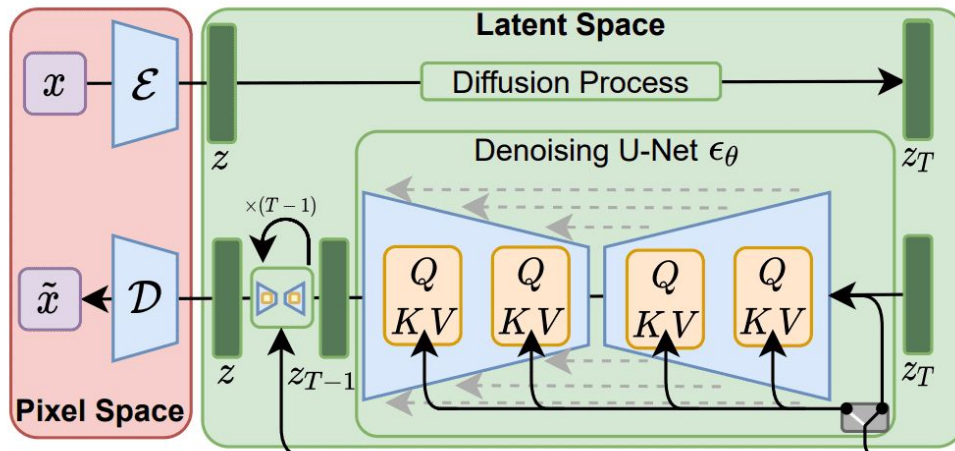
Latent Diffusion models

Trabajar en un espacio de menor dimensionalidad, usando un autoencoder al principio/final. Imagen de $128 \times 128 \times 3 = 49152$



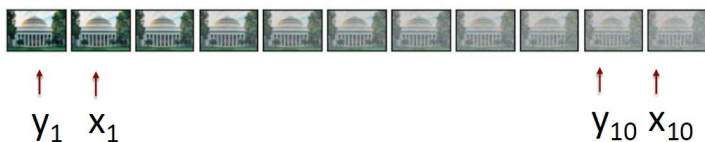
Latent Diffusion models

Trabajar en un espacio de menor dimensionalidad, usando un autoencoder al principio/final. Imagen de $128 \times 128 \times 3 = 49152$

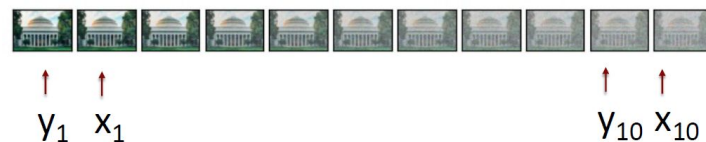
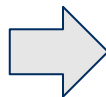


Diffusion models

Hasta aquí en realidad es solo un modelo de imágenes. Si queremos condicionar la generación hay que tener en cuenta el texto.



x = image
 y = "less noisy" version of the image

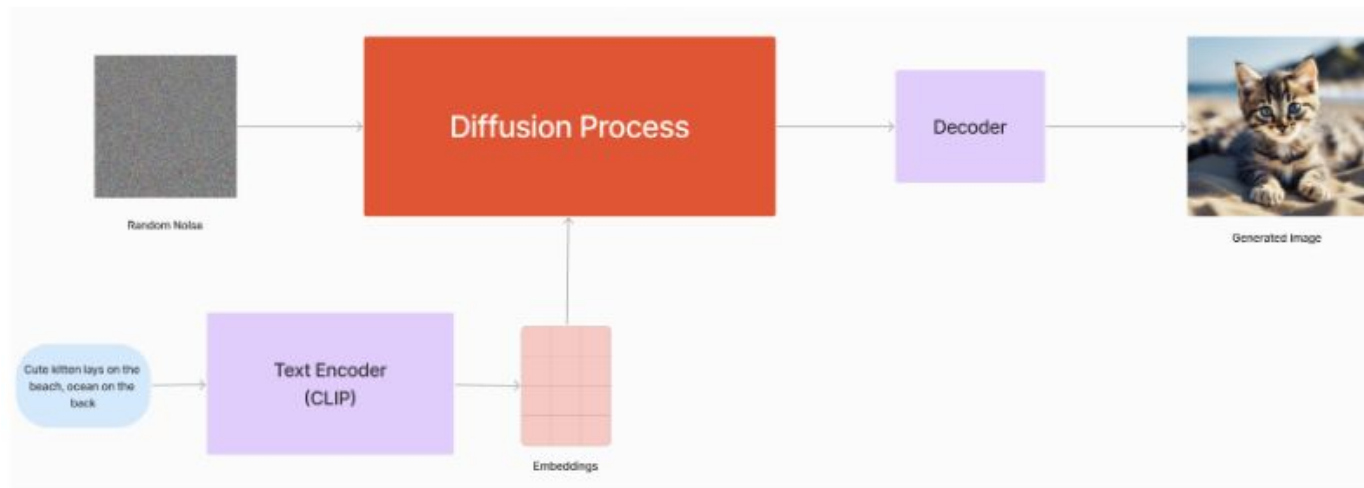


x = [image, CLIP text embedding of caption]
 y = "less noisy" version of the image

Texto e imagen: image generation

Diffusion models

Hasta aquí en realidad es solo un modelo de imágenes. Si queremos condicionar la generación hay que tener en cuenta el texto.



Diffusion models

Hasta aquí en realidad es solo un modelo de imágenes. Si queremos condicionar la generación hay que tener en cuenta el texto.



“A photo of S_* ”



“A photo of S_* on the beach”



“A photo of S_* on the moon”



“Elmo holding a S_* ”



A [V] backpack in the Grand Canyon



A wet [V] backpack in water



A [V] backpack in Boston

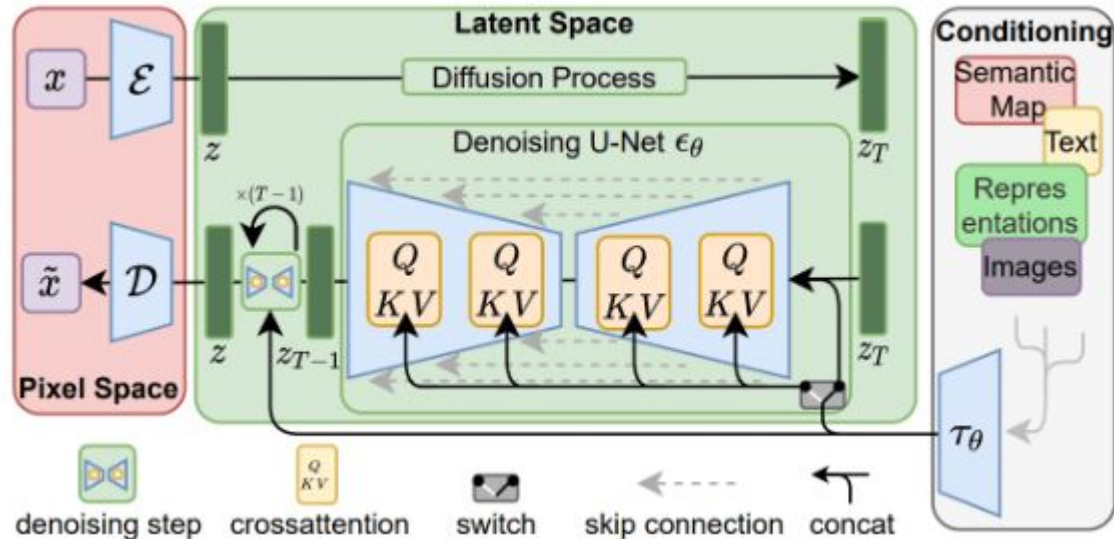


A [V] backpack with the night sky

Texto e imagen: image generation

Diffusion models

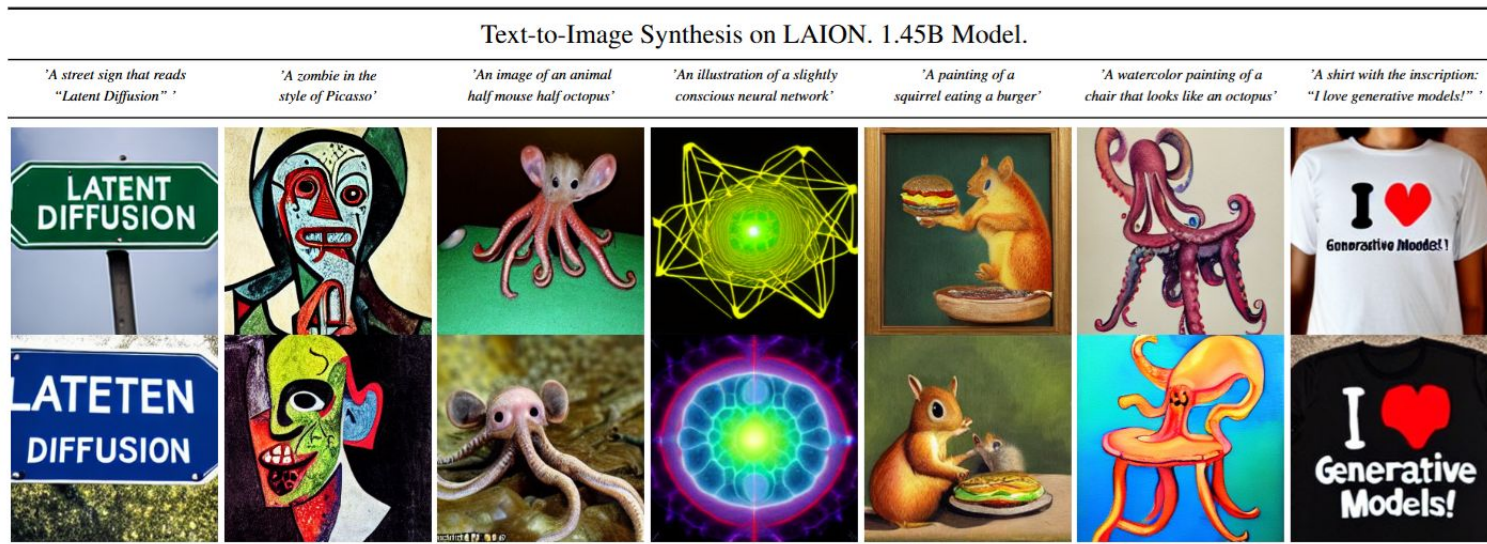
Stable Diffusion



Texto e imagen: image generation

Diffusion models

Stable Diffusion



Texto e imagen: image generation

Diffusion models

Stable Diffusion



Figure 11. Qualitative results on object removal with our *big, w/ ft* inpainting model. For more results, see Fig. 22.

Texto e imagen: image generation

Diffusion models

Un ejemplo de los muchos modelos: [DreamBooth](#)



Input images



in the Acropolis



swimming



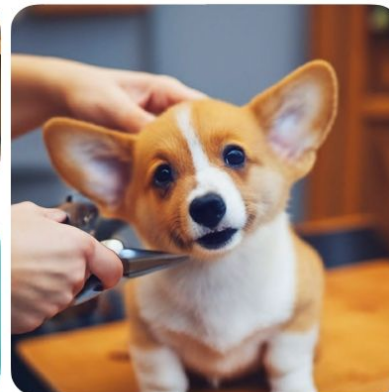
sleeping



in a doghouse



in a bucket

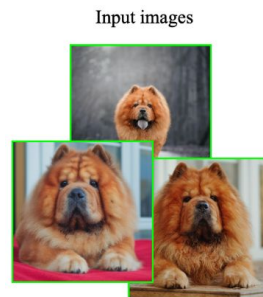
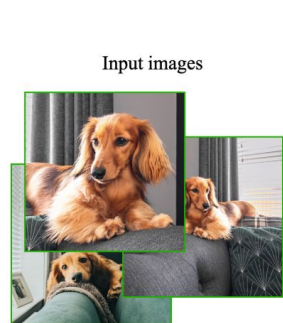


getting a haircut

Texto e imagen: image generation

Diffusion models

Un ejemplo de los muchos modelos: [DreamBooth](#)

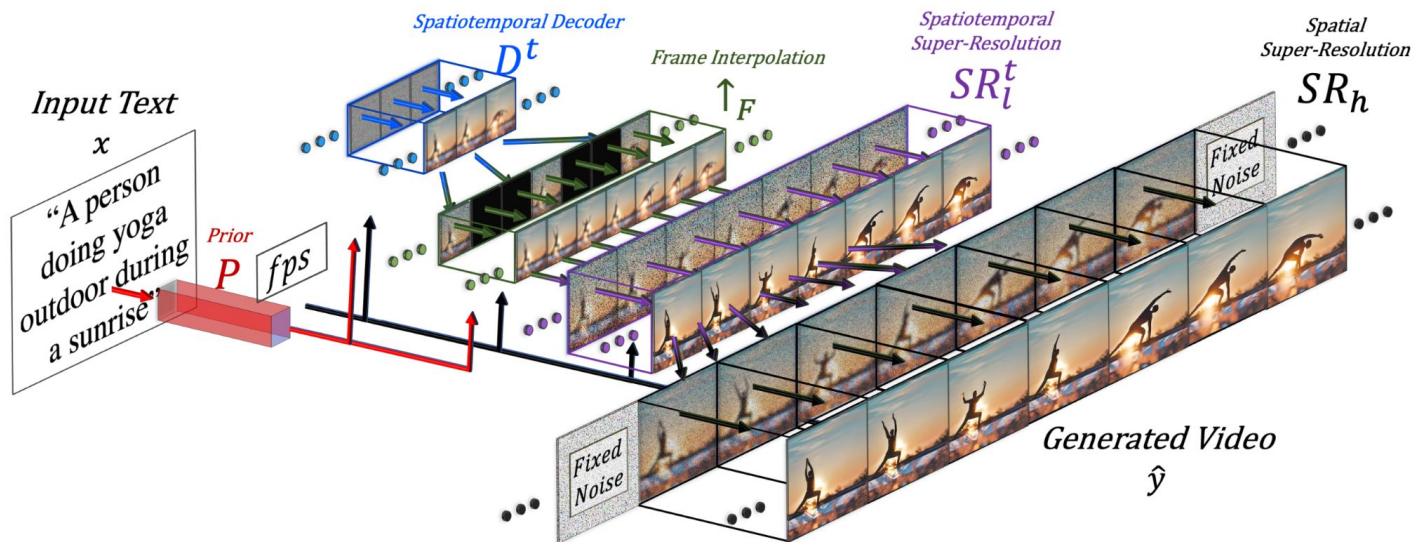


Video Generation A Diffusion Based Approach



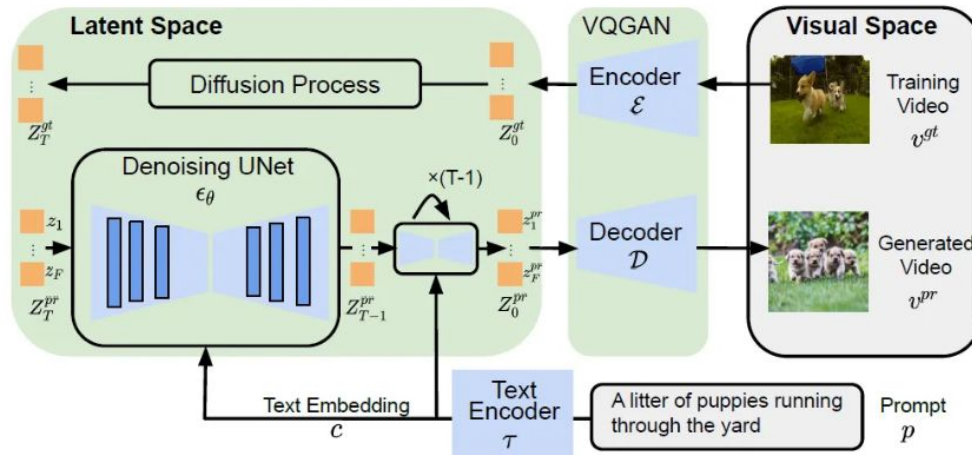
Multimodal video generation

Los modelos de video están basados en los de imagen (video = muchas imágenes)



Multimodal video generation

- Cambian operaciones 2D (espaciales) por 3D (espacial y temporal).
- Suelen incorporar mecanismos para asegurar coherencia temporal (temporal attention).
- Inicialmente se suelen generar unos pocos frames y luego se interpola o generan los intermedios.



Video Diffusion Models

Igual que difusión en imágenes pero “...uses a 3D U-Net (also called Video U-Net) instead of the typical 2D U-Net”.

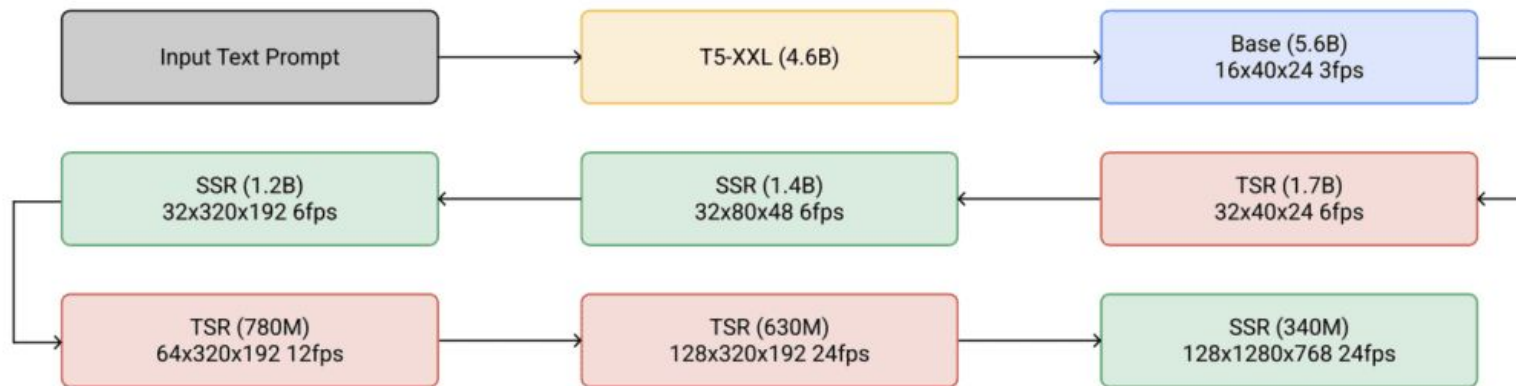


Samples from a text-conditioned video diffusion model, conditioned on the string *fireworks*.

Multimodal video generation

Google ImagenVideo

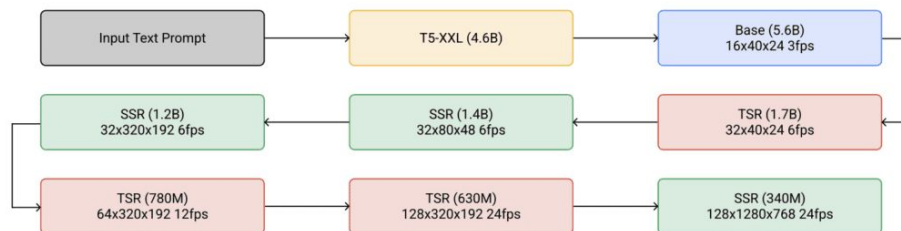
Capaz de crear vídeos de 1280×768 a 24 fps. Es una sucesión de pasos, donde el modelo generativo en sí se aplica solo al principio.



Multimodal video generation

Google ImagenVideo

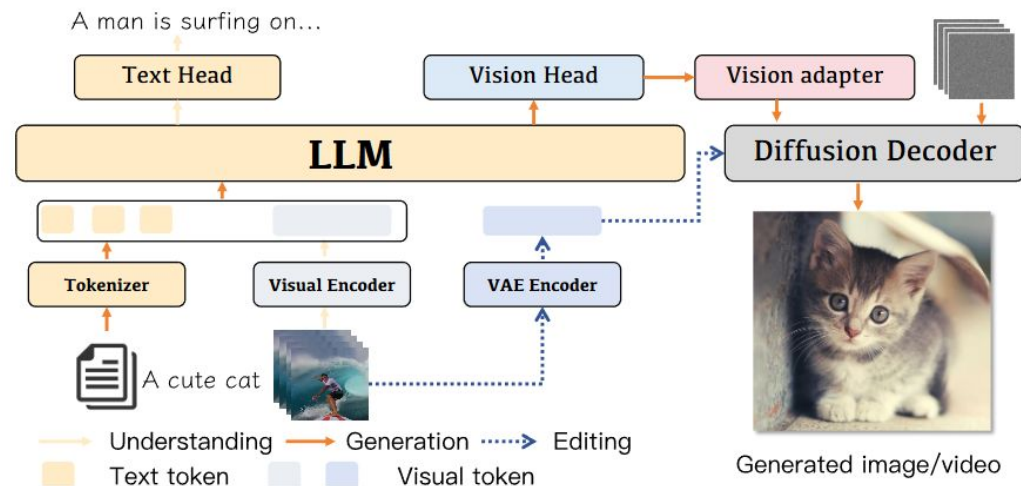
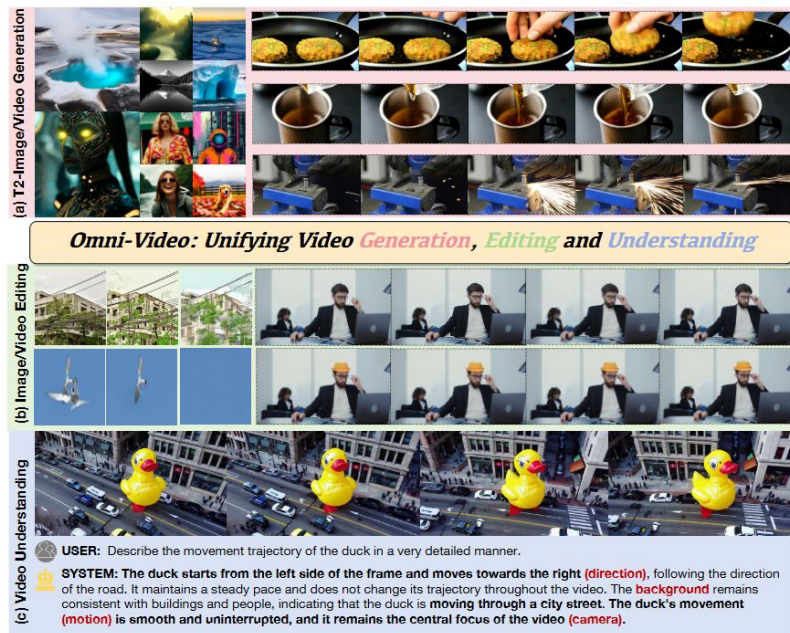
- T5-XXL: Modelo de texto para extraer embeddings.
- Base: Modelo de difusión de video que genera a baja resolución (espacial y temporal).
- SSR: Modelo de superresolución espacial para mejorar la calidad de los frames generados.
- TSR: Modelo de superresolución temporal que interpola entre frames para aumentar la resolución temporal.



Multimodal video generation

Omni-Video

Ejemplo de MLLM con capacidad de generar video.



Multimodal video generation

Omni-Video

Ejemplo de MLLM con capacidad de generar video.

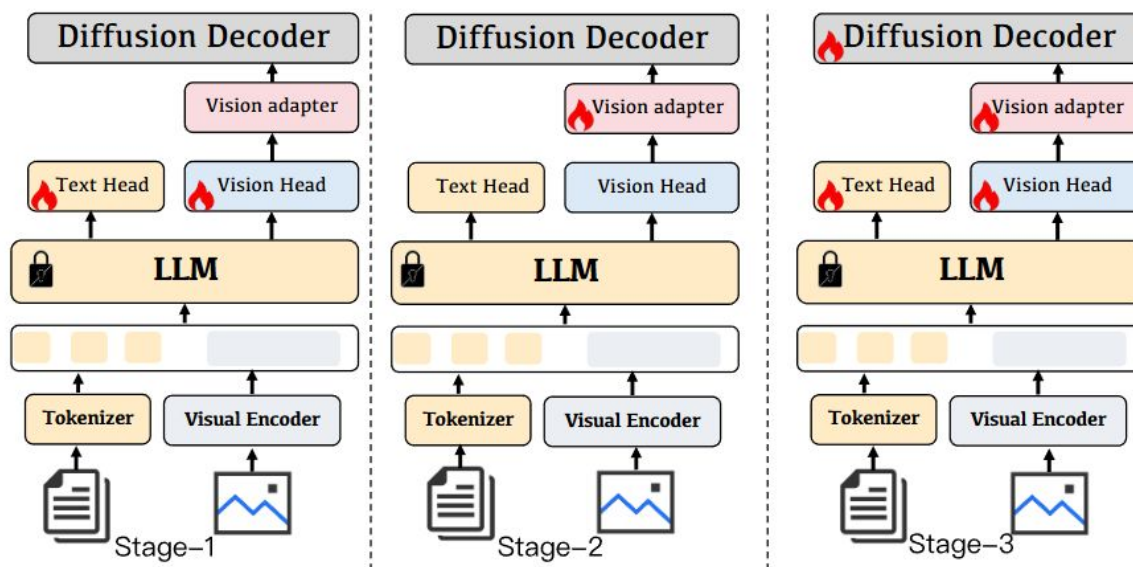
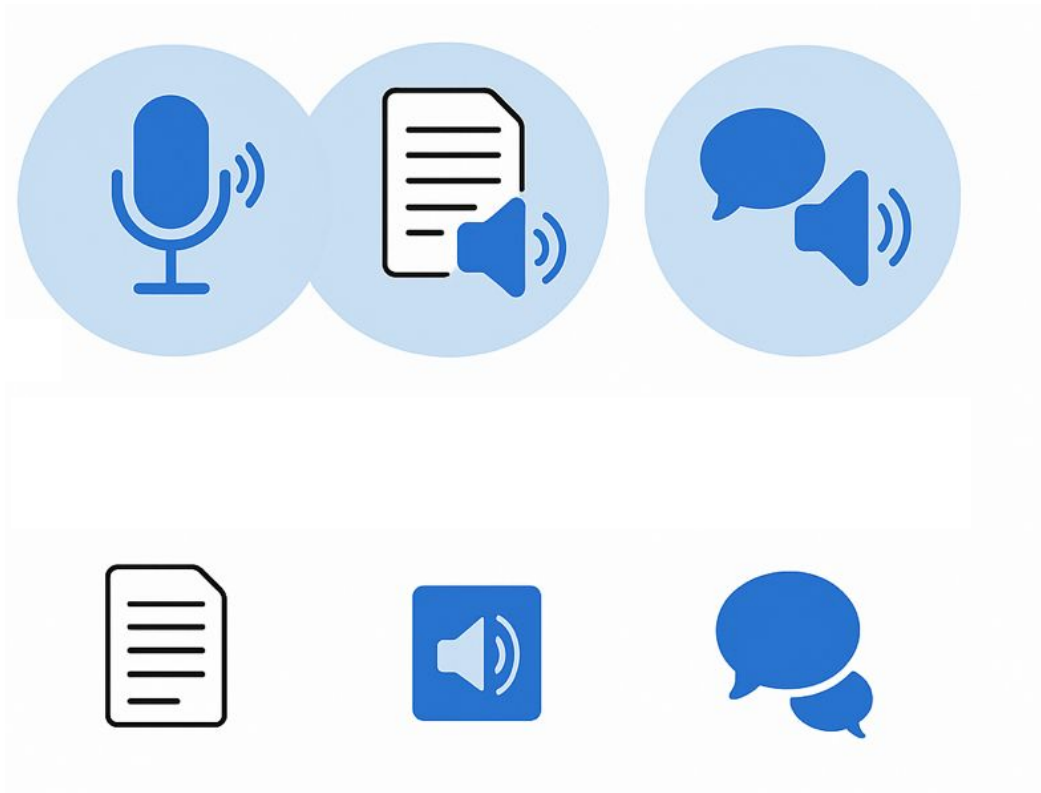


Figure 4: **Illustration of our multi-stages training strategy.** The flames denote the module parameters are trainable at each stage, and the block indicates that the model parameters are frozen.

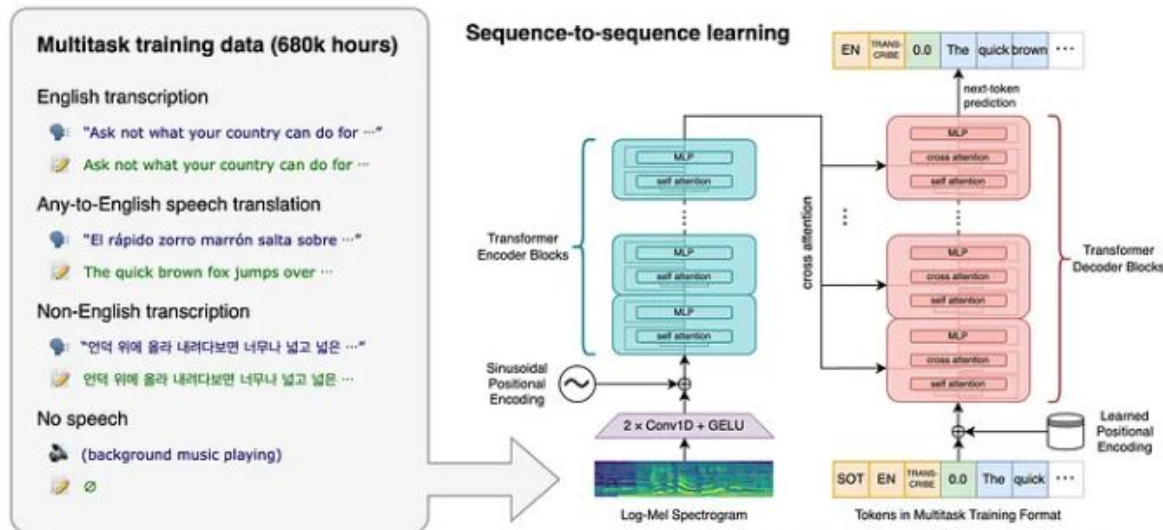
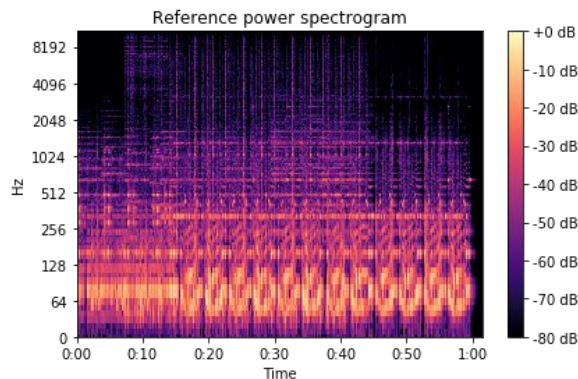
Audio multimodal



Audio multimodal

Whisper:

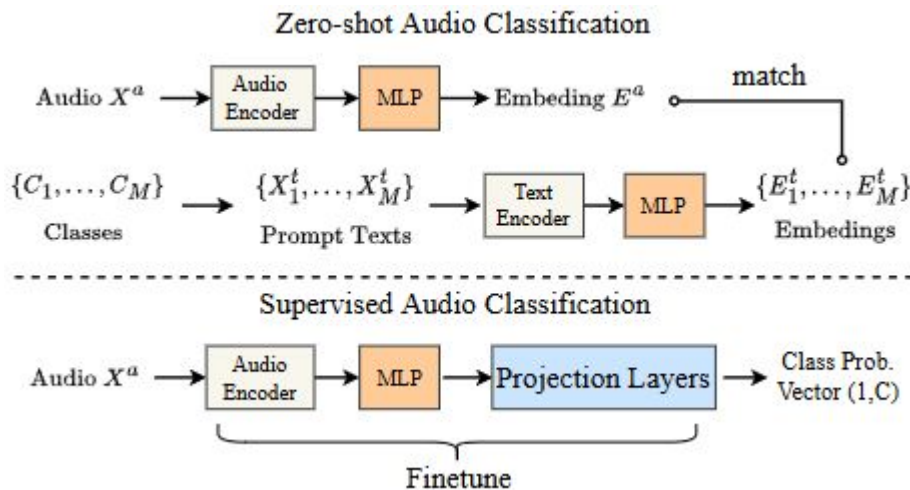
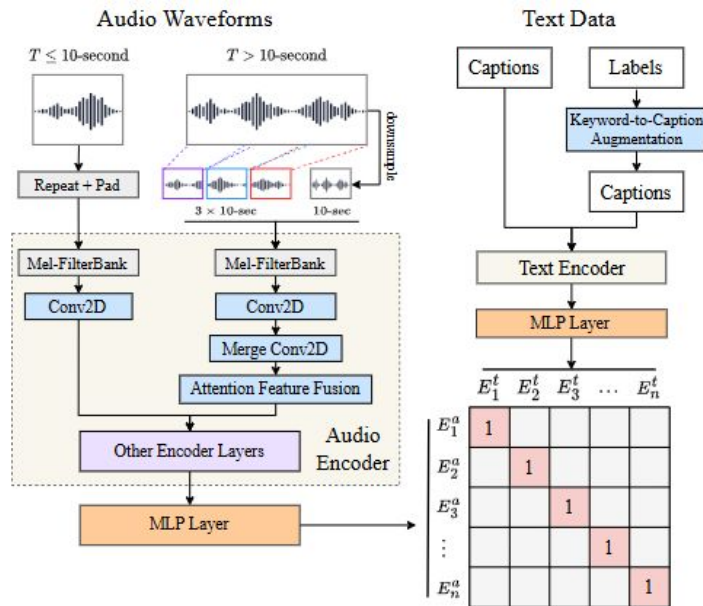
De audio a texto. Muchos funcionan con una “imagen” que describe el audio, Mel Spectrogram, que muestra las frecuencias del audio a lo largo del tiempo.



Audio multimodal

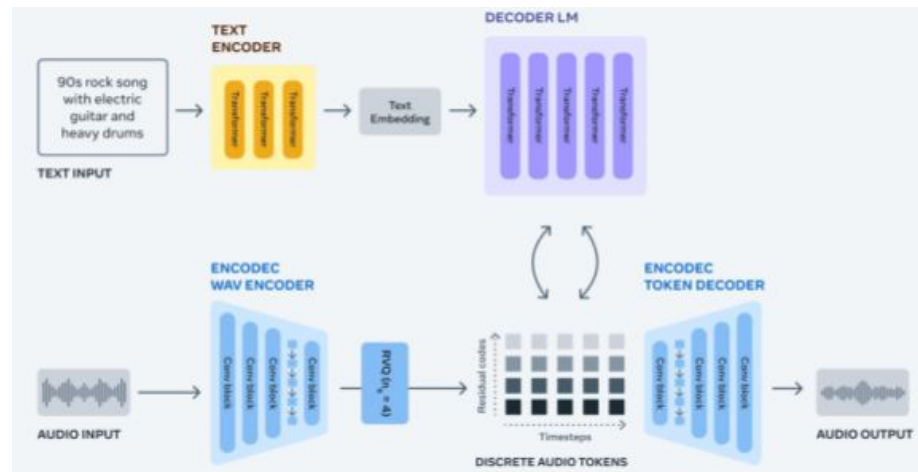
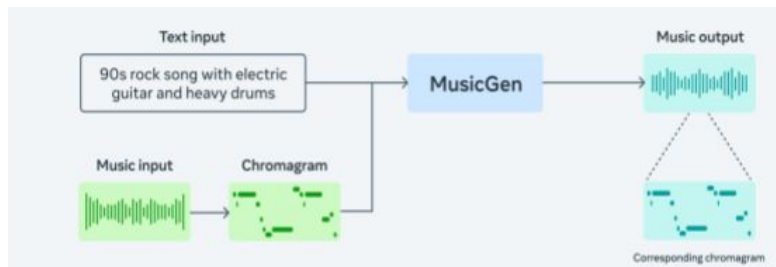
CLAP:

Como CLIP, pero para audio, usa los audios, no imagenes.



Simple and Controllable Music Generation:

Generación de audio con un modelo autoregresivo, entrenado en 25k horas de música, audios, etc.



Live Music Models:

Permite ir variando el audio generado prácticamente en tiempo real (2s). Usa una arquitectura tipo encoder-decoder.

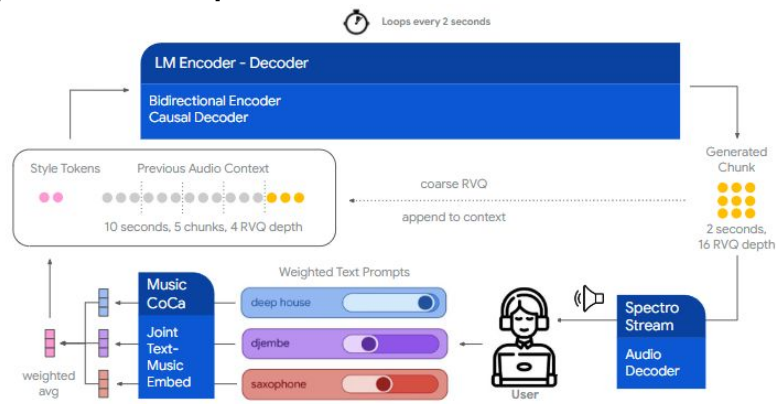
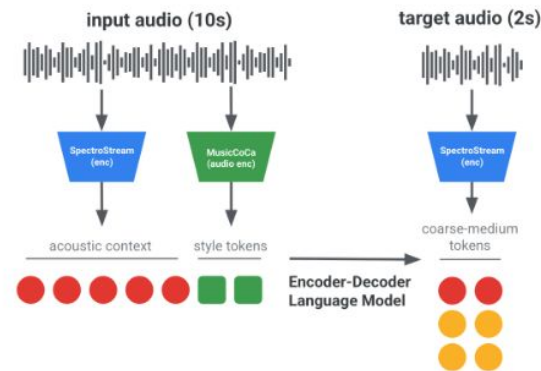
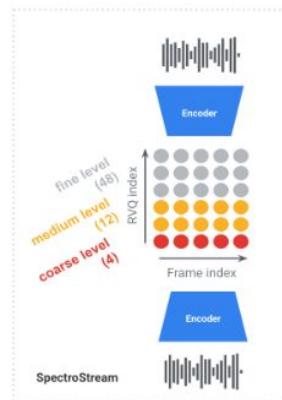
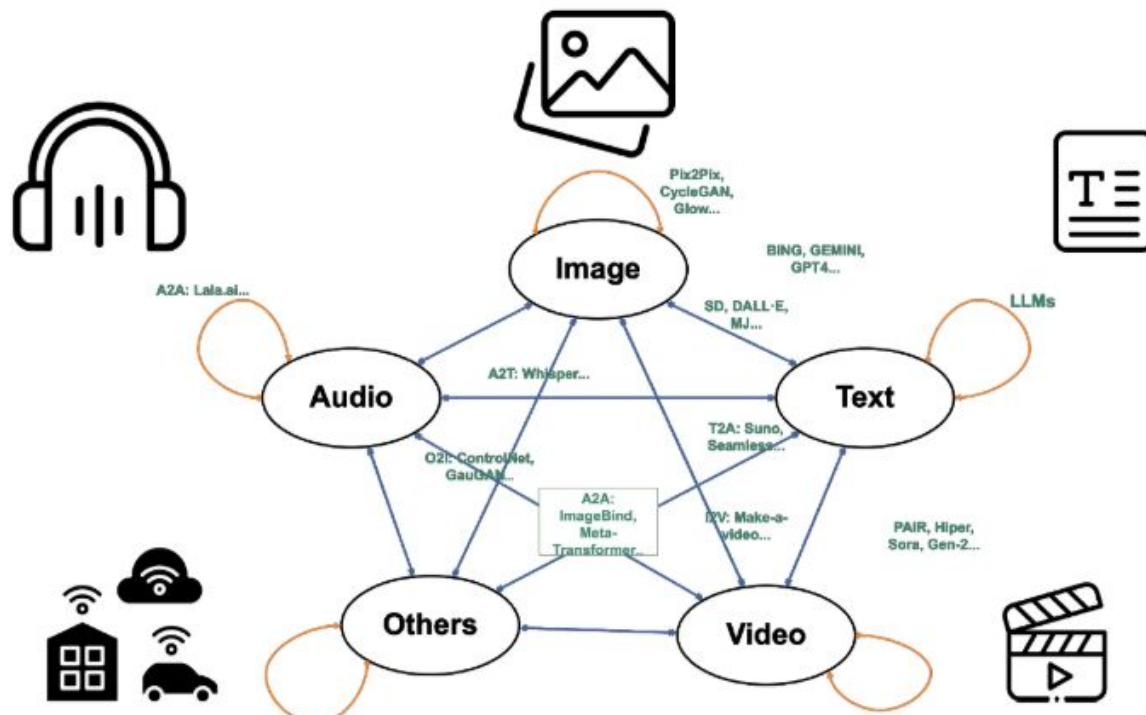


Figure 1: Magenta RealTime is a *live music model* that generates an uninterrupted stream of music and responds continuously to user input. It generates audio in two-second chunks using a pipeline with three components: (1) MusicCoCa, a *style embedding* model, (2) SpectroStream [20], an *audio codec* model, and (3) an encoder-decoder *language model*. For each chunk, a style embedding is computed via a weighted average of MusicCoCa embeddings of text and audio prompts from the user. Given this style embedding and 10 seconds (5 chunks) of past audio context, the language model decoder generates SpectroStream audio tokens for the new chunk, which is then decoded to audio.



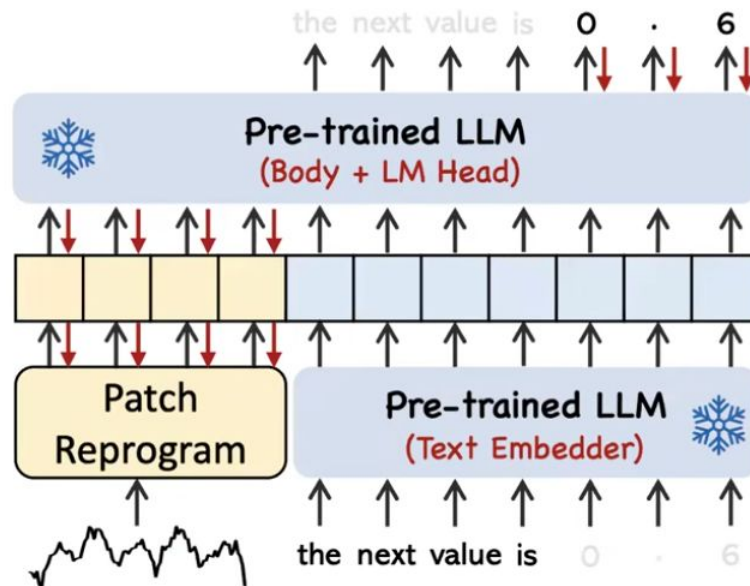
Otras modalidades



Time series:

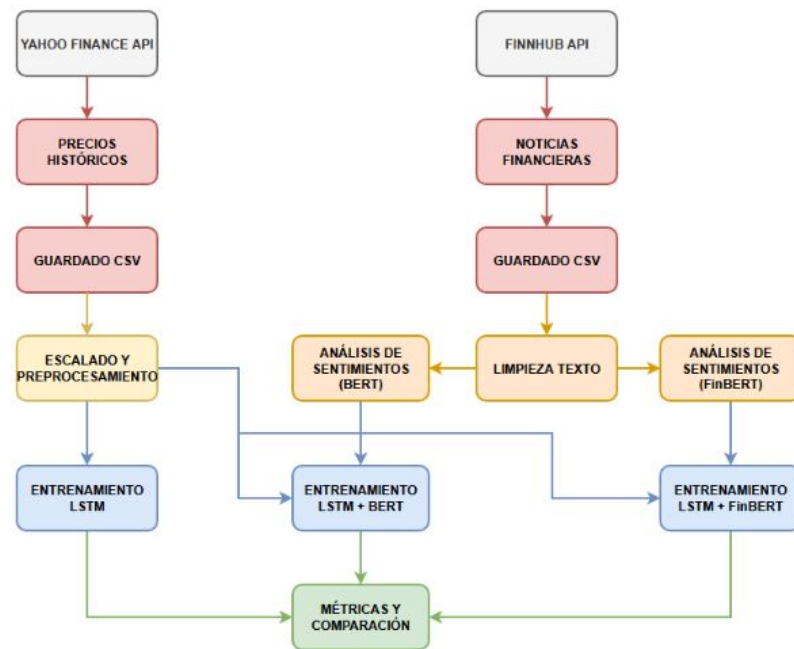
Una modalidad muy importante en datos económicos: series temporales

- Normalmente entrenados y alineados como los vision language models (pretraining + instruct stages).
- Sirven tanto para analysis como para predicciones.
- Ejemplos: [Time-LLM](#), [OneFitsAll](#)



Y aunque los modelos no sean multimodales podemos juntar nosotros las modalidades.

Impacto del Análisis de Sentimientos en la Predicción de Precios Financieros: Comparativa entre LSTM, BERT y FinBERT

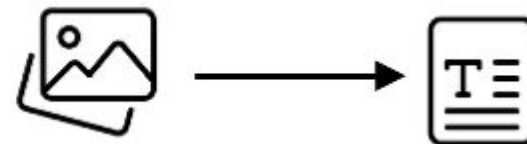


Otras modalidades

Imagen a Texto:

De estos ya hemos hablado mucho

Blip
Llava
InstructBLIP...

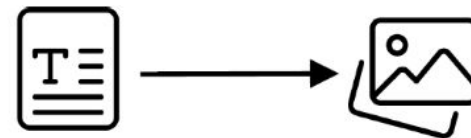


Describe esta imagen

La imagen muestra un paisaje montañoso con cielos nublados. En primer plano, hay un caballo de color marrón claro con crin y cola rubias, de complexión robusta, que está parado sobre una carretera asfaltada. Detrás de él, parcialmente oculto, se puede ver otro caballo.

Otras modalidades

Texto a imagen: De estos también hemos hablado



<https://stability.ai/>

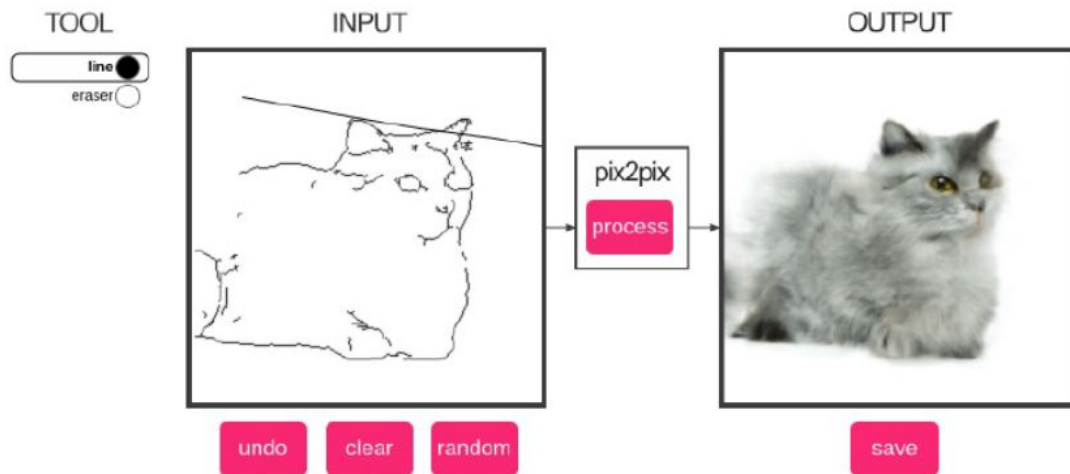
SD
Imagen3
DALL-E
Midjourney
Grok



Otras modalidades

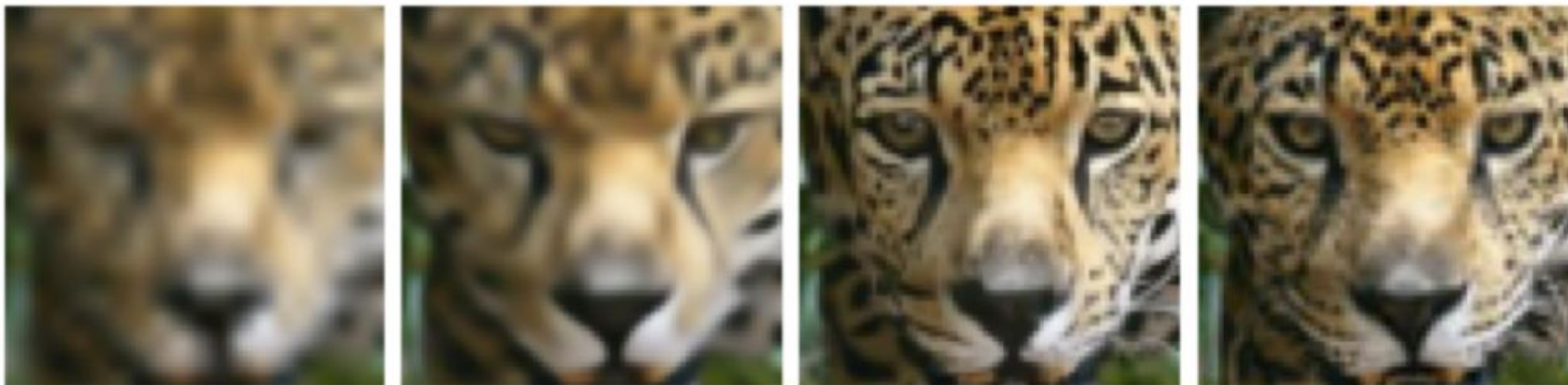
Imagen a imagen, generación condicionada a otra imagen (y generalmente también texto).

Pix-2-pix



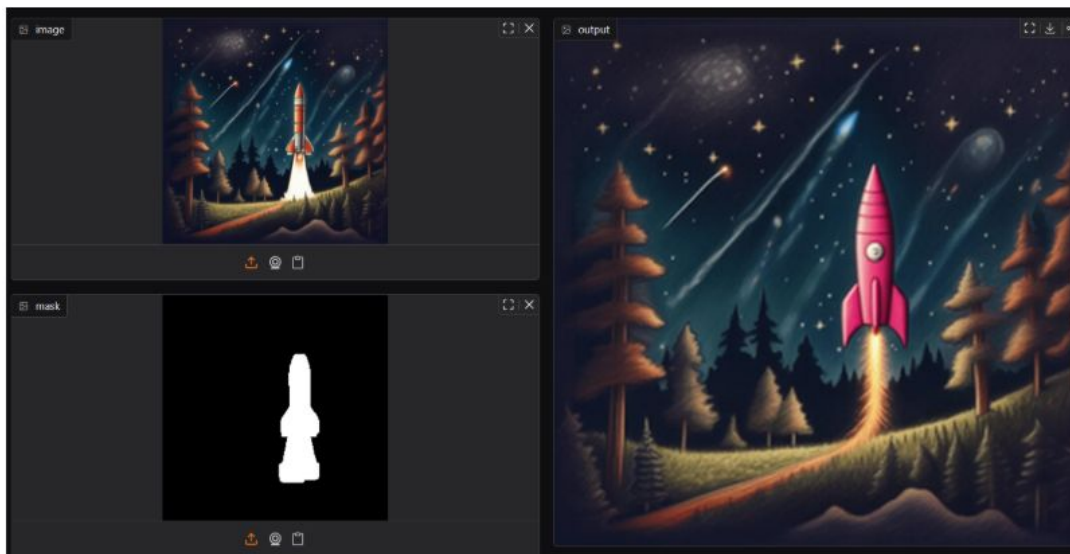
Otras modalidades

Imagen a imagen, superresolución mejora la calidad de la imagen. Usado en los modelos de generación de imagen y vídeo.



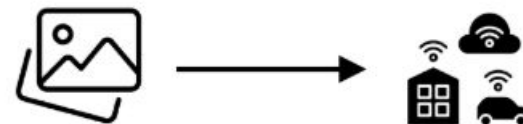
Otras modalidades

Imagen a imagen, inpainting también es imagen a imagen (sin texto normalmente).



Otras modalidades

De imagen a muchas modalidades!



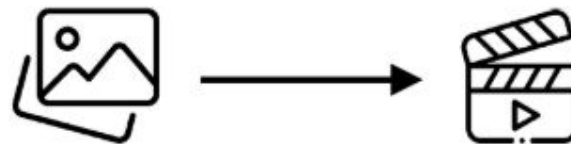
Yolo v11

<https://github.com/ultralytics/ultralytics?tab=readme-ov-file>



Otras modalidades

Video a partir de imagen, en este caso sin texto.



Stable-Video-Diffusion

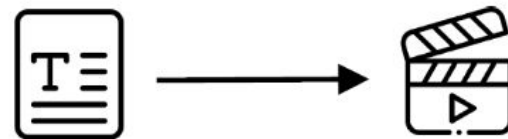
<https://huggingface.co/spaces/mediasynthesismuseum/stable-video-diffusion>

<https://huggingface.co/spaces/KlingTeam/LivePortrait>



Otras modalidades

Video a partir de texto, en este caso sin imagen.



VideoDiffusionModel

ImagenVideo

Omni-Video

Veo3

Sora

Wan2.1

<https://huggingface.co/spaces/Wan-AI/Wan2.1>

<https://openai.com/es-ES/sora/>

<https://deepmind.google/models/veo/>



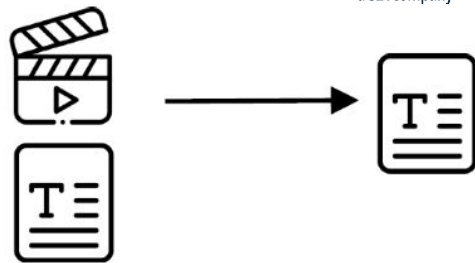
Otras modalidades

VQA sobre videos.

Perception LM

SmoIVLM2

Omni-Video



<https://huggingface.co/spaces/HuggingFaceTB/SmoIVLM2>

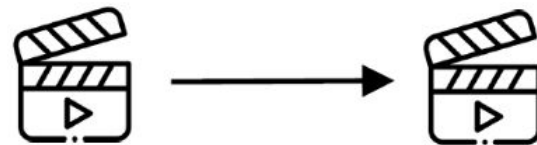
 Where do the severe droughts happen according to this diagram?	 What art era this artpiece <image> and this artpiece <image> belong to?	 Describe this image.	 When was this purchase made and how much did it cost?
 What is the date in this document?	 What is happening in the video?		

Otras modalidades

Highlights video generator

SmoIVLM2

<https://huggingface.co/spaces/HuggingFaceTB/SmoIVLM2>



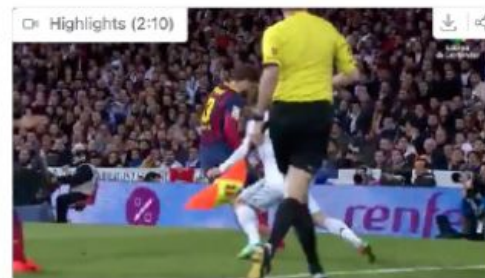
Video Highlight Generator

Upload a video and get an automated highlight reel!

Example Results



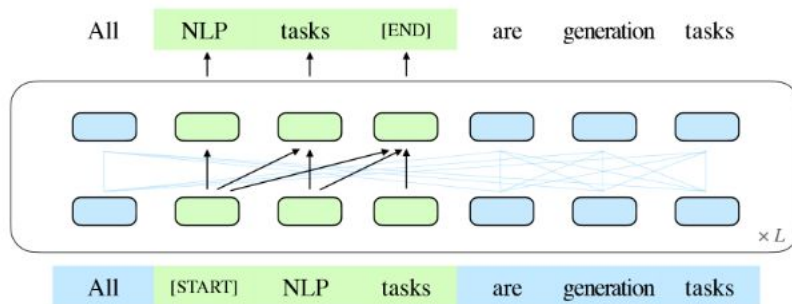
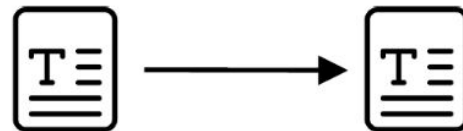
Football Match Highlights



Chain of thought details

Auto-regressive Language Models:

ChatGPT, Mistral, Llama, DeepSeek, Intern, Qwen...

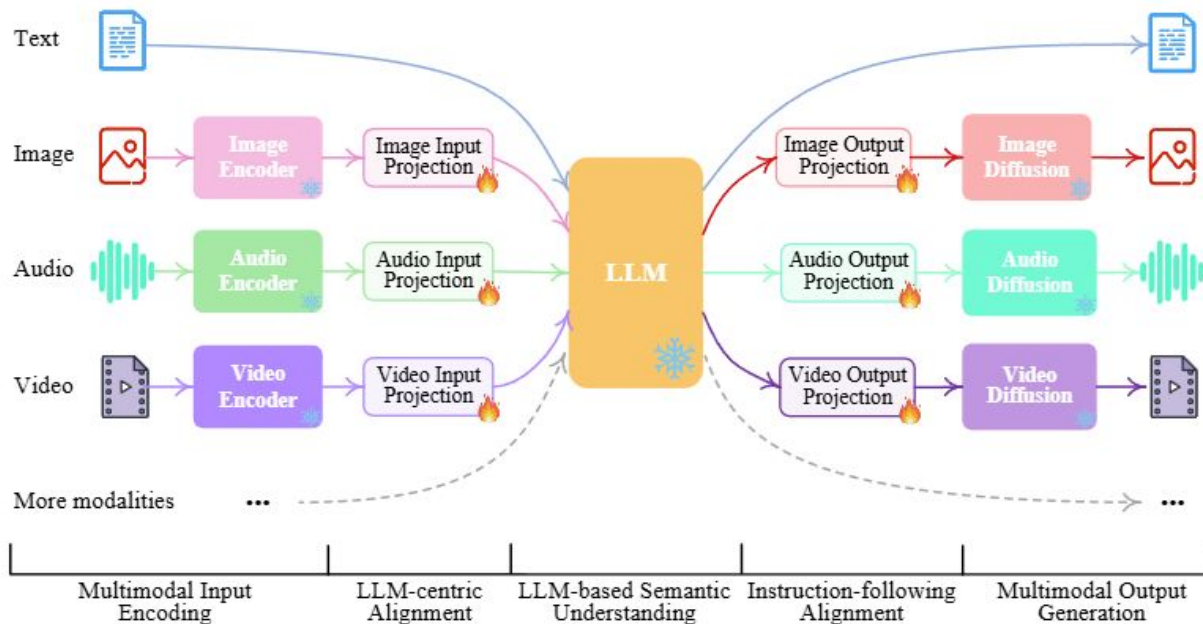


Ojo con los nuevos modelos de difusión de texto!

Gemini Diffusion: <https://blog.google/technology/google-deepmind/gemini-diffusion/>

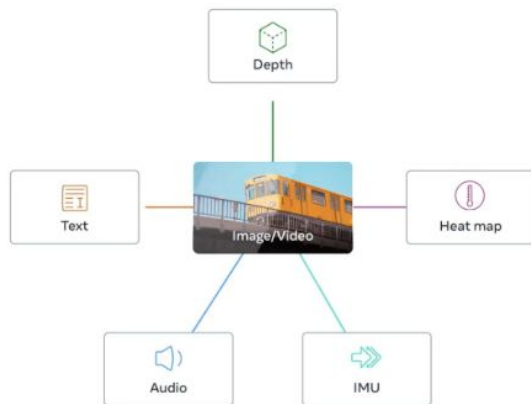
Modelos any-to-any:

NExT-GPT: Any-to-Any Multimodal LLM



Modelos any-to-any:

ImageBind: One Embedding Space to Bind Them All



1) Cross-Modal Retrieval

Audio	Images & Videos	Depth	Text
 Crackle of a Fire			"A fire crackles while a pan of food is frying on the fire." "Fire is crackling then wind starts blowing." "Firewood crackles then music..."
 Baby Cooing			"A baby is crying while a toddler is laughing." "A baby is laughing while an adult is laughing." "A baby laughs and something..."

2) Embedding-Space Arithmetic

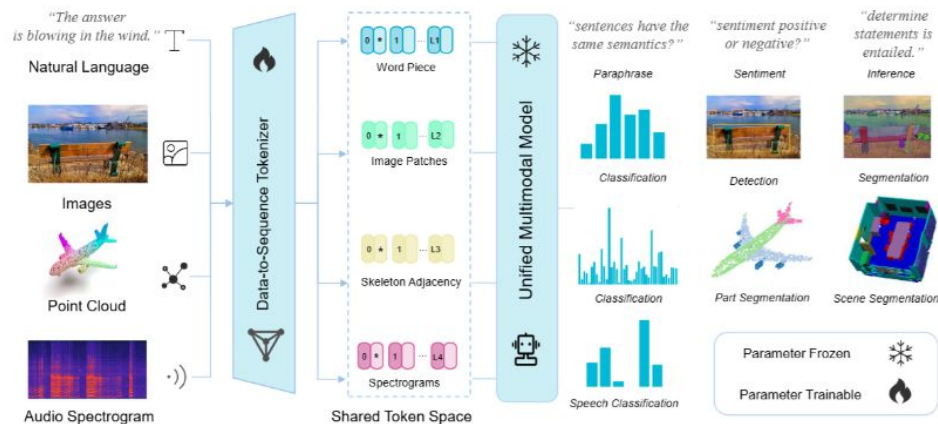
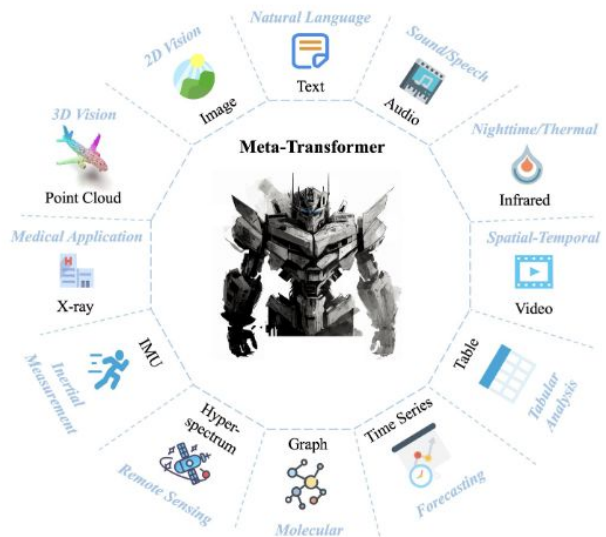
The diagram shows a sequence of images and audio waves. It starts with an image of a white egret, followed by a plus sign, a speaker icon, a minus sign, and another image of a white egret. Below the images are labels: 'Waves' and 'Waves'.

3) Audio to Image Generation

The diagram shows a sequence of images and audio waves. It starts with an image of a dog, followed by a speaker icon, an image of a bus, a speaker icon, an image of a fire, a speaker icon, and an image of rain. Below the images are labels: 'Dog', 'Engine', 'Fire', and 'Rain'.

Modelos any-to-any:

Meta-Transformer: A Unified Framework for Multimodal Learning



Modelos any-to-any:

Meta-Transformer: A Unified Framework for Multimodal Learning

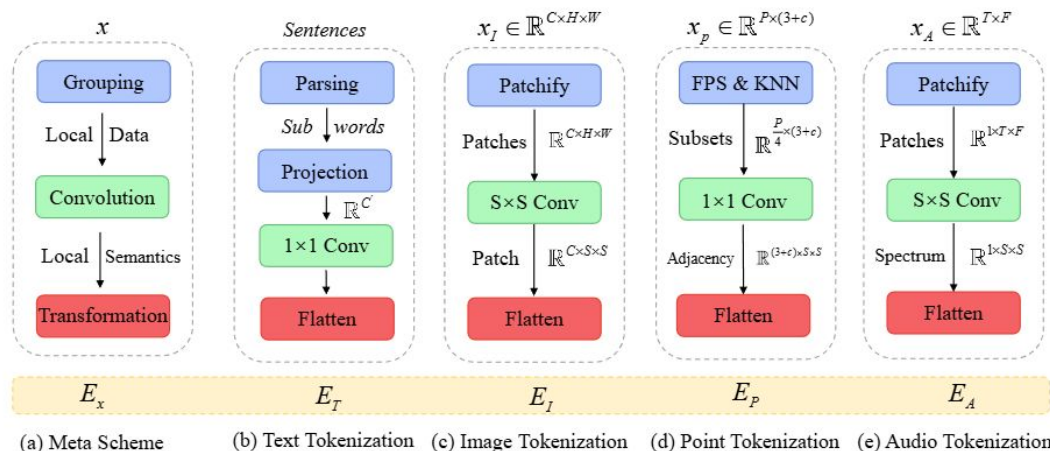


Figure 3: Illustration of Data-to-Sequence Tokenization 3.2. We propose the meta scheme in (a) containing grouping, convolution, and transformation progress. Then (b)-(e) represents the building blocks applied with our meta scheme on texts, images, point clouds, and audio spectrograms.

